Multimodal Graph Learning for Breast Cancer Survival Prediction

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Use of Generative AI

I acknowledge the use of ChatGPT (model: 6 Astra; OpenAI; https://chatgpt.com/) for proofreading and assistance with literature searches during the preparation of this thesis. The tool supported language refinement and the identification of potentially relevant publications. I also acknowledge the use of Claude (model: Fable 5.2; Anthropic; https://claude.ai/) for code proofreading. I remain responsible for the selection and verification of cited sources, the interpretation of the literature, and the accuracy and integrity of the final thesis.

Abstract

Spatial proteomics reveals the cellular organisation of breast tumours, yet summaries of individual tissue regions can obscure how cell states are paired across their boundaries. Capturing these relationships offers an opportunity to enrich survival prediction beyond clinical information and simpler tissue measurements.

This thesis introduces NIMSurv, a multimodal graph learning framework that integrates clinical predictors and predefined tissue summaries with learned representations of tissue regions and their interfaces. Its central methodological contribution is Niche-conditioned Interface Residual Pooling (NIRP), a readout that explicitly represents boundary-cell pairing by centring learned features separately on each side of an interface and pooling their paired cross-products. These representations are combined in an additive Cox-type model for patient-level survival prediction. A mathematical characterisation distinguishes within-interface pairing from between-interface variation, while an exact score decomposition traces fitted pairing contributions to individual boundary connections.

Development-stage evaluation used repeated patient-level cross-validation in 544 METABRIC patients to assess ten-year model-based net disease-specific survival. NIMSurv achieved Uno's concordance of 0.703 and an integrated Brier score of 0.113. Relative to a clinical reference incorporating cell composition and mean marker measurements, concordance increased by 0.035 (95% patient-bootstrap interval [0.006, 0.063]) and integrated Brier score decreased by approximately 0.008. Comparisons of refitted variants further supported the predictive value of paired-cell features beyond interface availability.

Matched genomic augmentation yielded a modest additional improvement in risk ordering. Complementary analyses in Basel assessed learning within a second cohort and frozen transfer of a separate common-input five-year overall-survival predictor. The transferred predictor achieved concordance of 0.725 in 276 patients without refitting or recalibration. Tissue maps localised selected fitted pairing contributions to cellular boundaries, while limited reconstruction by measured-marker surrogates clarified the distinction between exact computational tracing and explanatory fidelity.

Together, these contributions advance spatially informed survival modelling by making boundary-cell relationships an explicit part of multimodal prediction and linking fitted patient-level pairing scores directly to their tissue origins.

Publication Arising from This Thesis

The research reported in this thesis has given rise to the following peer-reviewed conference paper, which has been accepted for publication.

Binh Minh Tran, Lin Zhang, and Xiuying Wang. 2027. HiNT-Surv: hierarchical niche-aware multimodal survival prediction from imaging mass cytometry, clinicopathologic variables, and genomic data. In *2027 IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM)*, IEEE Symposium Series on Computational Intelligence (SSCI), Gold Coast, Australia. **Accepted, to appear**.

Chapters 4 and 5 are based in part on this paper. The majority of the methodology, text, figures and experiments in the paper originated from the first author, the author of this thesis.

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List of Abbreviations

Abbreviation or label	Meaning
General, statistical and biological terms	
API	Application programming interface
AUC	Area under the receiver operating characteristic curve
AUPR	Area under the precision–recall curve
Bi-LSTM	Bidirectional long short-term memory network
CI	Confidence interval
C-index	Concordance index
CK	Cytokeratin
CNN	Convolutional neural network
CODEX	Co-detection by indexing
CPH	Cox proportional hazards; also the benchmark with ER as a predictor
CSV	Comma-separated values
CV	Cross-validation
DNA	Deoxyribonucleic acid
DSS	Disease-specific survival; here, survival from breast-cancer death
ELU	Exponential linear unit
ER	Oestrogen receptor
ER^+ / ER^-	Oestrogen-receptor-positive / oestrogen-receptor-negative
GAT	Graph attention network
GNN	Graph neural network
H&E	Haematoxylin and eosin
HER2	Human epidermal growth factor receptor 2
HR	Hazard ratio
IBS	Integrated Brier score
IHC	Immunohistochemistry
IMC	Imaging mass cytometry
IPCW	Inverse probability of censoring weighting
IQR	Interquartile range
KL	Kullback–Leibler divergence
LN	Layer normalisation
MIBI	Multiplexed ion beam imaging
mIF	Multiplex immunofluorescence
MLP	Multilayer perceptron
n/a	Not available
OS	Overall survival
panCK	Pan-cytokeratin
PCA	Principal component analysis
PR	Progesterone receptor
Q1–Q5	Quintile groups of predicted risk in the Basel calibration analysis
ReLU	Rectified linear unit

Continued on next page

Abbreviation or label	Meaning
ResNet	Residual network; ResNet-50 has 50 layers
RMSE	Root mean squared error
ROC	Receiver operating characteristic
RQ	Research question; RQ1–RQ5 are stated in Section 1.3
RSF	Random survival forest
SD	Standard deviation
SMA	Smooth-muscle actin
SVM	Support vector machine
TME	Tumour microenvironment
X^{hi}, X^{med}, X^{lo}, X^{+}	High, intermediate, low or positive level of marker X in type labels
Cohorts, molecular classifiers and named methods	
COVET	Covariance environment
DMGN	Deep multimodal graph-based network
ENVI	Environmental variational inference
GraphSAGE	Graph sample and aggregate
IntClust	Integrative cluster
MCAT	Multimodal co-attention transformer
METABRIC	Molecular Taxonomy of Breast Cancer International Consortium
NIMSurv	Niche–Interface Multimodal Survival, the framework proposed in this thesis
NIRP	Niche-conditioned Interface Residual Pooling, the proposed interface readout
PAM50	Prediction Analysis of Microarray 50-gene classifier
SpaCEy	Spatial Clinical Explainability
SPIFEE	Graph-learning framework with spatially varying functional edge features
TCGA-LUAD	The Cancer Genome Atlas lung adenocarcinoma cohort
UTAG	Unsupervised discovery of tissue architecture with graphs
Model and component labels used in this thesis	
AttenSurv-MH	Locally adapted AttenSurv, an attention-based deep recurrent survival model
BroadSurv	Broad-learning survival model using pseudo conditional-probability targets
Clin	Clinical-only reference model
Clin+Comp	Clinical predictors with cell composition and mean marker levels
Clin+Flat	Clinical predictors, predefined tissue summaries and mean-pooled cell embeddings
Clin+Gen	Clinical predictors with genomic information
Clin+Node	Clinical predictors, predefined tissue summaries and the regional pathway
Clin+Sum	Clinical predictors with the complete predefined tissue summaries
CPH-strat	ER-stratified Cox proportional hazards benchmark
DAGSurv	Directed acyclic graph-based survival model
DeepHit	Neural survival model predicting probabilities over event times and causes
DeepSurv	Neural Cox proportional hazards model
DMGN-M	Locally adapted DMGN using marker-image regions and clinical predictors
Input-feature	NIRP variant using projected cell inputs instead of learned contextual embeddings
NIMSurv+Gen	NIMSurv with genomic augmentation, fitted using the staged procedure
Pooled-centred	NIRP variant centred after pooling endpoints by niche-type pair
Support-only	NIRP variant retaining availability indicators without the pairing matrices
Uncentred	NIRP variant using uncentred endpoint products

List of Notation

Symbol	Meaning	Section
Conventions and operators		
$\mathbf{a},\ \mathbf{A}$	bold lowercase and uppercase letters denote vectors and matrices, respectively	
$\widehat{x},\ \bar{x}$	estimated or fitted value of x; mean of x over the indicated set or endpoint list	
$\mathbf{a}^\top,\ [\mathbf{a};\mathbf{b}]$	transpose; vertical concatenation	
$\mathbb{I}(\cdot),\ \lvert\cdot\rvert$	indicator function; number of elements of a set	
$\langle \mathbf{A}, \mathbf{B}\rangle_{\mathrm{F}}$	Frobenius inner product, the sum of entrywise products	4.7.1
t^-	instant just before time t (left limit)	4.5.3
$\mathrm{READOUT},\ \mathrm{AGG}_{\mathrm{patient}}$	generic graph readout and patient-level aggregation	2.4
Indices, counts and dimensions		
$i,\ i'$	patients	2.2.2
$r = 1, \ldots, m_i$	tissue cores of patient i, with m_i retained cores	2.2.2
$u,\ v$	cells, the nodes of a spatial cell graph	2.2.1
$j,\ j'$	connected niche instances within a core	4.2
$a,\ b \in \{1, \ldots, K\}$	niche types; a type pair is written with $a < b$	4.2.3
$k \in \{0, 1, 2\}$	cell-encoder layer, with $k = 0$ the input	4.3.1
$p,\ p'$	measured marker channels	4.7.3
s	ER stratum, ER^+ or ER^-	2.3.3
t	time from the prediction origin, in years	2.3.1
$P,\ A,\ C,\ Q$	numbers of marker channels, phenotype categories, clinical coordinates and predefined tissue-summary coordinates	A.1
K	number of niche types ($K = 10$)	4.2.2
$d,\ d_{\mathrm{p}}$	cell-embedding dimension and projected-interface dimension	4.3
$n_{\mathrm{train}},\ n_{\mathrm{batch}}$	numbers of training patients and of batches per epoch	4.5.1
n_{eval}	number of evaluation patients	A.2.5
$N_r,\ E_r^{\mathrm{ret}}$	numbers of cells and of retained crossing edges in core r	A.2.3
Data, graphs and tissue objects		
$\mathcal{P}_{\mathrm{dev}},\ \mathcal{P}_{\mathrm{gen}},\ \mathcal{P}_{\mathrm{ext}}$	METABRIC development population, matched genomic subset and Basel overall-survival population	5.1.1
$\mathbf{x}_v^{\mathrm{marker}},\ \mathbf{p}_v,\ \ell_v$	marker vector, two-dimensional centroid and released phenotype label of cell v	2.2.1

Continued on next page

Symbol	Meaning	Section
$\mathcal{D}_{ir}$	IMC-derived cell features, coordinates and annotations for core r of patient i	2.4.3
$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$	spatial cell graph with cells $\mathcal{V}$, edges $\mathcal{E}$ and cell features $\mathbf{X}$	2.4.1
$\mathbf{z}_{\mathcal{G}},\ \mathbf{z}_{ir},\ \mathbf{z}_i$	generic graph-, core- and patient-level representations	2.4
f_θ	generic function mapping core data to a core representation	2.4.3
$G = (V, E)$	cell graph of one core, with cells V and undirected edges E	4.2
V_j	cell set of niche instance j	4.2
$t(j)$	niche type of instance j	A.2.2
V_{ir}	cells of core r of patient i	4.3.1
$\mathcal{N}(v),\ \mathcal{N}_{\text{niche}}(j)$	neighbours of cell v in the cell graph and of instance j among niche instances	4.3.1
$c_{jj'} = c_{j'j}$	number of crossing edges between instances j and j'	4.3.1
$I_{jj'},\ I$	interface, the oriented set of crossing edges between instances j and j'; I denotes a generic retained interface	4.2.3
$e = (u_e, v_e)$	crossing edge with left endpoint u_e and right endpoint v_e	4.3.2
n_I	number of crossing edges in interface I (n in Chapter 2)	4.3.2
$\mathcal{I}_{ab}^{(r)},\ \mathcal{I}_{i,ab}$	retained interfaces of type pair (a, b) in core r, and in all cores of patient i	4.3.3
$B,\ \mathbf{n}_L,\ \mathbf{n}_R$	interface biadjacency matrix and endpoint-appearance counts of its left and right cells	A.2.2
$F_L,\ F_R$	projected features of the distinct left and right endpoint cells	A.2.2
Cell encoder and regional pathway		
$\mathbf{h}_v^{(k)}$	embedding of cell v after encoder layer k; $\mathbf{h}_v^{(0)}$ joins processed markers and one-hot phenotype labels	4.3.1
$\mathbf{h}_v$	final cell embedding $\mathbf{h}_v^{(2)} \in \mathbb{R}^d$	4.3.1
$\bar{\mathbf{h}}_{\mathcal{N}(v)}^{(k)}$	mean embedding of the neighbours of cell v	4.3.1
$W_k^{\text{self}},\ W_k^{\text{nb}}$	self and neighbour weight matrices of encoder layer k	4.3.1
$\boldsymbol{\rho}_j^{(0)},\ \boldsymbol{\rho}_j^{(1)}$	regional state of instance j before and after the contact-weighted update	4.3.1
$U_{\text{self}},\ U_{\text{nb}}$	self and neighbour weight matrices of the regional layer, in $\mathbb{R}^{d\times(d+1)}$	4.3.1
$\alpha_{r,a}$	indicator that niche type a is absent from core r	4.3.1
w_{ir}	cell-count weight of core r for patient i	4.3.1
$\mathbf{z}_i^{\text{node}}$	patient regional-pathway block, with $K(d+1)$ coordinates	4.3.1
NIRP interface pathway		
$\boldsymbol{\Phi}$	shared bias-free endpoint projection, in $\mathbb{R}^{d_{\text{p}}\times d}$	4.3.2
$\mathbf{x}_e,\ \mathbf{y}_e$	projected left and right endpoint states, $\boldsymbol{\Phi}\mathbf{h}_{u_e}$ and $\boldsymbol{\Phi}\mathbf{h}_{v_e}$	4.3.2
$\bar{\mathbf{x}}_I,\ \bar{\mathbf{y}}_I$	endpoint means of interface I, counting every endpoint appearance ($\bar{\mathbf{x}}$, $\bar{\mathbf{y}}$ in Chapter 2)	4.3.2
$M_{xy},\ C_{xy}$	cross-moment and centred cross-moment of paired endpoint features	2.5.2
$\pi,\ \mathbb{E}_\pi$	uniformly random permutation of right endpoint slots, and expectation over it	2.5.2
$\mathbf{C}_I$	locally centred residual matrix of interface I	4.3.2
$N_{i,ab}$	number of retained crossing edges of type pair (a, b) for patient i	4.3.3

Continued on next page

Symbol	Meaning	Section
$m_{i,ab}$	unsupported-pair indicator $\mathbb{I}(N_{i,ab} = 0)$	4.3.3
$\mathbf{C}_{i,ab}$	edge-weighted patient residual matrix of type pair (a, b)	4.3.3
$\mathbf{z}_i^{\text{int}}$	patient interface-pathway block, with $\binom{K}{2}(d_{\text{p}}^2 + 1)$ coordinates	4.3.3
π_I	interface weight $n_I/N_{i,ab}$	4.3.4
N	shorthand for $N_{i,ab}$ in Appendix A.2.3	A.2.3
$\boldsymbol{\mu}_x,\ \boldsymbol{\mu}_y$	pooled endpoint means over a patient's retained interfaces of one type pair	4.3.4
$\mathbf{C}_{i,ab}^{\text{pool}},\ \mathbf{M}_{i,ab}^{\text{pool}}$	pooled-centred and uncentred pooled moments	4.3.4
$\mathcal{B}_{i,ab}$	between-interface term, $\mathbf{C}_{i,ab}^{\text{pool}} - \mathbf{C}_{i,ab}$	4.3.4
Patient predictors and survival model		
$T_i,\ U_i,\ \widetilde{T}_i,\ \Delta_i$	event time, censoring time, observed time $\min(T_i, U_i)$ and endpoint-specific event indicator	2.3.1
$S(t \mid \mathbf{w}),\ H(t \mid \mathbf{w})$	survival function and cumulative hazard for predictors $\mathbf{w}$	2.3.1
$h(t \mid \mathbf{w}),\ h_0(t)$	hazard and Cox baseline hazard	2.3.3
$H_1,\ S_{1,\text{net}}$	cumulative breast-cancer hazard and model-based net survival $\exp(-H_1)$	2.3.2
$\mathbf{c}_i,\ \mathbf{g}_i,\ \mathbf{q}_i$	clinical, genomic and predefined tissue-summary vectors	4.1.3
$\mathbf{v}_i$	tissue block $[\mathbf{z}_i^{\text{node}}; \mathbf{z}_i^{\text{int}}; \mathbf{q}_i]$	4.3.5
$\mathbf{w}_i$	patient predictors $[\mathbf{c}_i; \mathbf{g}_i; \mathbf{v}_i]$, with $\mathbf{g}_i$ omitted outside the genomic extension	4.3.5
$s_i,\ \nu_i$	ER stratum of patient i and ER-negative indicator $\mathbb{I}(s_i = \text{ER}^-)$	4.3.5
$\eta_i,\ \widehat{\eta}_i$	log-relative-hazard score and its fitted value	2.3.3
$h_{0,s}(t),\ h_1(t \mid \mathbf{w}_i, s_i)$	baseline hazard of ER stratum s and breast-cancer cause-specific hazard	4.3.5
$\boldsymbol{\beta}_C,\ \boldsymbol{\beta}_G,\ \boldsymbol{\beta}_V$	clinical, genomic and shared tissue coefficients ($\boldsymbol{\beta}$ in Chapter 2)	4.3.5
$\boldsymbol{\gamma}_V$	ER-negative deviation of the tissue coefficients	4.3.5
$R_s(t),\ D_s(t)$	risk set and tied target-event set of stratum s at time t; $R_s(t; \mathcal{A})$ and $D_s(t; \mathcal{A})$ within fitting set $\mathcal{A}$	4.3.5
$\widehat{H}_{0,s}(t),\ \widehat{S}_{1,\text{net}}(t \mid \mathbf{w}_i, s_i)$	Breslow baseline cumulative hazard of stratum s and estimated net survival	4.3.5
$\mathcal{L},\ \mathcal{L}_{\text{Cox}}$	penalised training objective and event-normalised negative Cox partial log-likelihood	4.5.1
$\lambda_\beta,\ \lambda_\gamma,\ \lambda_\omega$	penalties on the coefficients, the ER deviations and the neural weights	4.5.1
$\boldsymbol{\omega}$	neural parameters of the cell encoder, regional layer and projection $\boldsymbol{\Phi}$	4.5.1
$D_{\mathcal{A}},\ \Sigma_s^R(t),\ \Sigma_s^D(t)$	number of target events in $\mathcal{A}$, and sums of e^η over $R_s(t)$ and $D_s(t)$ in the Efron objective	A.2.4
Prediction and evaluation		
τ	prediction horizon	4.5.3
$\widehat{S}_i(t)$	fitted endpoint-specific survival curve of patient i	2.3.3
$\widehat{\psi}_i(\tau)$	horizon risk $1 - \widehat{S}_i(\tau)$, used for ranking	4.5.3
$\widehat{G}_s(t)$	reverse Kaplan–Meier censoring survival in ER stratum s	4.5.3
$w_{ii'}$	censoring weight of the comparable pair (i, i')	4.5.3

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Symbol	Meaning	Section
$\chi_{ii'}(\tau),\ \kappa_{ii'}$	comparable-pair indicator and concordance credit, with half credit for tied predictions	A.2.5
$C,\ \widehat{C}_\tau$	concordance, Uno's or Harrell's as labelled, and the truncated Uno-type estimate	4.5.3
$\widehat{\mathrm{BS}}(t)$	censoring-adjusted Brier score at time t; $\widehat{\mathrm{BS}}_{\mathrm{net}}(t)$ for DSS	A.2.5
$\widehat{\mathrm{IBS}}(0,\tau)$	integrated Brier score over $[0,\tau]$; $\widehat{\mathrm{IBS}}_{\mathrm{net}}(0,\tau)$ for DSS	A.2.5
$\Delta C,\ \Delta\mathrm{IBS}$	paired differences: NIMSURV minus comparator, or augmented minus unaugmented for genomics; positive ΔC and negative $\Delta\mathrm{IBS}$ favour the first model	4.5.4
Interpretation and explanatory fidelity		
$\boldsymbol{\Theta}_{ab},\ \boldsymbol{\Gamma}_{ab}$	pairing coefficient matrix of type pair (a,b) and its ER-negative deviation	4.7.1
$\eta^{\mathrm{pair}}_{i,ab}$	raw pairing log-risk contribution of type pair (a,b) for patient i	4.7.1
$a_{i,I,e}$	exact allocation of $\eta^{\mathrm{pair}}_{i,ab}$ to crossing edge e of interface I	4.7.1
$b_{ab,s},\ \mathcal{A}_s$	ER-specific reference contribution, the mean over the stratum-s outer-training patients $\mathcal{A}_s$	A.4
$\widetilde{\eta}^{\mathrm{pair}}_{i,ab}$	display-centred pairing contribution $\eta^{\mathrm{pair}}_{i,ab} - b_{ab,s_i}$	A.4
b_e	frozen local pairing score of edge e, equal to $N_{i,ab}\, a_{i,I,e}$	4.7.3
$t_{e,pp'}$	centred product of measured markers p and p' at the endpoints of edge e, used as surrogate input	4.7.3
R^2	coefficient of determination for surrogate fidelity	4.7.3

Chapter 1

Introduction

1.1 Clinical Motivation

Two patients with breast cancer can have similar clinical findings, receive the same treatment, and still experience very different outcomes. Clinical prognostic tools, such as the Nottingham Prognostic Index and PREDICT, help estimate a patient's outlook from clinical and pathological measurements (Galea et al., 1992; Wishart et al., 2010). Genomic studies of the METABRIC breast cancer cohort have refined this picture by identifying molecular subgroups with different survival patterns (Curtis et al., 2012; Pereira et al., 2016). Yet outcomes still vary within these groups. This variation motivates the search for information that clinical records and bulk molecular profiles, which combine material from many cells, may not capture.

Imaging mass cytometry (IMC) provides a way to examine this information at the cellular level. It measures dozens of proteins in individual cells while preserving their positions in the tissue (Giesen et al., 2014). In breast cancer, IMC studies have identified cellular environments associated with molecular subtype and survival (Jackson et al., 2020; Ali et al., 2020; Danenberg et al., 2022). These environments differ not only in which cells they contain, but also in how those cells are arranged. Two tumours, for example, may contain the same proportion of immune cells. In one, those cells may be dispersed among tumour cells; in the other, they may be concentrated in separate regions. Studies linking such arrangements to patient outcomes motivate preserving spatial information rather than reducing the tissue to counts and averages (Keren et al., 2018; Schürch et al., 2020; Danenberg et al., 2022).

The challenge is to turn these detailed tissue measurements into a prediction for an individual patient. IMC measures thousands of cells within small tissue samples, called *cores*, and a patient may contribute several cores. A model must combine information across these samples, relate it to the patient's clinical record, and estimate survival over time. It must also account for incomplete follow-up, where observation ends before the event of interest occurs (Cox, 1972). Combining these sources is a form of *multimodal survival prediction*: learning from more than one type of patient information to predict survival. Related approaches have combined features learned from tissue images with clinical and molecular data in

histology (Chen et al., 2022a, 2021b; Jaume et al., 2024) and, more recently, in multiplexed imaging (Fu et al., 2023).

1.2 Problem Statement

Biological association does not by itself establish predictive value. A tissue pattern may be associated with survival without improving prediction once other patient information is available. Assessing learned tissue representations therefore requires asking both what information they preserve and what they add to an informative comparison model. This thesis addresses three related problems.

Added value beyond clinical information and simple tissue measurements. DMGN combines IMC image regions with patient variables, whereas SpaCEy learns from cell graphs under a tissue-only survival prediction protocol (Fu et al., 2023; Rifaioglu et al., 2026). Their reported comparisons do not establish what learned tissue representations add when a reference model already includes clinical predictors and predefined tissue summaries, such as cell-type proportions and average protein measurements. This comparison matters because an improvement over a clinical-only model may come from information in the tissue that simpler measurements already provide, rather than from learning its spatial organisation.

Representing cell pairs at tissue boundaries. A cell graph represents a tissue core as a network of nearby cells. A graph model combines each cell's measurements with information from its neighbours to learn a numerical description, referred to here as its *learned cell state*. Regional approaches also describe recurring cellular environments and connected tissue regions (Bhate et al., 2022; Xie et al., 2026). In this thesis, recurring mixtures of nearby cell types define *niche types*. A connected region assigned to one type is a *niche instance*, and the graph connections crossing between two neighbouring instances form an *interface*.

At an interface, the question is not only which cell states occur on each side, but which states are paired across the boundary. Two interfaces can contain the same states on either side and the same number of connections, yet connect those states in different combinations. Simple summaries of each side do not reveal this distinction. Existing graph models already pass information across boundaries, and statistical pooling methods represent relationships between feature values (Haviv et al., 2025; Wang and Ji, 2023). The studies reviewed in Chapter 3 nevertheless leave open whether explicitly summarising paired cell states at individual interfaces improves survival prediction beyond clinical information, predefined tissue summaries and learned regional features. This motivates an interface-level *readout*: a step that turns boundary-cell pairings into features for patient-level prediction.

Scope and interpretation of learned tissue features. Spatial and genomic measurements describe related aspects of disease, so genomic information may overlap with what a tissue model already captures. A predictor trained in one patient cohort may also perform differently in another with different measurements or outcomes. Interpretation raises a further question: mapping a fitted contribution onto tissue shows where part of the calculation comes from, but not whether measured proteins can explain its value. These questions require separate comparisons. Better performance from the complete framework cannot answer them on its own.

> *This thesis argues that integrating clinical information with the spatial organisation of cells can improve patient-level survival prediction. It also shows how selected contributions to these predictions can be traced back to the underlying tissue, connecting patient-level modelling with cellular patterns.*

1.3 Research Aim and Questions

The aim of this thesis is to investigate whether integrating clinical information with the spatial organisation of cells can improve patient-level survival prediction in breast cancer. It develops and evaluates a multimodal graph learning framework that combines clinical information, predefined tissue summaries and learned spatial representations from imaging mass cytometry (IMC). These representations describe tissue regions and paired cell states across their boundaries. The study also examines the added value of genomic information, transfer across cohorts and the interpretation of selected model contributions.

The primary analysis evaluates ten-year model-based net disease-specific survival (DSS) in METABRIC (Danenberg et al., 2022), with breast-cancer death as the event. Evaluation considers both patient risk ordering and survival-probability error. Cross-cohort transfer is assessed using a separate five-year overall survival (OS) model restricted to measurements shared by METABRIC and Basel (Jackson et al., 2020). A complementary analysis trains and evaluates the framework within Basel (Section 5.3.2).

Five research questions guide the study:

RQ1 Does the combined clinical–IMC framework improve survival prediction over clinical information alone or clinical information combined with predefined tissue summaries? *(Section 5.2.1)*

RQ2 Does explicit boundary-cell pairing add predictive value beyond regional features and information about the availability of boundary connections? *(Section 5.2.2)*

RQ3 Does genomic information improve survival prediction beyond clinical and IMC data when models are compared on the same patients? *(Section 5.4)*

RQ4 How well does the shared-input five-year OS model transfer from METABRIC to Basel without refitting or recalibration? *(Section 5.5)*

RQ5 How can the pairing component of a fitted risk score be traced to niche-type pairs and individual interfaces? How faithfully can models based on measured proteins reproduce selected pairing scores? *(Section 5.6)*

1.4 Research Approach and Contributions

NIMSurv (Niche–Interface Multimodal Survival) represents each tissue core as a graph of nearby cells. Niche types and their connected instances are identified without using survival outcomes. A shared cell-graph encoder then learns cell features from the survival task. Two pathways summarise these features: the regional pathway describes niche instances and their surroundings, while Niche-conditioned Interface Residual Pooling (NIRP) describes how cell states are paired across each interface. Information from each patient's cores is combined with clinical predictors and predefined tissue summaries in an additive Cox-type survival model. The feature groups contribute to a patient-level risk score used to estimate survival over time.

The design builds on graph representation learning (Hamilton et al., 2017), Cox survival modelling (Cox, 1972), and statistical and graph-pooling ideas for representing relationships between features (Wang and Ji, 2023; Haviv et al., 2025; Barber, 2007). Evaluation keeps all cores and data sources from each patient together, and assesses predictions on patients held out from model fitting and selection. The thesis makes three linked contributions.

(1) **A multimodal graph learning framework for breast cancer survival prediction.** NIMSurv brings regional and boundary-pairing information into a common patient-level model alongside clinical predictors and predefined tissue summaries. This provides an integrated setting for assessing what learned tissue representations add to simpler measurements.

(2) **NIRP, an interface-level pairing readout with mathematical characterisation and exact score tracing.** NIRP measures cell features relative to the average on their own side of each interface, then pools products of the resulting paired deviations. It combines these summaries by niche-type pair across a patient's cores. The mathematical analysis accounts for cells participating in multiple

boundary connections, distinguishes variation within interfaces from differences between interface means, and clarifies what local, pooled and uncentred representations retain. For a fixed fitted model, an exact decomposition traces the pairing component of its log-risk score to individual boundary connections.

(3) **Empirical evidence on predictive value, cross-cohort transfer and explanatory fidelity.** Evaluation in METABRIC supports added predictive value beyond clinical predictors and predefined tissue summaries. Component comparisons examine pairing within refitted models. Matched genomic analyses and transfer of a separate OS predictor examine additional inputs and cross-cohort prediction. Tissue tracing locates selected fitted contributions, while simpler models based on measured proteins test how faithfully those scores can be reproduced. The tested models do not reconstruct the selected scores reliably, distinguishing exact computational tracing from a faithful measured-protein explanation.

1.5 Thesis Organisation

Chapter 2 introduces the biological, survival and graph-learning foundations. Chapter 3 reviews spatial proteomics, multimodal survival prediction and tissue representations, and identifies the gaps motivating RQ1–RQ5. Chapter 4 specifies NIMSurv, NIRP and the evaluation designs. Chapter 5 presents the findings by research question. Chapter 6 interprets their significance and limitations, and Chapter 7 summarises the contributions and priorities for further evaluation.

CHAPTER 2

Background

Predicting survival from tumour tissue requires connecting information collected at different scales. Clinical records describe the patient and their disease. Tissue samples reveal individual cells, the proteins they contain and the way those cells are arranged. Follow-up records describe what happens to the patient over time. The challenge is to combine these measurements into a prediction for the patient, rather than treating individual cells or tissue samples as independent outcomes.

This chapter follows three connected questions: what do the measurements tell us about the tissue, how can they be turned into a patient-level survival prediction, and what evidence is needed to evaluate and interpret that prediction? It begins with the biological setting and the available data, then introduces survival modelling, spatial graphs and the representation of cells at tissue boundaries. The final sections explain how predictions are assessed and how fitted model contributions can be traced back to tissue. Technical terms are introduced in context, and the equations are accompanied by explanations of what they mean. The List of Notation collects the symbols used throughout the thesis.

2.1 Breast Cancer Prognosis and the Tumour Microenvironment

2.1.1 Prognostic Heterogeneity and Oestrogen-Receptor Status

Prognosis describes the likely course of a disease. In breast cancer, it is informed by patient characteristics and tumour features, including tumour size, grade, lymph-node involvement and receptor status. Tools such as the Nottingham Prognostic Index and PREDICT combine these measurements into a score or survival estimate, while multigene assays provide additional molecular information (Galea et al., 1992; Wishart et al., 2010; Amin et al., 2017; Paik et al., 2004; Cardoso et al., 2016). Even so, patients with similar clinical characteristics can experience different outcomes (Curtis et al., 2012; Pereira et al., 2016). This variation motivates the search for information that routine measurements may not capture.

Oestrogen-receptor (ER) status is important because ER groups differ in treatment, risk and the timing of events. ER-positive disease, for example, can recur well beyond five years after diagnosis (Pan et al., 2017). A survival prediction therefore needs a stated *horizon*: how far into the future it describes. This thesis

uses ten years for its primary prediction target. The association between a tissue feature and outcome can also differ by ER status (Danenberg et al., 2022).

These differences raise two separate modelling questions. Do the ER groups have different patterns of events over time? Does the same tissue feature have a different association with risk in each group? Separate baseline hazards address the first question, while an interaction term addresses the second. Section 2.3.3 explains these choices after introducing the survival model.

2.1.2 Composition, Cell State and Spatial Organisation

A tumour contains more than malignant cells. Immune cells, supporting stromal cells and blood-vessel cells form its *tumour microenvironment* (TME). This thesis describes that environment through three complementary views. *Composition* records the proportions of different cell types. *Measured cell state* describes a cell through its protein profile: two T cells may share a type label but have different protein measurements (Jackson et al., 2020; Ali et al., 2020). *Spatial organisation* describes how the cells are arranged, including which cells lie next to one another.

These views answer different questions. Knowing how many immune cells a sample contains does not tell us where they are. In Figure 2.1, two tissue fields contain the same numbers of tumour, immune and stromal cells, but immune cells are dispersed among tumour cells in one and concentrated in a separate region in the other. Their composition is identical, while their organisation differs. Studies linking cellular neighbourhoods to patient outcomes motivate retaining this spatial information (Keren et al., 2018; Schürch et al., 2020; Danenberg et al., 2022). The cell graphs introduced in Section 2.4 provide a way to describe both the cells and their arrangement.

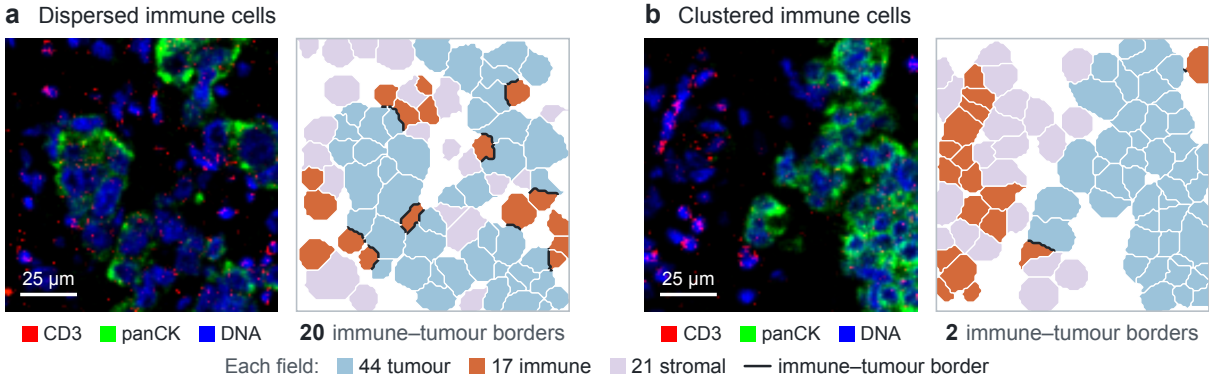


Figure 2.1. Same cellular composition, different spatial organisation in METABRIC–IMC cores 699 and 88 (Danenberg et al., 2022): immune cells are (a) dispersed or (b) clustered.

2.2 Multimodal Measurements and Units of Analysis

2.2.1 Clinical, Genomic and Spatial Proteomic Measurements

A *modality* is a source of measurements. This thesis draws on clinical, genomic and spatial proteomic data, which describe different aspects of the same patient. *Clinical records* include information such as age, tumour size, grade, lymph-node status and receptor status. *Bulk genomic measurements* describe molecular features, including gene expression and DNA alterations, in sampled tumour material. Because they combine material from many cells, they do not show which individual cells contain those features (Curtis et al., 2012; Pereira et al., 2016).

Imaging mass cytometry (IMC) measures selected proteins while preserving their positions in the tissue (Giesen et al., 2014). Antibodies carrying heavy-metal tags bind to chosen proteins in a thin tissue section. A laser removes the tissue point by point, and a mass cytometer detects the tags. The measurements form one image for each protein, showing where it was detected. Together, the selected proteins form the *marker panel*. Proteins outside this panel are not measured.

To use these images in a cell-based model, image processing first estimates the boundary of each cell. This step is called *segmentation*. Measurements inside the estimated boundary are then combined into the cell's protein profile. *Annotation* uses that profile to assign a phenotype, or cell-type label, such as T cell or fibroblast (Greenwald et al., 2022; Ali et al., 2020; Windhager et al., 2023). Figure 2.2 shows the stages leading from an image to a spatial cell graph.

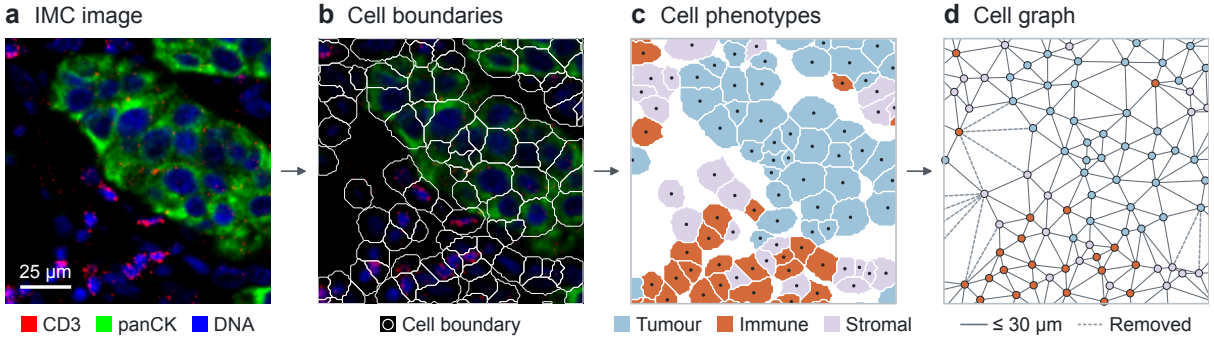


Figure 2.2. From an IMC image to a cell graph in METABRIC–IMC core 318 (Danenberg et al., 2022): (a) three of the measured channels; (b, c) released cell boundaries, centres and phenotype groups; (d) Delaunay edges, with edges longer than 30 μm removed (Section 4.2.2). Scale bar: 25 μm.

Each cell v is therefore described by three things: its marker vector $\mathbf{x}_v^{\text{marker}}$, an ordered list of protein measurements; its estimated centre $\mathbf{p}_v$; and its phenotype label ℓ_v. These describe what was measured, where the cell lies and how it was classified.

All three depend on processing decisions. Moving an estimated cell boundary changes the measurements assigned to that cell, while changing its label changes the composition of its neighbourhood. Chapter 4 specifies the marker panels and processing steps used in this study.

2.2.2 Patient-Level Prediction and Tissue Sampling

Thousands of measured cells do not provide thousands of independent survival outcomes. Patient i may contribute several small tissue samples, called *cores*, indexed by $r = 1, \ldots, m_i$. All of those cores belong to the same patient and share one survival record. This is a *multiple-instance* structure: the outcome belongs to a collection of instances, rather than being observed separately for each instance (Ilse et al., 2018; Campanella et al., 2019).

The prediction must respect this structure. Cell measurements are combined into descriptions of cores, and the available cores are combined into a description of the patient. The number of patients, rather than the number of cells, determines how many independent patient records are available for learning and evaluation.

Sampling also limits what the model can see. A core contains only part of a tumour, and different regions may contain different cellular environments (Jackson et al., 2020). Additional cores can provide a broader view, but the resulting representation still describes the sampled tissue, not the entire tumour. A cell population or spatial pattern that is not seen in a core may be present elsewhere. Chapter 6 returns to this limitation.

2.3 Foundations of Survival Prediction

Once the data have been linked to the patient, the next question is what the model should predict. Survival prediction concerns both whether an event occurs and when it occurs. Its defining difficulty is that follow-up is often incomplete: some patients are still alive when observation ends.

2.3.1 Event Times and Censoring

A survival outcome measures time from a defined starting point, such as diagnosis, to an event, such as death. The event is not observed for every patient. In Figure 2.3, patient A died of breast cancer after three years, so the event time is known. Patient B was last seen alive at six years. We know that B survived to that point, but not what happened afterwards. This incomplete observation is called *right censoring*.

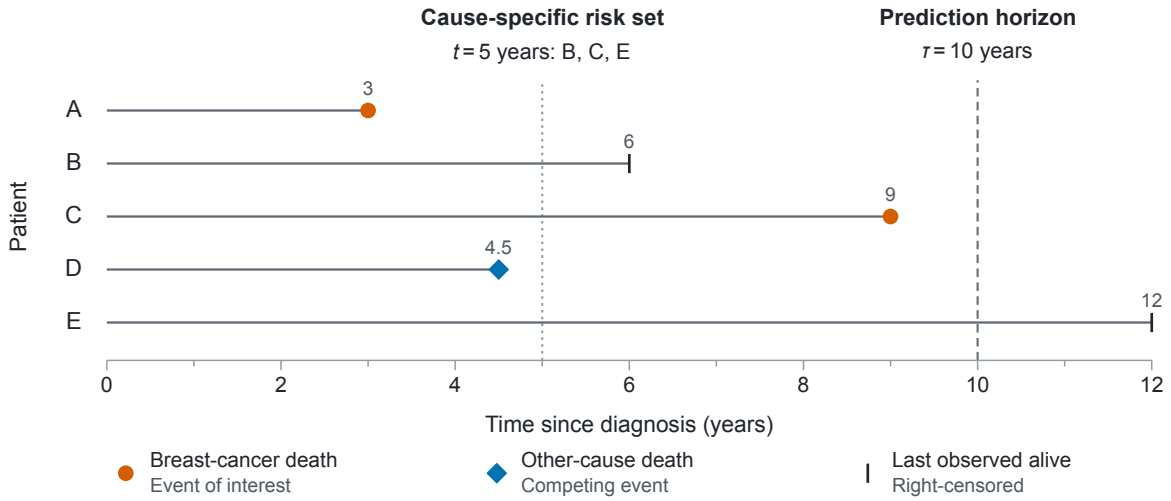


Figure 2.3. Follow-up, censoring and risk sets. Markers distinguish breast-cancer death, other-cause death and right-censoring; the cause-specific risk set at year five comprises B, C and E.

Let T_i be patient i's event time and U_i the time at which observation would end. The recorded data are

$$\widetilde{T}_i = \min(T_i, U_i), \qquad \Delta_i = \mathbb{I}(T_i \leq U_i), \tag{2.1}$$

where Δ_i is one if the event is observed and zero otherwise. In words, the record contains the earlier of the event and follow-up end, together with an indicator of which occurred. Survival methods use the known follow-up of censored patients. They neither discard those patients nor treat their last observed visit as a death.

At any time, the *risk set* contains the patients who are still under observation and able to experience the event. At year five in Figure 2.3, the risk set contains B, C and E. Patient A has already died of breast cancer, and D has died of another cause. B remains in the risk set until censoring at year six, contributing information to comparisons before that time but not afterwards. Censoring therefore removes later information without discarding earlier follow-up. The treatment of other-cause deaths is explained in the next subsection.

The Kaplan–Meier estimator uses the changing risk sets to estimate a survival curve without prescribing its shape (Kaplan and Meier, 1958). Its interpretation relies on *independent censoring*: within the groups being analysed, the end of observation should not itself provide information about when the event would occur (Kalbfleisch and Prentice, 2002). Follow-up ending because a patient has become too ill, for example, would violate this assumption.

Three related quantities describe survival. The *survival probability*, $S(t \mid \mathbf{w})$, is the chance of remaining event-free beyond time t for a patient with predictors $\mathbf{w}$. The *hazard*, $h(t \mid \mathbf{w})$, is the instantaneous event rate among patients who remain event-free. It describes how quickly events are occurring at that moment, rather than the probability of an event over a whole period. Accumulating that rate up to time t gives the *cumulative hazard*, $H(t \mid \mathbf{w})$. The relationship $S(t \mid \mathbf{w}) = \exp[-H(t \mid \mathbf{w})]$ converts cumulative hazard into survival probability: a larger accumulated hazard means lower survival (Kalbfleisch and Prentice, 2002).

2.3.2 Survival Endpoints and Competing Risks

The event definition determines the question being answered. *Overall survival* (OS) counts death from any cause, whereas *disease-specific survival* (DSS) in this thesis concerns death from breast cancer. In Figure 2.3, A, C and D are OS events, but only A and C are DSS events. D's death from another cause is a *competing event*: it prevents a later breast-cancer death.

The *cause-specific hazard* describes the rate of breast-cancer death among patients who are still alive. When this hazard is fitted, other-cause deaths are coded as censored at the time of death and remove those patients from subsequent risk sets (Austin et al., 2016; Putter et al., 2007). Writing H_1 for the cumulative breast-cancer hazard gives the associated survival quantity

$$S_{1,\text{net}}(t \mid \mathbf{w}) = \exp[-H_1(t \mid \mathbf{w})]. \tag{2.2}$$

This thesis calls it *model-based net survival from breast cancer*. It is the survival probability implied by the breast-cancer hazard alone, considered without the hazards of other causes of death.

This distinction matters when interpreting the prediction. One minus net survival is not the probability of breast-cancer death when other causes of death are also possible. That probability is the *cumulative incidence*, which accounts for patients dying of another cause first. For example, a patient who dies of heart disease cannot subsequently die of breast cancer. Cumulative incidence therefore depends on survival from all causes together. It is no larger than $1 - S_{1,\text{net}}(t \mid \mathbf{w})$ and is usually smaller (Austin et al., 2016; Putter et al., 2007; Latouche et al., 2013). The primary model in this thesis predicts net survival, not cumulative incidence.

Estimating the cause-specific hazard does not require hypothetical death times from different causes to be independent (Tsiatis, 1975; Austin et al., 2016). However, treating competing deaths as independent censoring when *scoring* net-survival predictions is a separate assumption. Thus, fitting the hazard and

evaluating net-survival predictions raise different questions. Chapter 4 states the scoring assumptions used here.

2.3.3 Cox Modelling and ER Stratification

The Cox model separates the way the event rate changes over time from differences between patients (Cox, 1972):

$$h(t \mid \mathbf{w}_i) = h_0(t) \exp(\eta_i), \qquad \eta_i = \boldsymbol{\beta}^\top \mathbf{w}_i. \tag{2.3}$$

The *baseline hazard*, $h_0(t)$, supplies the shared time pattern. The patient-specific factor $\exp(\eta_i)$ scales that pattern up or down. Here, the *log-risk score* η_i is a weighted sum of the patient's predictors $\mathbf{w}_i$, with weights $\boldsymbol{\beta}$ estimated from the data.

The score is not a probability. It describes relative hazard and must be combined with the baseline to obtain a survival probability. For patients sharing a baseline, a score difference of $\ln 2 \approx 0.69$ corresponds to twice the hazard, while equal scores give equal hazards. Their hazard ratio, $\exp(\eta_i - \eta_{i'})$, remains constant over time. This is the *proportional-hazards* assumption: predictors change the level of the hazard curve without changing its shape.

The coefficients are estimated using *partial likelihood*. At each observed event, it compares the patient who experienced the event with others still in the risk set. In Figure 2.3, A's death at year three is compared within the risk set of all five patients. At C's death at year nine, only C and E remain. These comparisons allow the coefficients to be fitted without first specifying the baseline hazard. The score has no separate intercept because adding the same constant to every score can be absorbed into the baseline. Neural Cox models replace the linear score with a learned function (Kvamme et al., 2019). To obtain survival probabilities, the baseline cumulative hazard must also be estimated, commonly using the Breslow estimator (Breslow, 1972; Therneau and Grambsch, 2000).

A *stratified* Cox model allows different groups to have different baseline hazards. For ER stratum s_i,

$$h(t \mid \mathbf{w}_i, s_i) = h_{0,s_i}(t) \exp(\eta_i). \tag{2.4}$$

Each ER group now has its own time pattern, while the predictor coefficients are shared. Events are compared only with patients in the same stratum. Proportional hazards is therefore assumed within ER groups, not across them (Therneau and Grambsch, 2000).

Stratification alone does not allow the same feature to have a different association with risk in each group. That requires an *interaction term*, such as an additional coefficient applying to one ER group.

The two choices have distinct roles: separate baselines allow different time patterns, while interactions allow different feature associations. Because the baselines differ, equal scores across ER groups do not necessarily imply equal survival probabilities (Therneau and Grambsch, 2000).

Once the score and baseline have been fitted, the predicted survival curve is

$$\widehat{S}_i(t) = \exp\left[-\widehat{H}_{0,s_i}(t)\exp(\widehat{\eta}_i)\right]. \tag{2.5}$$

The hats indicate estimated quantities. The patient's risk factor scales the ER-specific baseline cumulative hazard, and the result is converted into a survival probability. Fitting to all-cause death gives OS; fitting to breast-cancer death gives the model-based net survival defined above. Within a stratum, a higher score means lower predicted survival. Comparisons across strata also require the baselines. Chapter 4 uses this stratified form with an ER interaction for tissue coefficients.

2.3.4 Multimodal Fusion and Additive Survival Scores

The survival model must also combine the different sources of patient information. This step is called *multimodal fusion*. Models can join the original measurements, combine numerical representations learned from each source, or combine predictions from separately fitted models (Lipkova et al., 2022; Steyaert et al., 2023).

An *additive survival score* combines clinical, tissue and optional genomic contributions on the log-risk scale. Each source supplies a contribution, and those contributions are added to form the patient's score. The tissue contribution may itself be a complex function of thousands of cells and their arrangement. Additivity applies to the final combination, not necessarily to the calculations within each source. Survival probabilities are then obtained from the combined score and the baseline hazard; separate probabilities are not added together.

This approach separates two decisions. *Representation* determines which information is retained from the tissue, while *fusion* determines how that information is combined with other sources. The sources may contain overlapping information, so adding their contributions does not imply that the sources are independent (Lipkova et al., 2022; Steyaert et al., 2023). The trade-off is that the final sum does not capture arbitrary interactions between sources. Such interactions must be specified, as with the ER-dependent tissue coefficients used here. Chapter 3 discusses this choice in relation to other fusion approaches, and Chapter 4 specifies the additive model.

2.3.5 Events, Dimensionality and Penalised Estimation

Detailed tissue measurements can produce many candidate predictors from relatively few patients. The number of observed *events* is especially important because events provide the risk-set comparisons used to learn relative risk. For example, a cohort of 500 patients with 50 observed deaths still provides only 50 observed events, although patients without an observed death contribute through their follow-up and risk-set membership. Model complexity must therefore reflect both the available events and the follow-up information (Harrell, 2015; Riley et al., 2019, 2020).

Regularisation limits how closely a model can adapt to its training data. One approach penalises large coefficients. Ridge penalties shrink coefficients towards zero, while lasso and elastic-net penalties can also set some coefficients to zero, selecting variables (Verweij and van Houwelingen, 1994; Tibshirani, 1997; Simon et al., 2011). These controls help stabilise estimation when the tissue representation contains many related features. The next section explains how those features can preserve the tissue's spatial organisation.

2.4 Spatial Graphs and Hierarchical Tissue Representation

A survival model needs a numerical description of each patient's tissue. Spatial graphs provide one way to build that description while retaining information about neighbouring cells. The representation can then be summarised at several levels: cells, tissue regions, cores and patients.

2.4.1 Spatial Cell Graphs and Message Passing

A *spatial cell graph* represents a tissue core as a network, $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$. The cells are its *nodes*, $\mathcal{V}$; connections between nearby cells are its *edges*, $\mathcal{E}$; and the rows of $\mathbf{X}$ contain the cell measurements. Neighbours can be defined using a distance threshold or Delaunay triangulation of the cell centres (Figure 2.2d). An edge means that two cells are spatially close under the chosen rule. It does not establish physical contact or biological signalling.

A graph neural network (GNN) uses this network to learn an *embedding*, $\mathbf{h}_v$: a vector, or ordered list of numbers, that describes a cell together with its local context (Gilmer et al., 2017; Kipf and Welling, 2017). The underlying operation is *message passing*. At each step, the model gathers the current vectors of a cell's neighbours and combines them with the cell's own vector through a learned transformation.

For a cell with three neighbours, mean aggregation first averages the three neighbour vectors. That average is then combined with the cell's own vector to produce its updated representation. This is the mean-aggregation scheme used in GraphSAGE (Hamilton et al., 2017). Updating all cells once forms one layer. A second layer uses the updated neighbour vectors, which already contain information from their own neighbours. Information can therefore travel up to two graph steps. More generally, k layers can bring information from cells up to k steps away. Figure 2.4 shows how the cell's own vector and its neighbours' vectors enter each update.

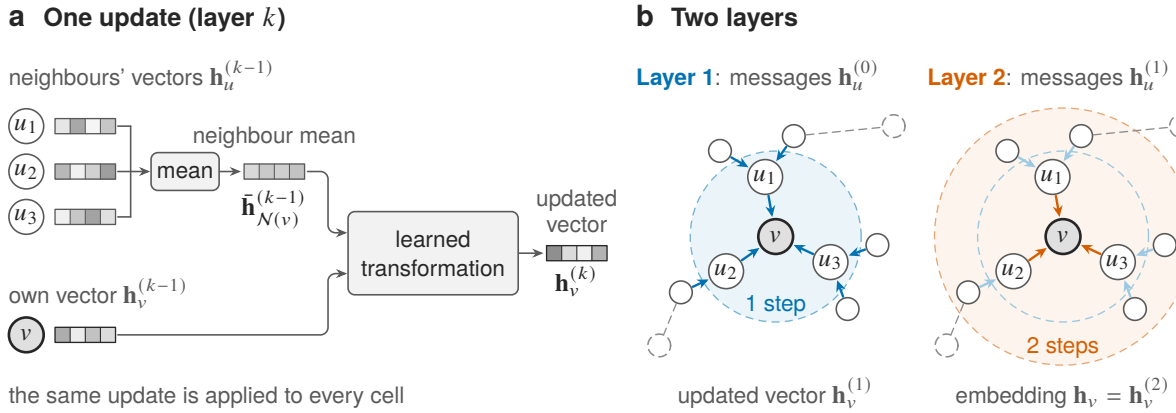


Figure 2.4. Two-layer message passing. (a) Each update combines a cell's own vector with its neighbours' mean vector through a learned transformation. (b) After two layers, the embedding draws on cells up to two graph steps away (shaded). Pale arrows show layer-1 messages; dashed cells lie beyond this range.

A learned embedding is different from the measured cell state introduced in Section 2.1.2. The measured state is the protein profile assigned to the cell; the embedding is a task-trained numerical representation built from the cell's inputs and its neighbours. Its coordinates need not correspond to individual proteins. The same learned transformations are applied to every cell, allowing the model to process new graphs with compatible inputs.

After learning the cell vectors, the model needs to summarise them. A *readout* converts a variable number of cell embeddings into a fixed-length description of the core:

$$\mathbf{z}_{\mathcal{G}} = \mathrm{READOUT}\big(\{\mathbf{h}_v : v \in \mathcal{V}\}\big). \tag{2.6}$$

The result does not depend on the order in which the cells are listed. Different readouts retain different information. A mean describes the typical embedding, whereas a sum also reflects how many embeddings were combined. With the embeddings held fixed, repeating every vector twice leaves the mean unchanged but doubles the sum (Zaheer et al., 2017; Xu et al., 2019).

Message passing and readout therefore serve different purposes: the former builds contextual cell features, while the latter decides how those features are summarised. Because messages can cross regional boundaries, even a whole-core readout may retain information from both sides of a boundary. The pairing readout in Section 2.5 instead makes selected boundary relationships explicit in the summary.

2.4.2 Niche Types, Instances and Interfaces

Between a single cell and an entire core lies the local cellular environment. A *niche type* describes a recurring mixture of cell populations. Such types can be identified by calculating the proportions of cell types around each cell and grouping neighbourhoods with similar compositions (Schürch et al., 2020).

A type describes a kind of environment, not one particular location. In this thesis, a *niche instance* is one connected occurrence of a niche type. It contains the cells assigned to that type that can be reached from one another along graph edges without leaving the type. The same type can therefore occur in several disconnected regions: A_1 and A_2 in Figure 2.5 share a type but occupy different locations (Bhate et al., 2022). A city offers an analogy: "commercial district" describes a kind of area, while each separate commercial district is a particular instance. Sharing a type does not mean belonging to the same connected region.

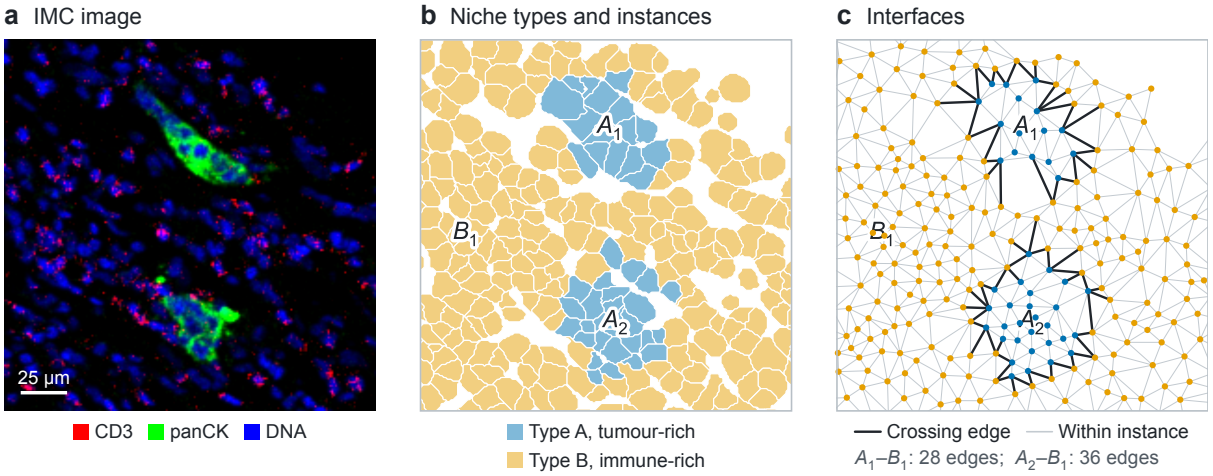


Figure 2.5. Niche instances and interfaces in METABRIC–IMC core 160 (Danenberg et al., 2022): (a) three of the measured channels; (b) niche types and instances constructed as in Sections 4.2.2–4.2.3, where A_1 and A_2 are separate instances of the tumour-rich type inside the immune-rich instance B_1; (c) crossing edges of the interfaces A_1–B_1 and A_2–B_1. Scale bar: 25 μm.

An *interface* describes where two neighbouring instances meet. It consists of the graph edges crossing between them, and therefore records the cell pairs connected across their boundary. It does not include every possible pairing of cells from the two regions. Figure 2.5c shows two interfaces containing 28 and

36 crossing edges. This distinction between type, instance and interface allows the model to describe what environments occur, where they occur and which cells connect them. Chapter 4 specifies how these objects are constructed and retained.

2.4.3 Hierarchical Readouts and Patient-Level Aggregation

A *hierarchical readout* summarises the tissue in stages. It first combines cells within regions, then combines the regional descriptions. Regions can also become nodes in a coarser graph, with edges indicating which regions are adjacent. The hierarchy may follow predefined tissue regions or be learned through graph-pooling operations (Ying et al., 2018; Lee et al., 2019; Bhate et al., 2022).

Adding a level to the hierarchy does not automatically preserve more information. If regional means are simply averaged using cell-count weights, the result is the global cell mean. For example, a region of three cells with mean 2 and another region containing one cell with value 6 give $(3\times 2+1\times 6)/4=3$, the mean of all four cells (Wang and Ji, 2023). The hierarchy changes the representation when it transforms the regional features or represents relationships between regions, rather than merely repeating the same averaging operation.

Core descriptions must then be combined for each patient. For the data $\mathcal{D}_{ir}$ from core r of patient i,

$$\mathbf{z}_{ir} = f_\theta(\mathcal{D}_{ir}), \qquad \mathbf{z}_i = \mathrm{AGG}_{\text{patient}}\big(\{\mathbf{z}_{ir}\}_{r=1}^{m_i}\big), \tag{2.7}$$

where f_θ produces a numerical description of a core and $\mathrm{AGG}_{\text{patient}}$ combines the patient's available core descriptions. In practical terms, the model summarises each core before combining the summaries into a patient vector. The weights can reflect the amount of sampled tissue, using cell counts, or be learned through attention (Ilse et al., 2018; Lu et al., 2021). Combining separate cores in this way does not imply that cells in different samples are spatial neighbours.

This structure enables *weak supervision*: the model learns from patient outcomes without requiring a survival label for each cell. The prediction depends on the patient vector, which depends on core vectors and cell embeddings. Training against the patient's survival record adjusts the model along this chain. Cells and interfaces can therefore contribute to the patient prediction without being assigned separate survival labels (Ilse et al., 2018; Campanella et al., 2019).

Regional features describe areas and their surroundings. The next section focuses on a more specific relationship: how the features of neighbouring cells are paired where two regions meet.

2.5 Boundary-Cell Pairing and Centring

A description of each region does not, by itself, specify which cells are connected across their shared boundary. This section introduces the feature-product calculations used to describe those pairings and the reference averages used to interpret them. These ideas provide the foundation for Niche-conditioned Interface Residual Pooling (NIRP). Chapter 3 discusses the related statistical and representation methods; Chapter 4 develops the interface-specific readout and its patient-level aggregation.

2.5.1 Marginal and Paired Representations

A *marginal summary* describes one side of a boundary on its own. Even complete lists of the values on each side do not tell us which values are paired, just as two guest lists do not tell us who spoke to whom. Figure 2.6 illustrates the difference. Each side contains one value of $+1$ and one of -1, so the two examples have the same values, the same means and the same number of pairs. Only the pairing changes.

When equal signs are paired, both products are positive: $(+1) \times (+1) = +1$ and $(-1) \times (-1) = +1$. When opposite signs are paired, both products are -1. The average product therefore distinguishes the two examples even though neither side's values have changed. Products of feature values are called *second-order statistics*. Here, they retain information about the pairing that separate summaries of the two sides leave out.

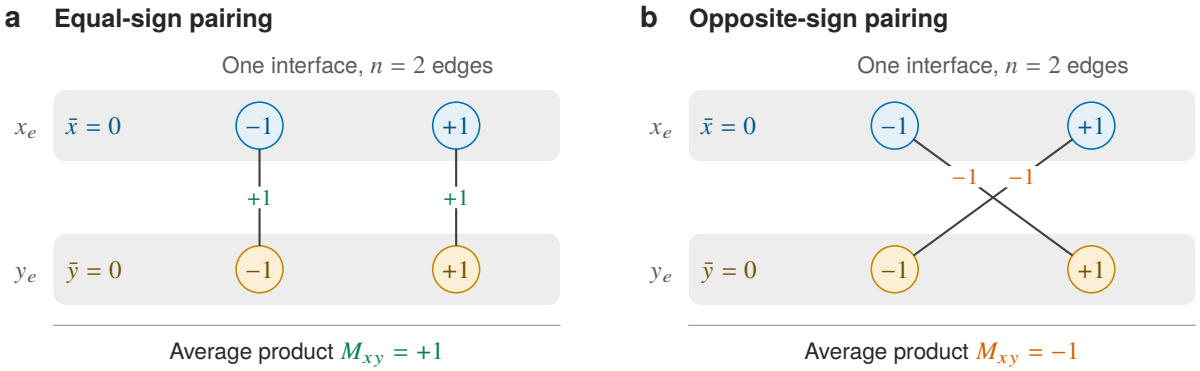


Figure 2.6. Identical endpoint marginals with (a) equal-sign and (b) opposite-sign pairing, yielding mean products $M_{xy} = +1$ and -1, respectively.

2.5.2 Cross-Moments and Re-pairing References

Cells are usually described by several features rather than one number. An *outer product* extends the paired-product idea to these feature vectors. It forms a table in which each entry multiplies one feature on the left by one feature on the right. For an illustrative pair of vectors $\mathbf{x} = (1, 2)^\top$ and $\mathbf{y} = (3, 4)^\top$,

$$\mathbf{x}\mathbf{y}^\top = \begin{pmatrix} 1\times 3 & 1\times 4 \\ 2\times 3 & 2\times 4 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}.$$

The rows correspond to left-side features and the columns to right-side features. Averaging these tables across crossing edges summarises which feature values are paired across the interface.

Let $(\mathbf{x}_e, \mathbf{y}_e)$ be the measured or learned feature vectors at the two endpoints of crossing edge e. For $n > 0$ edges, the means of the two endpoint lists are $\bar{\mathbf{x}} = n^{-1} \sum_e \mathbf{x}_e$ and $\bar{\mathbf{y}} = n^{-1} \sum_e \mathbf{y}_e$. The average outer product is the *cross-moment*, M_{xy}. Subtracting the outer product of the means gives the *centred cross-moment*, C_{xy}:

$$M_{xy} = \frac{1}{n} \sum_{e=1}^{n} \mathbf{x}_e \mathbf{y}_e^\top, \qquad C_{xy} = M_{xy} - \bar{\mathbf{x}}\bar{\mathbf{y}}^\top. \tag{2.8}$$

The first quantity summarises the paired values themselves. The second summarises their departures from the average on each side: whether above-average values tend to meet above-average or below-average values. Its entries are not correlations. Centring subtracts the means but does not divide by standard deviations, so the entries need not lie between -1 and $+1$.

The subtraction also has a useful interpretation as a comparison with random pairing. Keep both endpoint lists fixed, but randomly shuffle which right endpoint accompanies each left endpoint. Under a uniformly random permutation π, each right endpoint is equally likely to occupy any position. Consequently, $\mathbb{E}_\pi[\mathbf{y}_{\pi(e)}] = \bar{\mathbf{y}}$ for each position e, and

$$\mathbb{E}_\pi \left[\frac{1}{n} \sum_{e=1}^{n} \mathbf{x}_e \mathbf{y}_{\pi(e)}^\top \right] = \bar{\mathbf{x}}\bar{\mathbf{y}}^\top. \tag{2.9}$$

Thus, the product of means is exactly the expected cross-moment under random re-pairing. Subtracting this *re-pairing reference* compares the observed pairing with that expectation. The comparison is *conditional*: the endpoint values and the interface to which they belong remain fixed; only their pairing changes. It therefore concerns how the same values are paired, not a change in either side's measurements.

The endpoint lists count appearances, not distinct cells. A cell that belongs to three crossing edges appears three times. Re-pairing preserves those appearances, but not the original spatial geometry.

A value of zero also needs careful interpretation. With only one edge, the centred cross-moment is always zero because each endpoint equals its own side's mean. With no edges, the means cannot be calculated at all. An unavailable readout is therefore different from an observed readout whose value happens to be zero. Keeping *availability* separate from *value* is central to RQ2.

2.5.3 Local and Pooled Centring

Centring compares feature values with an average, so the choice of average changes what the representation retains. *Local centring* uses the two endpoint means of each individual interface. It describes variation within that interface. *Pooled centring* instead calculates the means after combining endpoint appearances across interfaces of the same niche-type pair. It can retain both within-interface variation and differences between interface means.

Consider two interfaces of the same type pair with equal numbers of edges (Figure 2.7). Every endpoint on both sides is +1 in the first interface and −1 in the second. With local centring, the mean on each side is +1 for the first interface and −1 for the second. Every value equals its local mean, so every centred value is zero and both interface statistics are zero.

With pooled centring, the positive and negative values balance, giving a mean of zero on each side. Every observed pair still has matching signs, so every paired product is +1 and the pooled statistic is +1. This statistic captures the contrast between an interface where high values meet high values and another where low values meet low values, even though neither interface contains internal variation.

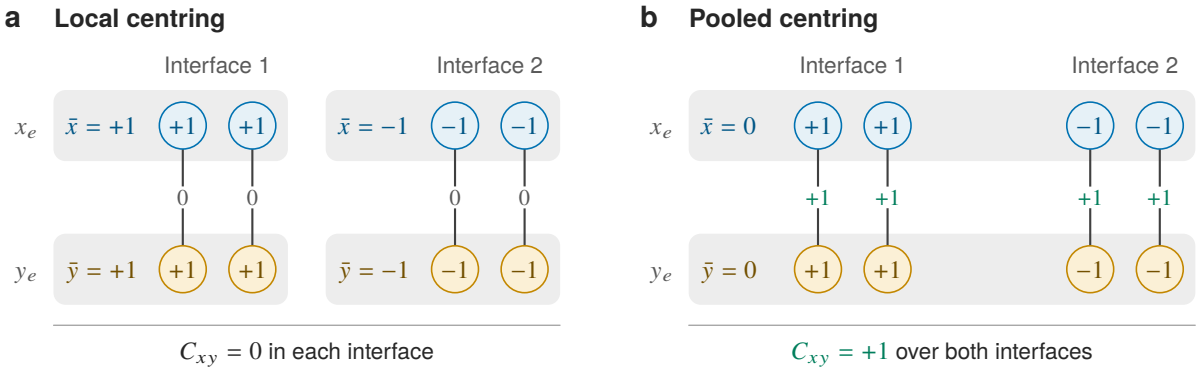


Figure 2.7. Local (a) and pooled (b) centring of identical endpoint values and pairings. Shading groups values sharing a mean; edge labels show centred products.

Centring is therefore a choice about which relationships to represent, not simply a way to remove noise. Local centring focuses on variation within boundaries; pooled centring also retains differences between them. Chapter 4 develops the interface readout and the comparisons used to investigate these choices.

2.6 Predictive Evaluation and Model Interpretation

Building a model raises two further questions. Does it predict well for patients who were not used to develop it? How does it use the information it receives? The first concerns predictive evaluation; the second concerns interpretation. A convincing answer to one does not replace evidence for the other.

2.6.1 Validation and Added Predictive Value

Performance measured on the training patients is optimistic because the model has already adapted to those data. *Internal validation*, using cross-validation or bootstrapping, estimates performance beyond the training sample. *Internal–external validation* holds out each cohort in turn, allowing the model to be developed on the others (Steyerberg, 2019; Steyerberg and Harrell, 2016). *Independent external validation* evaluates a fixed predictor on data that played no part in its development. It tests *transportability*: whether the predictor works in another setting (Royston and Altman, 2013; Riley et al., 2024; Debray et al., 2015).

For tissue-based prediction, the patient must be the unit of separation. All cells and cores from the same patient stay together in any data split. Training data are used to estimate model parameters. Validation data guide choices such as the model configuration and when to stop training. Test data assess the procedure after those choices have been made.

Nested cross-validation separates model selection from final assessment. Within each outer training partition, an inner procedure chooses the configuration without access to the outer test patients. The outer test partition then evaluates the selected procedure, rather than helping to select it. Data-dependent transformations, including scaling, are also fitted using training data only and then applied to the held-out patients.

Applying a fixed predictor to another cohort tests *transfer*. Training the same model architecture again in that cohort tests whether it can *learn* from those data. These are different questions. Transfer also requires compatible marker panels, phenotype definitions and measurement scales. Two measurements with the same marker name are not necessarily on the same scale, and aligning the inputs alone does not establish that predictions will transport (Riley et al., 2024; Debray et al., 2015).

The value of additional information also depends on the comparison model. Adding tissue data to clinical variables asks whether that extra source helps. Comparing different tissue representations asks how effectively the same source is used. A *matched comparison* keeps the patients, data partitions and outcomes identical, so a performance difference is not caused by changing the evaluation population.

Chapter 5 uses these matched comparisons to assess *incremental value*, or what one model adds beyond another.

2.6.2 Discrimination, Probability Error and Calibration

Prediction quality has several aspects. A model can put patients in the right risk order while still assigning inaccurate survival probabilities. Evaluation therefore considers ordering, probability error and agreement between predicted and observed outcomes.

Discrimination concerns risk ordering. Concordance measures how often the patient predicted to be at higher risk experiences the event first among pairs whose ordering can be determined (Harrell et al., 1982, 1996). In Figure 2.3, A and C are comparable because both deaths are observed. A and B are also comparable because B was still alive when A died. B and C are not: B's follow-up ended before C's death, leaving their event ordering unknown. Concordance of 0.5 corresponds to random ordering, while 1 corresponds to perfect ordering.

Uno's concordance uses *inverse probability of censoring weighting* (IPCW) within a chosen horizon and under the relevant censoring assumptions (Uno et al., 2011). This gives greater weight to observed comparisons that were less likely to remain observable. It adjusts for missing follow-up without assigning invented event times to censored patients. Time-dependent receiver operating characteristic (ROC) analysis instead assesses risk separation at a specified time (Heagerty and Zheng, 2005).

Probability error asks how far the predicted probabilities are from the outcomes. The *Brier score* measures squared error in predicted survival status at a chosen time, with adjustment for censoring. For a predicted survival probability of 0.7, the unweighted squared error is $(1 - 0.7)^2 = 0.09$ for a patient known to survive beyond that time. For a patient with an observed event before that time, it is $(0 - 0.7)^2 = 0.49$. The *integrated Brier score* (IBS) averages this error over a time interval; lower values are better (Graf et al., 1999; Gerds and Schumacher, 2006). IBS reflects both discrimination and calibration, rather than either alone. As a simple illustration of the scale, predicting 0.5 for every patient gives a squared error of 0.25 at one time point before censoring adjustment.

Calibration asks whether predicted probabilities agree with observed frequencies. For OS, a group assigned 70% survival at a given horizon should show approximately 70% survival at that time. Calibration curves examine this agreement across the range of predictions. Average agreement describes whether predictions are systematically too high or too low, while the calibration slope summarises whether they are too extreme or too compressed (Van Calster et al., 2016; Austin et al., 2020; Crowson et al., 2016).

DSS evaluation must instead match the net-survival target in Section 2.3.2. When competing deaths occur, net survival is not a directly observable survival frequency, so its assessment depends on the stated assumptions.

Finally, a comparison needs an uncertainty estimate for the *paired performance difference*, not just a performance estimate for each model separately. Resampling patients with replacement and recomputing both models' metrics on the same resample preserves that pairing. When the predictions have already been fitted and saved, this procedure measures variation arising from the evaluation sample, conditional on those fitted models. Chapter 5 reports paired differences with intervals of this kind.

2.6.3 Prediction Decomposition, Fidelity and Stability

Interpreting a prediction begins by deciding what is to be explained (Rudin, 2019; Yuan et al., 2023). *Computational decomposition* identifies the terms that raise or lower a patient's fitted log-risk score. Like an itemised bill, it shows how the parts add up to the total. It describes a calculation that has already been made. Changing inputs and recomputing the prediction asks a different question: what would the model calculate with different information? (Ying et al., 2019; Luo et al., 2020).

Tissue correspondence connects a fitted contribution to measured cells, for example by locating the boundary cells that produced it. This shows where part of the calculation comes from. *Fidelity* asks whether a simpler description can reproduce that calculation. For example, a model using measured proteins can be tested on held-out patients to see whether it reconstructs the fitted contributions. The target in that test is the original model's output, not patient survival. A tissue map can therefore locate a contribution without establishing that the measured proteins explain its value.

Stability asks whether a tissue pattern retains its predictive role when the model is fitted again on different data. Recognisable tissue, faithful reconstruction and stable contributions are distinct forms of evidence. Chapter 5 examines tissue correspondence and fidelity, while Chapter 6 considers stability and the limits of the interpretation.

2.7 Chapter Summary

Connecting tissue measurements to survival requires a clear prediction target, a representation of the sampled tissue and evaluation at the patient level. Clinical and molecular data describe different aspects of disease. Spatial graphs preserve cellular context, while regional and interface readouts summarise different aspects of tissue organisation. At a boundary, paired products retain relationships that separate summaries

of each side may miss. The choice of centring reference determines which of those relationships remain in the representation.

These choices must be evaluated against the question the model is intended to answer. Predictive evaluation tests performance for held-out patients; interpretation examines how a fitted prediction was calculated. Table 2.1 connects the concepts introduced here to their use in later chapters. Chapter 3 reviews the related work, and Chapter 4 specifies the model and evaluation design.

Table 2.1. Background concepts across the thesis.

Concept	Introduced	Used in
Composition, cell state and organisation	§2.1.2	Predefined summaries and learned representations (Section 4.1.3)
Cores, patients and multiple-instance structure	§2.2.2	Patient-level aggregation and partitioning (Sections 4.3.1 and 4.5.2)
Censoring, risk sets and net survival	§2.3.1–2.3.2	Survival target and scoring (Sections 4.3.5 and 4.5.3)
Stratified Cox model and additive score	§2.3.3–2.3.4	Survival head and training objective (Sections 4.3.5 and 4.5.1)
Cell graphs, message passing and readout	§2.4.1	Shared cell encoder (Section 4.3.1)
Niche types, instances and interfaces	§2.4.2	Tissue construction (Sections 4.2 and 4.2.3)
Cross-moments, re-pairing reference and centring	§2.5.2–2.5.3	NIRP and its properties (Sections 4.3.2 and 4.3.4); RQ2 (Section 5.2.2)
Validation, transfer and matched comparison	§2.6.1	Evaluation design and comparators (Sections 4.5.2 and 4.4)
Concordance, Brier score and calibration	§2.6.2	Performance metrics and uncertainty (Sections 4.5.3 and 4.5.4)
Decomposition, fidelity and stability	§2.6.3	Tissue tracing and surrogate tests (Section 5.6)

CHAPTER 3

Literature Review

This review asks whether learned spatial tissue representations improve survival prediction beyond clinical information and predefined tissue summaries, and whether explicit boundary-cell pairing could add information beyond existing representations. It distinguishes biological association, prediction on held-out patients and methodological precedents. These provide a rationale, evidence of generalisation and modelling tools, respectively. The synthesis positions NIMSURV and Niche-conditioned Interface Residual Pooling (NIRP) within this evidence.

3.1 Review Methods and Evidence Base

3.1.1 Question and Scope

The primary question is: *what evidence supports the additional prognostic value of learned spatial tissue representations, and what remains unresolved about tissue-boundary representations?* The empirical focus is breast cancer studied through spatially resolved, multiplexed single-cell protein imaging, principally imaging mass cytometry (IMC). Comparators include clinical predictors, predefined tissue summaries and other learned representations. Outcomes comprise discrimination, probability error, calibration, transfer and explanation, with each study's endpoint and horizon retained.

An empirical strand separates outcome association from held-out prediction; a companion methodological strand covers transferable representations, evaluation and interpretation. Other cancers and modalities provide adjacent rather than direct breast-survival evidence. Linked reports and repeated analyses of a cohort are not treated as independent studies or populations. Reporting draws on PRISMA 2020 and PRISMA-S (Page et al., 2021; Rethlefsen et al., 2021).

3.1.2 Sources and Search Strategy

Searches covered PubMed/MEDLINE, Scopus, arXiv, bioRxiv and medRxiv, with backward reference checking and forward citation tracking through Google Scholar. Imaging searches began on 15–16 March 2026 and methodological searches on 20–21 March; the final update was 1 November 2026. Imaging

coverage extended from 1 January 2014 to that cutoff, with English-language restrictions. Foundational methodological papers were not subject to the lower date limit.

The imaging search joins platform, representation and outcome blocks with AND, and terms within blocks with OR (Table 3.1). Four separate methodological searches omit the platform block. Citation tracking starts from eight seed papers covering breast IMC atlases, prediction, second-order representations and connected regions. Appendix C records source-specific queries, seeds and retrieval counts. Availability at the cutoff determines eligibility; preprint and published versions, including HiGINE, are linked rather than counted twice.

Table 3.1. Search concepts; exact source-specific queries are recorded in Appendix C.

Block	Terms or methodological focus
Platform	imaging mass cytometry; multiplexed imaging; multiplex immunofluorescence; MIBI; CODEX; cyclic immunofluorescence; spatial proteomics
Representation	graph neural network; cell graph; cellular neighbourhood; niche; tissue microenvironment; spatial statistics; cell–cell interaction
Outcome	survival; prognosis; prognostic; hazard; recurrence; outcome
Separate searches	Feature products and edge-based learning; competing risks; cross-validation uncertainty; explanation fidelity and faithfulness

3.1.3 Eligibility, Selection and Coverage

Eligibility follows three routes: human spatial single-cell tumour imaging linked to patient outcomes; transferable representation or spatial-statistical methods; and methods for survival evaluation, transfer or explanation. Grade and response classification remain adjacent evidence, not survival prediction. Descriptive single-marker, non-spatial or non-human studies are excluded unless eligible through a methodological route. Non-spatial comparators within eligible studies remain relevant. Reports without sufficient full-text methods and superseded versions are excluded.

The author screened titles and abstracts and assessed full texts. The search record reports 586 identified records, 174 duplicates or overlaps, 412 screened records, 138 full-text assessments and 91 retained records after 47 full-text exclusions (Figure 3.1). The retained set comprises 17 association studies, 13 survival or progression prediction studies and 61 methodological or clinical foundations. Appendix C provides the detailed accounting and exclusion reasons.

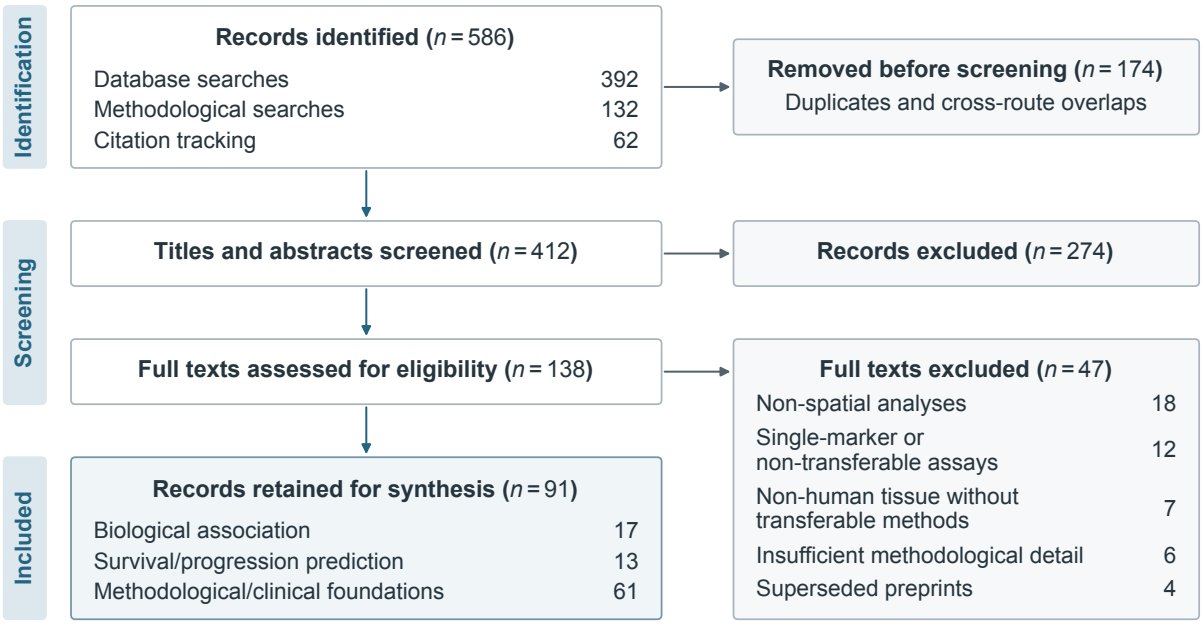


Figure 3.1. Literature-selection flow.

3.1.4 Extraction, Appraisal and Synthesis

Comparison fields, informed by CHARMS, cover report version, disease, cohort, platform, patient/specimen/event counts, inputs, representation, endpoint, horizon, comparators, partitioning, model selection, performance, uncertainty, external evaluation and explanation tests (Moons et al., 2014). Cohort size is distinguished from the analysis population and specimen count.

Prediction-study appraisal considers participants, predictor measurement, outcomes and analysis, drawing on CHARMS and PROBAST+AI (Moons et al., 2014, 2025). Particular attention is given to patient-level separation, preprocessing and tuning boundaries, comparator strength, calibration and external evaluation. Association studies are considered in terms of sampling, adjustment, multiplicity and validation; methodological studies by their operations and assumptions. Table 3.3 provides a focused design appraisal.

Narrative synthesis groups studies by task, representation and evaluation design. Matched within-study contrasts inform incremental value; hazard ratios and log-rank tests remain association evidence. Both positive and inconclusive findings are retained without significance-based vote counting. Differences in targets, inputs and validation, together with cohort reuse, preclude interpreting pooled concordance values as the effect of a common representation. Split-level standard deviations are distinguished from confidence intervals for paired gains.

3.2 Spatial Proteomics and Multimodal Survival Prediction

3.2.1 Biological Rationale

Breast IMC studies relate cellular organisation to outcomes, molecular subtypes and genomic alterations (Jackson et al., 2020; Ali et al., 2020; Curtis et al., 2012; Pereira et al., 2016). Danenberg and colleagues identify recurring tumour-microenvironment structures associated with disease-specific survival (DSS), with differences by oestrogen-receptor (ER) status (Danenberg et al., 2022). These findings motivate retaining organisation alongside abundance.

Multiplexed ion beam imaging, CODEX and cyclic immunofluorescence extend spatial protein measurement (Angelo et al., 2014; Goltsev et al., 2018; Lin et al., 2018; Lewis et al., 2021). Studies of ductal carcinoma in situ, lung cancer and melanoma relate spatial features to progression or treatment response (Risom et al., 2022; Sorin et al., 2023; Moldoveanu et al., 2022). SpatialScore illustrates a relational feature: a ratio of distances from $CD4^+$ T cells to tumour and regulatory T cells (Phillips et al., 2021). Its small, same-patient cross-platform assessment limits generalisation (Table 3.2). Spatial–genomic correspondence likewise motivates testing complementarity but does not establish added predictive value.

3.2.2 Breast IMC Prediction and Core Evidence

DMGN combines IMC image-region features with patient-variable embeddings through relational fusion and Cox-based prediction; an auxiliary reconstruction task preserves patient information (Fu et al., 2023). Its inputs include clinical and treatment variables and selected gene indicators. Importantly, its Basel comparisons include patient variables plus cell-type proportions. The unresolved question is therefore what an interface-conditioned readout adds under a matched representation and survival-model comparison.

SpaCEy learns protein-feature cell graphs and uses a neural Cox objective, comparing against tissue summaries with patient-separated splits and training-data tuning (Rifaioglu et al., 2026). Clinical metadata are not prediction inputs. It therefore evaluates tissue representation learning, whereas DMGN evaluates integrated image–patient modelling. Table 3.2 combines study characteristics and empirical findings; Table 3.3 makes their inferential limits explicit.

Table 3.2. Core studies, designs and findings. CV: cross-validation; OS: overall survival; SD: standard deviation.

Study and population	Representation, target and evaluation	Finding or evidential scope
Prediction and adjacent classification		
DMGN (Fu et al., 2023) Breast IMC; METABRIC 456 patients/518 samples; Basel 285/376. Deaths: 45.2%/28.1%.	Image regions plus patient variables; OS; five-fold patient CV and frozen common-input transfer. Basel includes a patient-variable/composition baseline.	Native-panel Harrell's C: 0.7484/0.7479. Common-input transfer: 0.6312.
SpaCEy (Rifaioglu et al., 2026) Breast IMC; JacksonFischer 352 patients/720 images; METABRIC 405/460.	Protein-feature cell graphs; neural Cox OS. Ten matched patient-separated splits with internal tuning; tissue-only comparators. Cohort models fitted separately.	GNN versus best summary, C (mean ± SD): JacksonFischer 0.720 ± 0.061 versus 0.617 ± 0.092; METABRIC 0.688 ± 0.038 versus 0.570 ± 0.071.
HiGINE (Edgren Schüllerqvist et al., 2025, 2027) Lung; mIF 298 patients/542 images; IMC 401 of 416 analysed.	Hierarchical cell graph plus stage; short/long-survival classification at 1,730 days (mIF) or 36 months (IMC). Patient-stratified ten-/five-fold CV.	Adjacent evidence: binary survival classification. The two reports describe one study.
SPIFEE (Tsang et al., 2026) Lung histology; 395 patients.	Phenotype-cluster nodes and spatial-function edges; survival-rank learning followed by Cox fitting. Patient-grouped five-fold CV repeated ten times.	A relational precedent at category level.
Sorin et al. (2023) Lung IMC; 416 discovery and 60 independent validation patients.	Image-derived ResNet-50 features, sparse PCA and SVM progression classification. Five-fold CV; stage-I model uses 286 discovery patients and 60 validation patients.	Independent progression assessment; a different endpoint and disease from breast DSS.
Tissue characterisation and association		
Danenberg et al. (2022) Breast IMC; 693 patients.	Cell-type neighbourhoods and TME structures; ER-specific DSS associations; structure-label transfer.	Biological rationale and category correspondence.
HRCHY-CytoCommunity (Xie et al., 2026) Breast; 263 patients, 73 deaths.	Unsupervised nested regions; adjusted OS association; parameter and annotation perturbations.	TC-13 presence: HR 0.20, 95% CI [0.047, 0.86], $p = 0.031$. Association and structural consistency.

Continued on next page

Table 3.2 continued

Study and population	Representation, target and evaluation	Finding or evidential scope
BiGraph (Wang et al., 2025) Breast IMC; Danenberg 579, Jackson 266, Wang 279 (four exclusions reported).	Cell/patient-similarity graphs; outcome-free groups; 379/200 discovery/internal split and external survival/response mapping.	Log-rank p: 0.00056 discovery, 0.24 internal, 0.007 external. Niche associations were not uniformly significant externally.
Phillips et al. (2021) Cutaneous T-cell lymphoma; 14 patients/70 regions.	SpatialScore distance ratio; pembrolizumab response. Same-patient Vectra assessment, subsampling and leave-one-patient-out sensitivity.	Relational and cross-platform evidence.
Grade classification		
Ali et al. (2025) Jackson and METABRIC.	GNN compared with unordered-cell models for grade classification.	ΔAUPR: Jackson 0.036 ($p = 0.086$); METABRIC 0.014 ($p = 0.56$). Spatial improvement was not established.

Table 3.3. Focused appraisal: credibility of the reported comparison is distinguished from relevance to the thesis.

Study	Design strength	Limit on inference
DMGN	Patient-level CV; integrated inputs; composition baseline; fixed-model transfer.	Patient block includes treatment and gene indicators; full-system comparisons do not isolate interface readout.
SpaCEy	Matched patient-separated splits, training-data tuning and tissue-summary comparisons.	No clinical prediction inputs; split SDs do not establish paired-gain uncertainty; breast models are fitted separately.
SPIFEE	Repeated patient-grouped assessment of explicitly relational tissue features.	Lung histology, category-level edges and a two-stage survival procedure provide indirect evidence for the present target.

3.2.3 Fusion and Survival-Model Choices

Multimodal methods differ in how sources interact (Lipkova et al., 2022; Steyaert et al., 2023). Pathomic Fusion and PORPOISE form cross-modal feature products; MCAT and SurvPath use genomic or pathway information to attend to image patches (Chen et al., 2022a,b, 2021b; Jaume et al., 2024). Alignment methods use optimal transport or translation, while MultiSurv accommodates missing sources and the ProtoPathway preprint links morphology with pathways (Xu and Chen, 2023; Zhou and Chen, 2023; Vale-Silva and Rohr, 2021; Gallagher-Syed et al., 2026). These address relationships across modalities,

whereas NIRP concerns pairing within tissue. Additive fusion offers an inspectable setting for that question, while restricting general cross-modal interactions and allowing specified ER-dependent effects.

The survival model is a separate choice (Wiegrebe et al., 2024). Cox and DeepSurv retain proportional hazards; random survival forests, DeepHit and BroadSurv represent survival differently (Cox, 1972; Katzman et al., 2018; Ishwaran et al., 2008; Lee et al., 2018; Wu et al., 2021). Other graph or attention models relate patient variables, visits or slide patches rather than cells (Sharma et al., 2021; Sun et al., 2021; Li et al., 2018; Chen et al., 2021a). Changing both representation and predictor tests a complete system. A shared survival family and matched inputs focus the comparison.

3.3 Spatial Tissue Representations

3.3.1 Summaries and Contextual Cell Features

Contact summaries distinguish mixed from compartmentalised breast architectures, while neighbourhood analyses relate recurring environments to expression and outcomes (Keren et al., 2018; Goltsev et al., 2018; Schürch et al., 2020). Ripley's K, pair-correlation and nearest-neighbour functions describe distance-dependent organisation (Ripley, 1977; Baddeley et al., 2015). spicyR, SPIAT and CytoMAP support localisation, spatial summaries and regional analysis; histoCAT and Squidpy compare contacts with label-permutation references (Canete et al., 2022; Feng et al., 2023; Stoltzfus et al., 2020; Schapiro et al., 2017; Palla et al., 2022). Category-based contacts need not retain the continuous states of observed cell pairs: similar category counts can conceal different pairings.

Cell-graph and hierarchical models instead learn contextual tissue features (Zhou et al., 2019; Pati et al., 2022; Lee et al., 2022). SpaCEy uses protein-feature cells, distance-pruned Delaunay edges, message passing and mean pooling (Rifaioglu et al., 2026). Messages can already cross regional boundaries; an interface readout explicitly groups selected relationships when constructing the patient representation. The KRONOS preprint and VirTues study additionally provide pretrained tissue features (Shaban et al., 2025; Wenckstern et al., 2026), separating the source of cell features from the choice of patient readout.

3.3.2 Regional and Hierarchical Models

Regional methods differ in what defines coherence, whether outcomes guide partitioning and whether disconnected occurrences remain distinct. Composition-based neighbourhoods and Spatial-LDA group similar cell mixtures, whereas UTAG, CellCharter and BANKSY incorporate continuous features and

spatial context (Schürch et al., 2020; Chen et al., 2020; Kim et al., 2022; Varrone et al., 2024; Singhal et al., 2024). These can identify structure without patient outcomes.

Task-guided alternatives let labels influence the representation: CytoCommunity can include sample labels, SPACE-GM learns local subgraphs and NaroNet learns regional hierarchies, supported by general graph-pooling methods (Hu et al., 2024; Wu et al., 2022; Jiménez-Sánchez et al., 2022; Bianchi et al., 2020). Tissue schematics preserves connected instances and adjacency motifs; HRCHY-CytoCommunity develops nested neighbourhoods and compartments (Bhate et al., 2022; Xie et al., 2026). Outcome-free regions may miss prognostic distinctions, whereas task-guided regions require separate assessment of biological consistency. The thesis chooses composition-based types followed by connected-instance extraction to define both regions and interfaces.

3.3.3 Regional Relations and Boundary Pairing

SPIFEE represents category-level relations with sampled spatial functions, such as cross-G, rather than a single contact count (Tsang et al., 2026). Its lung survival analysis uses censoring-informed ranking before Cox fitting. The contrast with NIRP is the observational unit: category-level edges describe spatial relationships across a sample; interface-specific readouts retain endpoint states at a particular connected-region boundary. Both are relational, but they preserve different information.

3.4 Statistical Precedents for Boundary-Cell Pairing

3.4.1 Products, Edges and Readouts

Bilinear pooling forms outer products of feature streams at image locations; compact bilinear pooling approximates the associated kernel representation (Lin et al., 2015; Gao et al., 2016). Second-order graph pooling uses products within contextual node embeddings, including $H^\top H = \sum_v h_v h_v^\top$ (Wang and Ji, 2023). COVET similarly combines within-cell feature products over neighbourhoods for ENVI (Haviv et al., 2025). NIRP instead multiplies projected states from distinct crossing-edge endpoints, grouped by interface. Its learned projection constrains score rank, unlike random-feature kernel approximation.

Graph-network and edge-conditioned formulations already make relations explicit through edge updates or attribute-dependent transformations (Battaglia et al., 2018; Simonovsky and Komodakis, 2017). NIRP's specificity is the selection of crossing edges, interface/type-pair grouping and centring reference (Table 3.4).

Table 3.4. Statistical precedents for NIRP: differences in observation unit, grouping and reference.

Precedent	Operation and distinction
Bilinear/compact pooling	Products between feature streams at an image location; aggregation or kernel approximation across locations.
Second-order graph pooling	Within-node feature products; global or hierarchical grouping of contextual embeddings.
COVET/ENVI	Within-cell products over neighbourhoods; a dataset-wide mean retains local covariance and mean shifts.
Graph/edge-conditioned blocks	Explicit edge updates or attribute-dependent filters before global readout.
Bipartite modularity	Observed–expected connections with a degree-based reference accounting for endpoint participation.
NIRP (proposed)	Products between crossing-edge endpoints, centred within each interface and pooled by type pair and patient.

3.4.2 References, Multiplicity and the Unresolved Benefit

Centring changes the represented information. COVET retains shifts from a dataset-wide mean, while graph-pooling experiments show that centring can remove useful distinctions (Haviv et al., 2025; Wang and Ji, 2023). Cell-label permutation holds locations fixed and tests category contacts; NIRP's re-pairing reference holds endpoint lists fixed and tests paired feature products (Baddeley et al., 2015; Schapiro et al., 2017). Repeated endpoint appearances remain part of that reference. Barber's degree-based bipartite reference provides a related treatment of participation (Barber, 2007).

Together with connected-instance representations (Bhate et al., 2022), these precedents motivate an explicit interface readout. They leave competing possibilities: pairing may preserve useful distinctions, but contextual embeddings may already capture them and local centring may remove prognostic between-interface variation. Its benefit therefore requires comparison. Chapter 4 specifies those tests.

3.5 Added Predictive Value and Cross-Cohort Transfer

3.5.1 Incremental Value and Uncertainty

Table 3.2 supports a task-dependent interpretation: DMGN evaluates integrated prediction, SpaCEy improves mean concordance over tissue-only summaries, and grade-classification comparisons do not establish spatial improvement. BiGraph and HRCHY-CytoCommunity provide association rather than individual-risk calibration evidence. Because several studies reuse METABRIC, methodological variety

exceeds population diversity. Added value, including genomics, therefore requires matched patients, inputs, endpoints and partitions.

Cross-validation folds share training patients, making naive independent-fold tests inappropriate; no universal unbiased variance estimator exists for k-fold CV (Dietterich, 1998; Bengio and Grandvalet, 2004). Corrected resampling and 5×2 designs address particular settings, while multi-dataset comparisons answer a different question (Nadeau and Bengio, 2003; Demšar, 2006). Patient-paired bootstrap of saved predictions conditions on fitted models. Multiple testing adds another limitation: Holm controls family-wise error, and the original Benjamini–Hochberg result assumes independence (Holm, 1979; Benjamini and Hochberg, 1995).

3.5.2 Targets, Competing Risks and Calibration

Competing mortality changes the target. Fine–Gray models cumulative incidence, which can also be obtained by combining cause-specific hazards; using only the breast-cancer hazard as real-world event probability can overstate risk (Fine and Gray, 1999; Wolbers et al., 2009). Discrimination, Brier scores and calibration must match the target and censoring assumptions (Schoop et al., 2011; Blanche et al., 2013; van Geloven et al., 2022). Good ordering or average calibration does not establish accurate probabilities across the prediction range (Van Calster et al., 2019).

The thesis predicts model-based net DSS (Section 2.3.2). Matched net-survival comparisons address relative predictive information under stated assumptions.

3.5.3 Frozen Transfer

Reproducibility in similar patients differs from transportability across settings (Justice et al., 1999). DMGN applies a fixed METABRIC predictor to Basel using common inputs; its restricted-input source analyses, not native-panel results, provide the relevant reference (Fu et al., 2023). By contrast, Danenberg transfers tissue labels, BiGraph maps subgroups, and HRCHY-CytoCommunity aligns profiles (Danenberg et al., 2022; Wang et al., 2025; Xie et al., 2026). SpaCEy trains separate breast models and tests cross-batch CODEX classification; VirTues transfers response-derived signatures to disease-free survival (Rifaioglu et al., 2026; Wenckstern et al., 2026).

These results concern different transferable objects. Frozen-predictor evaluation requires compatible measurements and outcomes, unchanged prediction and disclosure of the target cohort's role in development

(Riley et al., 2024; Collins et al., 2024). Accordingly, Basel-native learning and common-input OS transfer are separate experiments.

3.6 Tissue-Based Model Interpretation

3.6.1 Explanatory Targets

Interpretation must specify what is explained (Rudin, 2019; Jaume et al., 2021). DMGN, ProtoPathway, SPIFEE and PORPOISE expose image–patient, morphological–pathway or other fitted relationships (Fu et al., 2023; Gallagher-Syed et al., 2026; Tsang et al., 2026; Chen et al., 2022b). GNNExplainer and SpaCEy instead seek compact masks that preserve a prediction (Ying et al., 2019; Rifaioglu et al., 2026); such masks do not automatically allocate signed contributions to the original score.

Shapley methods allocate predictions relative to a reference (Lundberg and Lee, 2017). Where the model is already additive, its block contributions can be read algebraically. The thesis extends this tracing to crossing edges within a pairing term.

3.6.2 Fidelity and Stability

Measured phenotypes and proteins make highlighted tissue recognisable (Jackson et al., 2020; Danenberg et al., 2022), but plausibility differs from faithfulness to the model (Jacovi and Goldberg, 2020). NaroNet and SpaCEy test explanations with synthetic signals or matched removal controls (Jiménez-Sánchez et al., 2022; Rifaioglu et al., 2026); sanity checks examine dependence on weights and labels (Adebayo et al., 2018). The thesis's held-out marker surrogate similarly targets a fitted pairing score.

Sampling sensitivity and structural consistency are also distinct from reproducibility after refitting: SpaCEy uses tissue crops, whereas HRCHY-CytoCommunity perturbs parameters and annotations (Rifaioglu et al., 2026; Xie et al., 2026). Localising a pattern, reconstructing its calculation and reproducing its predictive role therefore require separate evidence.

3.7 Synthesis, Limitations and Research Gaps

3.7.1 Synthesis and Review Limitations

The evidence supports spatial organisation as biologically relevant, but its predictive benefit depends on the target, comparator and validation design. Different representations preserve different relationships; transfer and explanation are meaningful only for a specified predictor, tissue category or calculation.

Single-author screening, English-language restrictions, database coverage and citation-tracking seeds may omit relevant studies. Requiring outcome terms can miss representation papers, partly addressed by the methodological searches. Disease, modality and endpoint heterogeneity limit performance comparisons; cohort reuse and evolving preprints further restrict independence and maturity of evidence. The focused appraisal is not a complete study-level risk-of-bias or certainty assessment. Interpretation remains qualitative, distinguishing methodological precedents from clinical validation.

3.7.2 Two Primary Gaps and Three Scope Questions

RQ1 asks what learned tissue adds to informative clinical–tissue references for the primary DSS target. RQ2 asks whether explicit pairing adds value beyond regional context and readout availability, and how centring and endpoint features affect that value.

Table 3.5 positions reviewed studies by their principal research task and tissue representation. The highlighted intersection motivates NIMSurv: a clinical–tissue survival framework that combines contextual and regional features with explicit interface pairing through NIRP.

Genomic augmentation (RQ3), frozen common-input OS transfer (RQ4) and tissue tracing with reconstruction fidelity (RQ5) test the scope of the contributions defined in Chapter 1. Chapter 4 develops the evaluation framework.

Table 3.5. Positioning of NIMSurv in the reviewed literature.

	Tissue representation		
Research task	**Cell/image features**	**Regional organisation and category relations**	**Interface-specific cell pairing**
Representation learning and tissue organisation	KRONOS (2025) VirTues (2026)	UTAG (2022) CellCharter (2024) BANKSY (2024) CytoCommunity (2024) Tissue schematics (2022)	
Outcome association	BiGraph (2025)	Danenberg et al. (2022) Schürch et al. (2020) HRCHY-CytoCommunity (2026)	
Tissue-only survival modelling	Graph CNN (2018) Patch-GCN (2021a) SpaCEy (2026)	SPACE-GM (2022) SPIFEE (2026)	
Clinical multimodal survival modelling	DMGN (2023) MultiSurv (2021)		**NIMSurv** *This thesis* via NIRP

Note. Blank cells indicate combinations not directly addressed by the studies shown.

CHAPTER 4

Methodology

NIMSURV predicts patient survival by combining clinical information with two complementary representations of tumour tissue. The regional pathway describes connected cellular environments and their neighbours; the interface pathway captures paired cell states where those regions meet. The proposed interface readout, Niche-conditioned Interface Residual Pooling (NIRP), operates within a framework of graph encoding and Cox survival modelling.

Sections 4.1–4.2 define the cohorts, inputs and tissue structure; Section 4.3 specifies the predictor. Sections 4.4–4.5 describe comparisons, training and evaluation. The remaining sections cover genomic augmentation and transfer (RQ3–RQ4), followed by tissue tracing and score reconstruction (RQ5).

4.1 Study Design and Data

4.1.1 Prediction Pipeline Overview

Each tissue core becomes a graph of nearby cells. Local cell-type mixtures define recurring *niche types*; each connected occurrence is a *niche instance*, or region, and edges between neighbouring instances form an *interface*. Niche types are fitted on training patients without survival outcomes. A shared cell encoder then learns contextual features through the survival-prediction task.

The two tissue pathways aggregate these features across each patient's cores. Their outputs join clinical predictors and predefined tissue summaries in an additive log-risk score, which an estimated baseline hazard converts to a survival curve (Figure 4.1). All cores and modalities from a patient remain together in every split. Primary clinical–IMC models are fitted jointly; genomic augmentation uses the staged procedure in Section 4.6.1.

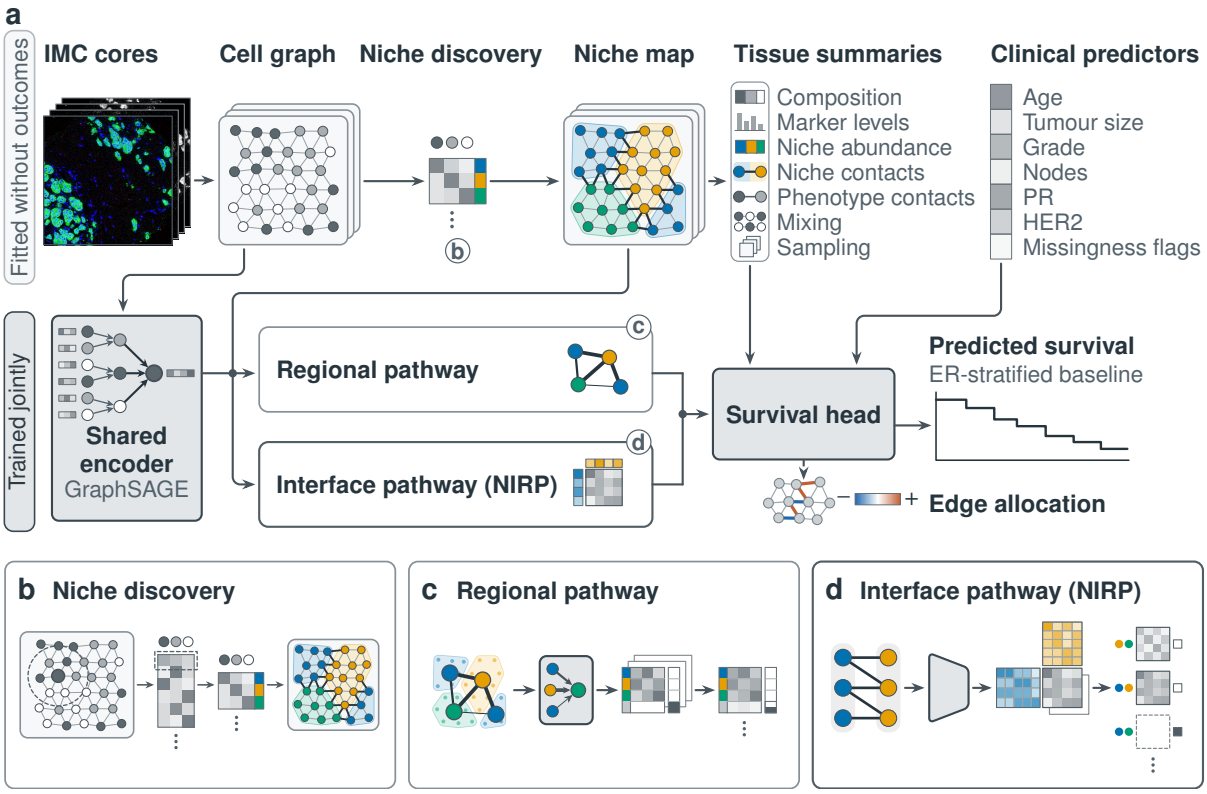


Figure 4.1. NIMSurv architecture: (a) overview; (b) niche discovery; (c) regional pathway; (d) NIRP. IMC image: MB-0288 (Danenberg et al., 2022); CD3 (red), panCK (green), DNA (blue).

4.1.2 Cohorts, Eligibility and Survival Targets

METABRIC–IMC is the development cohort. Released cell measurements, coordinates and phenotype annotations are linked by patient identifier to cBioPortal's `brca_metabric` clinical records (Danenberg et al., 2022; Ali et al., 2020). The second cohort comprises released Basel tumour samples (Jackson et al., 2020), excluding Zurich, liver controls and non-tumour samples. Both use de-identified secondary data; Appendix A.1.1 describes source annotations and linkage. Figure 4.2 distinguishes within-cohort evaluation from frozen transfer, and Table 4.1 lists the populations, targets and panels.

METABRIC eligibility requires a retained core, positive ascertainable follow-up, known ER status and at most one missing value among age, tumour size, grade and nodal status. Missing PR or HER2 does not count towards exclusion. Continuous and categorical missing values are imputed using training-patient medians and modes, with missingness indicators retained. Genomic completeness is required only for the matched extension. Basel eligibility requires a retained tumour core, positive OS follow-up and known ER status.

Table 4.1. Analytic populations and learning settings.

Setting	Patients; target	Markers / phenotypes	Evaluation role
METABRIC primary	544; ten-year DSS	36 / 32	Clinical–IMC prediction and components; RQ1–RQ2
Matched genomics	527; ten-year DSS	36 / 32	Same patients and partitions, with and without genomics; RQ3
Basel-native	276; five-year OS	30 / 27	Learning within the second cohort
Frozen transfer	276 Basel patients; five-year OS	10 / 8	Separate OS systems trained on 544 METABRIC patients and applied unchanged; RQ4

Disease-specific survival (DSS) counts confirmed breast-cancer deaths as events and censors confirmed other-cause deaths at death. Patients whose cause of death is unresolved are excluded from DSS but remain eligible for overall survival (OS), which counts all deaths. DSS predictions represent *model-based net survival* from the breast-cancer cause-specific hazard (Section 2.3.2). OS models are fitted separately.

Follow-up is measured in years: from diagnosis in METABRIC, and by converting the released Basel durations from months (Appendix A.3). Ten-year METABRIC net DSS is primary; five-year DSS and ten-year OS are secondary. Basel uses five-year OS because fewer than 20% remain alive and observed at ten years, both overall and among ER-negative patients. Section 4.5.3 defines scoring-support requirements.

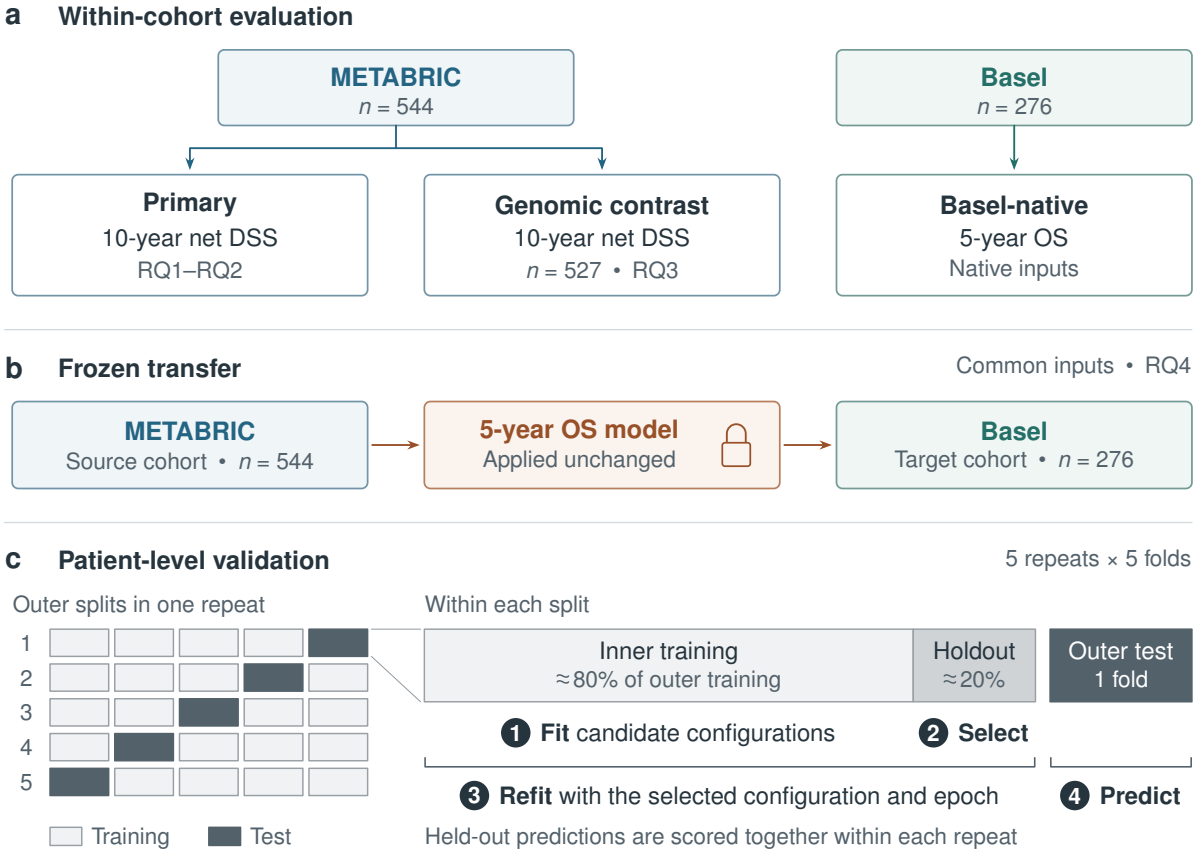


Figure 4.2. Study design: (a) within-cohort analyses; (b) frozen common-input OS transfer from METABRIC to Basel; (c) patient-level training, validation and testing.

4.1.3 Patient Predictors and Tissue Inputs

For patient i, $\mathbf{c}_i$, $\mathbf{g}_i$ and $\mathbf{q}_i$ denote clinical predictors, optional genomic predictors and predefined tissue summaries. The 13-coordinate METABRIC clinical block contains age, log tumour size, grade coded 1–3, indicators for one to three and at least four positive nodes, PR, HER2 and six missingness flags. Basel-native and common-input models use age, grade, PR, HER2 and four flags: eight coordinates. ER defines shared-head baseline strata and tissue interactions; applicable unstratified benchmarks receive ER as a predictor. Treatment is omitted because its availability at the prediction origin is uncertain.

The genomic block contains PAM50 and IntClust indicators, with Normal-like and IntClust 1 as references, plus mutation and copy-number candidates (Curtis et al., 2012; Pereira et al., 2016). Mutation candidates are TP53, PIK3CA, GATA3, CDH1, MAP3K1 and KMT2C; each requires at least 5% prevalence in the relevant training patients. Copy-number candidates are ERBB2, MYC, CCND1 and FGFR1 gains and

PTEN deep loss. The block has at most 26 coordinates. Wild type requires an assay-supported negative mutation call; missing or non-callable results are not negatives.

Imputation precedes transformation and scaling. Continuous and ordinal clinical coordinates use training-fitted means and standard deviations; binary predictors and missingness flags remain unscaled. Appendix A.1.2 specifies encodings and genomic eligibility.

Cell inputs combine processed markers and one-hot phenotype labels: 68 coordinates for METABRIC, 57 for Basel-native learning and 18 for common-input transfer. Segmentation and annotations are supplied measurements. Section 4.6.2 lists the common marker order and phenotype groups.

4.2 Spatial Graph and Niche Construction

Tissue preparation yields a cell graph, niche labels, connected instances and interfaces for each core. Let V and E denote cells and undirected edges, and V_j the cells in instance j. Graph neighbourhoods define connectivity and message passing; a separate nearest-neighbour set describes cell-type composition. Data-dependent transformations follow the training boundaries in Section 4.5.2.

4.2.1 Tissue Quality Control and Marker Preprocessing

Retained cores have at least 200 cells after source-flagged artefact removal and no flagged imaging or segmentation failure. Source flags exclude normal and in-situ tissue. Partial cores remain separate, and unobserved tissue is not imputed. Usable-core count records sampling extent; physical-area density is omitted.

Within each training partition, markers are clipped at their 99.9th percentile, transformed by $\operatorname{arcsinh}(x)$ with cofactor one, and standardised using transformed training means and standard deviations (Bendall et al., 2011). These steps limit extremes and align channel scales. Constants are applied unchanged to validation and test cells. Basel-native learning fits its own constants; frozen transfer uses METABRIC constants. Appendix A.1.3 specifies source filters and numerical handling.

4.2.2 Cell Graph Construction and Niche Discovery

A simple, undirected Delaunay graph connects cell centroids within each core, excluding edges longer than 30 μm. Messages pass in both directions; boundary statistics count each edge once. Separate cores remain disconnected. Distances use the 1 μm IMC pixel scale.

Niche discovery uses each cell and its 19 nearest neighbours (Figure 4.1b). Their phenotype proportions are square-root transformed and clustered into $K = 10$ prototypes from training cells (Lloyd, 1982; Arthur and Vassilvitskii, 2007). Fitting samples at most 300 phenotype-stratified cells per training core and uses ten k-means++ initialisations. Cells receive their nearest prototype's label. Neighbour-distance ties use cell identifiers; prototype-distance ties use the lowest index.

Thus, the 20-cell neighbourhood describes composition, while Delaunay edges govern message passing and regional connectivity. Both pathways share the niche vocabulary without requiring identical neighbourhoods. The per-core cap applies only to prototype fitting, not marker scaling. Appendix A.3.1 details degenerate coordinates, sampling and clustering.

4.2.3 Connected Niche Instances and Interfaces

One synchronous smoothing pass assigns each cell the most common label among itself and its graph neighbours, using pre-pass labels. Ties preserve the current label where possible, otherwise the lowest-indexed maximiser. Connected cells with the same label form an instance; disconnected occurrences remain separate.

Instances with fewer than five cells are processed in increasing size and merged into the adjacent instance with the most crossing edges. Labels and counts update after each merge; isolated instances remain. Five cells is therefore a merging threshold, not a guaranteed final minimum. Component identifiers resolve merge-order and destination ties (Appendix A.3.2).

For neighbouring instances j and j' with niche types $a < b$, their interface contains the oriented pairs

$$I_{jj'} = \{(u, v) : \{u, v\} \in E,\ u \in V_j,\ v \in V_{j'}\}.$$

The lower type index defines the left endpoint, giving $\binom{10}{2} = 45$ possible type pairs. NIRP requires at least five crossing edges per interface; smaller interfaces remain in the cell graph, regional adjacency and contact summaries.

The distance rule, composition neighbourhood, niche count and merging and interface thresholds determine representation scale and readout availability. They are fixed in the reported comparisons, outside the inner grid in Table 4.5.

4.3 NIMSurv Architecture

Shared cell embeddings $\mathbf{h}_v$ feed both pathways (Figure 4.1a). Regional instance states form $\mathbf{z}_i^{\text{node}}$; NIRP matrices pooled by niche-type pair form $\mathbf{z}_i^{\text{int}}$. Together with predefined summaries $\mathbf{q}_i$, they constitute the tissue block $\mathbf{v}_i$ used by the survival head.

4.3.1 Cell Encoding and Regional Aggregation

Cell encoding. Two mean-aggregation GraphSAGE layers combine each cell's marker and phenotype inputs $\mathbf{h}_v^{(0)}$ with its neighbours' features (Hamilton et al., 2017):

$$\mathbf{h}_v^{(k)} = \mathrm{LN}\Big(\mathrm{ReLU}\Big[W_k^{\text{self}}\mathbf{h}_v^{(k-1)} + W_k^{\text{nb}}\bar{\mathbf{h}}_{\mathcal{N}(v)}^{(k-1)} + \mathbf{b}_k\Big]\Big), \qquad k = 1, 2.$$

The neighbour mean $\bar{\mathbf{h}}_{\mathcal{N}(v)}^{(k-1)}$ uses all retained neighbours and is zero for isolated cells. LN denotes layer normalisation with learned scale and shift (Ba et al., 2016). Each self branch has bias $\mathbf{b}_k \in \mathbb{R}^d$; neighbour maps are bias-free. Dropout of 0.1 follows first-layer normalisation during training only; the second layer has no dropout. The contextual embedding $\mathbf{h}_v \in \mathbb{R}^d$ may include messages across niche boundaries. Appendix A.3.3 gives normalisation settings and parameter counts.

Regional encoding. Each instance joins its mean cell embedding with its log cell count (Figure 4.1c):

$$\boldsymbol{\rho}_j^{(0)} = \left[\,|V_j|^{-1}\sum_{v\in V_j}\mathbf{h}_v;\ \log|V_j|\,\right] \in \mathbb{R}^{d+1}.$$

Weights $U_{\text{self}}, U_{\text{nb}} \in \mathbb{R}^{d\times(d+1)}$ then combine this state with neighbouring instances:

$$\boldsymbol{\rho}_j^{(1)} = \mathrm{ReLU}\left(U_{\text{self}}\boldsymbol{\rho}_j^{(0)} + U_{\text{nb}}\sum_{j'\in\mathcal{N}_{\text{niche}}(j)}\frac{c_{jj'}}{\sum_{j''\in\mathcal{N}_{\text{niche}}(j)}c_{jj''}}\boldsymbol{\rho}_{j'}^{(0)} + \mathbf{b}_{\text{reg}}\right) \in \mathbb{R}^d,$$

Here $\mathbf{b}_{\text{reg}} \in \mathbb{R}^d$ is the self-branch bias; the neighbour map is bias-free. Crossing-edge counts $c_{jj'}$ weight regional contacts. Isolated instances receive a zero neighbour term.

Patient pooling. Within each core, states $\boldsymbol{\rho}_j^{(1)}$ are averaged by niche type, weighted by instance cell count. An absent type receives a zero state and indicator $\alpha_{r,a} = 1$. Core states and indicators are combined using

$$w_{ir} = \frac{|V_{ir}|}{\sum_{r'}|V_{ir'}|}.$$

The resulting $\mathbf{z}_i^{\text{node}}$ preserves type identity with $K(d+1)$ coordinates: d states and one absence indicator per type. Pooled indicators may be fractional because they describe absence across sampled cores. Weights use total core cell counts, not only cells of the selected type.

4.3.2 NIRP: Interface-Level Residual Pooling

NIRP forms a $d_{\text{p}} \times d_{\text{p}}$ matrix $\mathbf{C}_I$ from paired deviations on opposite sides of a retained interface (Figure 4.1d). It projects contextual endpoint features, centres them within the interface and averages their outer products. Section 4.3.3 then combines these matrices across the patient.

Endpoint projection. For crossing edge $e = (u_e, v_e)$, a shared bias-free projection gives

$$\mathbf{x}_e = \mathbf{\Phi}\mathbf{h}_{u_e}, \qquad \mathbf{y}_e = \mathbf{\Phi}\mathbf{h}_{v_e}, \qquad \mathbf{\Phi} \in \mathbb{R}^{d_{\text{p}} \times d}.$$

Interface-local centring. For interface I, let n_I be its edge count and let $\bar{\mathbf{x}}_I = n_I^{-1} \sum_{e \in I} \mathbf{x}_e$ and $\bar{\mathbf{y}}_I = n_I^{-1} \sum_{e \in I} \mathbf{y}_e$. Its residual matrix is

$$\begin{aligned} \mathbf{C}_I &= \frac{1}{n_I} \sum_{e \in I} (\mathbf{x}_e - \bar{\mathbf{x}}_I)(\mathbf{y}_e - \bar{\mathbf{y}}_I)^\top \\ &= \frac{1}{n_I} \sum_{e \in I} \mathbf{x}_e \mathbf{y}_e^\top - \bar{\mathbf{x}}_I \bar{\mathbf{y}}_I^\top. \end{aligned} \tag{4.1}$$

All left-feature/right-feature entries are retained. Each cell appears once per crossing edge, with the same multiplicities used in products and reference means (Figure 4.3).

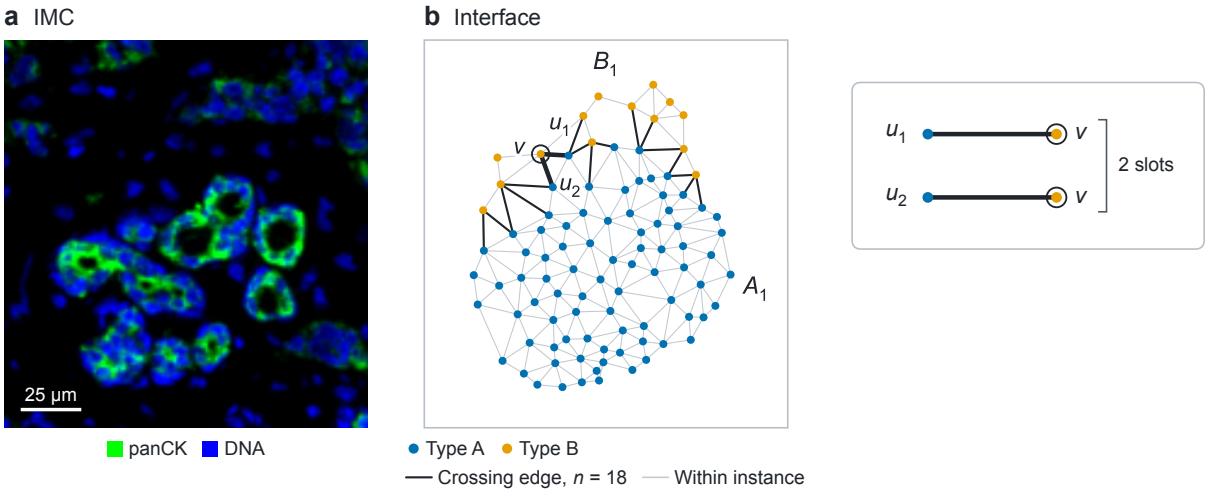


Figure 4.3. Repeated cell appearances in NIRP: cell v contributes twice in METABRIC–IMC core 714 (Danenberg et al., 2022). Scale bar: 25 μm.

The product of means is the expected cross-product under uniform re-pairing of the endpoint lists. This reference preserves cell multiplicities, not geometry, and relates to degree-based bipartite references (Barber, 2007). No permutation experiment is needed to compute it. Gradients pass through the projection and both means. Here, *residual* denotes a centred paired product, not a prediction error.

4.3.3 Patient-Level Interface Aggregation

Patient aggregation combines locally centred matrices without recomputing their means. Let $\mathcal{I}_{ab}^{(r)}$ contain patient i's retained interfaces of type pair (a, b) in core r, and $N_{i,ab} = \sum_r \sum_{I \in \mathcal{I}_{ab}^{(r)}} n_I$. Then

$$\mathbf{C}_{i,ab} = \begin{cases} \dfrac{1}{N_{i,ab}} \displaystyle\sum_r \sum_{I \in \mathcal{I}_{ab}^{(r)}} n_I \mathbf{C}_I, & N_{i,ab} > 0, \\ \mathbf{0}, & N_{i,ab} = 0. \end{cases} \tag{4.2}$$

Each retained crossing edge receives equal weight within its patient and type pair. This is equivalent to edge-count pooling within and across cores, unlike the regional pathway's total-cell weights. Shared endpoints mean these counts are weights, not independent sample sizes.

The indicator $m_{i,ab} = \mathbb{I}(N_{i,ab} = 0)$ distinguishes an unavailable readout from an observed zero residual. Flattening each matrix and appending its indicator forms $\mathbf{z}_i^{\text{int}}$ with $\binom{K}{2}(d_{\text{p}}^2 + 1)$ coordinates. Matrices and coefficients share type-pair, left-feature and right-feature order, with the right feature varying fastest.

Worked example. Two retained interfaces of the same type pair, in one or different cores, have five and ten crossing edges. Their patient matrix is

$$\mathbf{C}_{i,ab} = \frac{5\mathbf{C}_1 + 10\mathbf{C}_2}{15} = \frac{1}{3}\mathbf{C}_1 + \frac{2}{3}\mathbf{C}_2, \qquad m_{i,ab} = 0.$$

Each interface is centred before pooling. Illustrative entry values 0.2 and 0.5 give $(5\times0.2+10\times0.5)/15 = 0.4$. The flattened matrix and indicator enter the corresponding survival-coefficient block. With no retained interface, the matrix is zero and the indicator is one.

4.3.4 Pooled-Centring Decomposition

Section 2.5.3 illustrates how centring changes retained variation. For a supported type pair ($N_{i,ab} > 0$), define $\pi_I = n_I/N_{i,ab}$ and pooled means $\boldsymbol{\mu}_x = \sum_I \pi_I \bar{\mathbf{x}}_I$ and $\boldsymbol{\mu}_y = \sum_I \pi_I \bar{\mathbf{y}}_I$ across all cores. Pooled-centred and uncentred moments are

$$\mathbf{C}^{\text{pool}}_{i,ab} = \frac{1}{N_{i,ab}} \sum_r \sum_{I \in \mathcal{I}^{(r)}_{ab}} \sum_{e \in I} (\mathbf{x}_e - \boldsymbol{\mu}_x)(\mathbf{y}_e - \boldsymbol{\mu}_y)^\top,$$

$$\mathbf{M}^{\text{pool}}_{i,ab} = \frac{1}{N_{i,ab}} \sum_r \sum_{I \in \mathcal{I}^{(r)}_{ab}} \sum_{e \in I} \mathbf{x}_e \mathbf{y}_e^\top.$$

Proposition 4.1 (Decomposition of pooled centring). *For fixed endpoint features, interfaces and edge weights, with $N_{i,ab} > 0$,*

$$\begin{aligned} \mathbf{C}^{\text{pool}}_{i,ab} &= \mathbf{C}_{i,ab} + \mathcal{B}_{i,ab}, \\ \mathcal{B}_{i,ab} &= \sum_I \pi_I (\bar{\mathbf{x}}_I - \boldsymbol{\mu}_x)(\bar{\mathbf{y}}_I - \boldsymbol{\mu}_y)^\top. \end{aligned} \tag{4.3}$$

The uncentred pooled moment additionally contains $\boldsymbol{\mu}_x \boldsymbol{\mu}_y^\top$: $\mathbf{M}^{\text{pool}}_{i,ab} = \mathbf{C}_{i,ab} + \mathcal{B}_{i,ab} + \boldsymbol{\mu}_x \boldsymbol{\mu}_y^\top$.

Proof. Split each deviation from the pooled mean into a within-interface deviation and a difference between means. Mixed terms vanish because within-interface deviations sum to zero. Since $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$ are the means over all endpoint appearances, $\mathbf{M}^{\text{pool}}_{i,ab} = \mathbf{C}^{\text{pool}}_{i,ab} + \boldsymbol{\mu}_x \boldsymbol{\mu}_y^\top$. □

Local centring removes between-interface variation $\mathcal{B}_{i,ab}$; pooled centring retains it, and the uncentred moment also retains the mean product. Section 4.4.2 compares the learned procedures.

4.3.5 Multimodal Fusion and Survival Prediction

Patient-level representation. Predefined summaries describe composition, marker levels, contacts and sampling alongside the learned tissue pathways (Table 4.2). Niche summaries use the same training-fitted vocabulary. Clinical and, in the matched extension, genomic predictors supply patient-level information.

Table 4.2. Tissue summaries and pooling weights.

Summary	Core-level measurement	Weight
Composition	Phenotype fractions	Cell count
Marker levels	Mean transformed marker values	Cell count
Niche abundance	Fractions in each final niche type	Cell count
Niche contacts	Fraction of graph edges crossing each type pair	All-edge count
Phenotype contacts	Coarse-type contact z-scores, including same-type pairs	Cell count
Mixing	Epithelial–immune / immune–immune edge ratio	All-edge count
Sampling	Usable-core count and no-edge flag	Patient level

Phenotype-contact z-scores use 500 within-core label permutations (Schapiro et al., 2017; Palla et al., 2022). Mixing compares epithelial–immune with immune–immune edges, treating T cells, B cells, myeloid and other immune cells as immune (Keren et al., 2018). Undefined values, including zero permutation variance, receive zero and a separate flag. Patient mixing averages core ratios by all-edge count rather than dividing pooled contact counts. Summary columns are standardised on training patients; learned blocks receive no post-hoc scaling. Appendix A.1.4 specifies column order and pooling of undefined-value indicators.

The complete tissue vector is

$$\mathbf{v}_i = [\mathbf{z}_i^{\text{node}}; \mathbf{z}_i^{\text{int}}; \mathbf{q}_i].$$

Table 4.3 records its dimensions.

Table 4.3. Tissue feature dimensions ($K = 10$). $Q = A + P + 131$, where A and P count phenotypes and markers.

Block	Dimension	Evaluated sizes
Regional states and absence indicators	$K(d + 1)$	330 or 650
Pairing matrices and availability indicators	$\binom{K}{2}(d_{\text{p}}^2 + 1)$	765 or 2,925
Summaries: METABRIC / Basel / common	Q	199 / 188 / 149

The full-panel METABRIC tissue vector ranges from $330+765+199 = 1{,}294$ to $650+2{,}925+199 = 3{,}774$ coordinates. Across the (d, d_{p}) grid, the model has 11,465–33,737 trainable parameters, including neural weights, biases, normalisation parameters and survival coefficients (Table A.4).

Table 5.1 reports 174 breast-cancer deaths: 128 ER-positive and 46 ER-negative. Five-fold ER/event stratification implies 102–103 and 36–37 events, respectively, per outer-training partition.

ER-stratified risk model. Let s_i denote ER stratum, let $\mathbf{w}_i$ collect the applicable inputs, and define $\nu_i = \mathbb{I}(s_i = \mathrm{ER}^-)$. The patient score and DSS cause-specific hazard are

$$\begin{aligned} \eta_i &= \boldsymbol{\beta}_C^\top \mathbf{c}_i + \boldsymbol{\beta}_G^\top \mathbf{g}_i + (\boldsymbol{\beta}_V + \nu_i \boldsymbol{\gamma}_V)^\top \mathbf{v}_i, \\ h_1(t \mid \mathbf{w}_i, s_i) &= h_{0,s_i}(t) \exp(\eta_i). \end{aligned} \tag{4.4}$$

Baseline hazards differ by stratum, with proportional hazards assumed within strata. Clinical and genomic coefficients are shared; tissue coefficients combine a shared effect with a regularised ER-negative deviation. The score has no free intercept or tissue-selection gate, and omits genomics outside the matched extension.

Tissue functions may be nonlinear, but contributions are additive on the log-risk scale. Survival probabilities use the combined score and baseline, not sums of modality-specific probabilities. The ER interaction applies only to tissue, without unrestricted cross-modal interactions.

Survival estimation. After fitting, complete training risk sets estimate the ER-specific Breslow baseline. At event time t, its increment is

$$\Delta \widehat{H}_{0,s}(t) = \frac{|D_s(t)|}{\sum_{i' \in R_s(t)} \exp(\widehat{\eta}_{i'})},$$

where $D_s(t)$ contains stratum-s training patients with target events at t, $R_s(t)$ contains those still at risk, and $\widehat{\eta}$ is the fitted score. Net survival is

$$\widehat{S}_{1,\mathrm{net}}(t \mid \mathbf{w}_i, s_i) = \exp\{-\widehat{H}_{0,s_i}(t) \exp(\widehat{\eta}_i)\}. \tag{4.5}$$

OS uses its own event coding, fitted score and all-cause baseline.

4.4 Comparators and Component Evaluation

RQ1 compares the complete predictor with clinical, summary-based and learned references; RQ2 varies the interface representation. Comparisons match patients, endpoints and partitions, but differ in their control of the survival model, representation and fitting procedure.

4.4.1 Framework Comparators and Benchmarks

Shared-head comparisons. These retain the additive survival-model family while varying tissue inputs (Table 4.4). Each is fitted separately.

The flat reference uses the same cell-encoder family and width grid, averaging embeddings within cores and pooling cores by cell count alongside predefined summaries. Its embeddings already contain spatial context; it lacks explicit regional and interface pooling.

Table 4.4. Shared-head models and component comparisons.

Model	Tissue information or change	Question addressed
Complete-framework comparisons (RQ1)		
Clin	None	Clinical reference
Clin+Comp	Composition, marker means and applicable flags	Value of direct measurements
Clin+Sum	Complete predefined summaries	Value of summary features
Clin+Flat	Summaries and mean-pooled cell embeddings	Flat learned representation
Clin+Node	Summaries and regional pathway	Regional representation
NIMSurv	Summaries, regions and interfaces	Complete framework
Interface-component comparisons (RQ2)		
Support-only	Availability indicators without pairing matrices	Pairing beyond availability
Pooled-centred	Centre after pooling endpoints by type pair	Retaining between-interface variation
Uncentred	Uncentred endpoint products	Retaining mean-state information
Input-feature	Project cell inputs rather than encoder outputs	Endpoint feature construction

System-level benchmarks. CPH, CPH-strat, RSF, DeepSurv, DeepHit, BroadSurv, DAGSurv and AttenSurv-MH receive clinical variables and tissue summaries, retain their own prediction functions and use inner-validation IBS for selection. CPH includes ER as a predictor; CPH-strat uses ER-specific baselines. DMGN-M locally adapts DMGN by fusing marker-image regions and clinical embeddings through a graph and Cox-type predictor (Fu et al., 2023).

Architecture, optimisation and compute budgets are unmatched. CPH and CPH-strat search six and five ridge penalties, respectively. RSF searches six combinations of minimum leaf size and feature subsampling; DMGN-M searches six combinations of learning rate and auxiliary-loss weight. Selection uses inner-validation IBS. Appendix A.3.6 specifies libraries, input handling and DMGN-M images; Table A.6 lists search spaces.

Published context. SpaCEy is not fitted locally. Its breast experiments use tissue-only neural Cox OS prediction and unweighted comparable-pair concordance (Rifaioglu et al., 2026), unlike the clinical–tissue, ER-stratified net-DSS protocol here (Section 4.5.3).

4.4.2 Boundary-Pairing Component Comparisons

Clin+Node tests the complete interface block; Support-only asks whether pairing matrices add value beyond availability indicators. Pooled-centred and Uncentred replace local residuals with the representations in Proposition 4.1.

Input-feature retains learned projection but uses original cell inputs instead of contextual embeddings. It removes both neighbourhood aggregation and preceding nonlinear transformations, testing endpoint construction rather than message passing alone. All variants are refitted, allowing encoders, features, coefficients and configuration choices to adapt. These contrasts compare learning procedures, not interventions on a fixed readout; the proposition describes the separate fixed-feature relationship.

4.5 Training and Predictive Evaluation

Evaluation separates parameter fitting, configuration and epoch selection, and held-out assessment. Shared-head models use patient batches for optimisation, full inner-validation risk sets for selection and full training risk sets for baseline estimation. Preprocessing follows the same training boundaries.

Architecture and tissue construction were developed through preliminary METABRIC experiments. Model definitions, thresholds, search spaces, partitions and scoring rules were then fixed for final repeated cross-validation, Basel-native evaluation and frozen transfer. METABRIC therefore provides internal evaluation after development on the same cohort (Section 6.6.1).

4.5.1 Survival Objective and Optimisation

Primary clinical–IMC models jointly learn the encoder, readouts and survival coefficients; genomic tissue fitting updates only unfixed parameters. The objective combines event-normalised negative Cox partial log-likelihood and squared-norm penalties (Cox, 1972; Efron, 1977):

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{Cox}} &+ \lambda_\beta\left(\|\boldsymbol{\beta}_C\|_2^2 + \|\boldsymbol{\beta}_G\|_2^2 + \|\boldsymbol{\beta}_V\|_2^2\right) \\ &+ \lambda_\gamma\|\boldsymbol{\gamma}_V\|_2^2 + \lambda_\omega\|\boldsymbol{\omega}\|_2^2.\end{aligned} \tag{4.6}$$

Neural parameters $\boldsymbol{\omega}$ comprise the cell encoder, regional transformation and interface projection. Absent blocks have no penalty. Efron's approximation handles tied events; genomic preliminary scores remain fixed offsets.

Training updates. A batch-restricted Cox surrogate compares each event with patients at risk in the same ER stratum and batch, rather than using full-cohort risk sets at every update (Kvamme et al., 2019, Section 3.2).

Each epoch shuffles patients without replacement within ER strata. For n_{train} patients, the batch count is

$$n_{\text{batch}} = \lceil n_{\text{train}}/64 \rceil .$$

Each stratum is divided into this many approximately equal subsets. Corresponding subsets are combined and shuffled, targeting roughly 64 patients per batch without stratifying by event status. Incomplete batches are retained; event-free batches are logged and skipped. If only one stratum has events, only it contributes event terms (Equation A.13). Appendix A.3.7 gives batch counts implied by the partitions.

Adam uses explicit squared-norm penalties and zero optimiser weight decay (Kingma and Ba, 2015); Table 4.5 lists the settings.

Validation criterion. Shared-head selection and early stopping minimise unpenalised, event-normalised Cox loss on complete inner-validation risk sets. Checkpoints replace the incumbent only for improvement greater than 10^{-4}. The selected epoch determines refit duration; Appendix A.3 records checkpoint and tie conventions.

Table 4.5. Shared-head training settings.

Quantity	Setting
Selected within the inner split, when applicable	
Encoder width d	$\{32, 64\}$
Projection dimension d_{p}	$\{4, 8\}$
Main coefficient penalty λ_β	$\{10^{-3}, 10^{-2}, 10^{-1}\}$
Fixed settings and penalty rules	
ER-deviation / neural penalties	$10\lambda_\beta$ / 10^{-4}
Encoder depth / first-layer dropout	2 / 0.1
Adam learning rate / gradient-norm limit	10^{-3} / 1.0
Batch-count rule	$n_{\text{batch}} = \lceil n_{\text{train}}/64 \rceil$; stratum-wise allocation
Epoch limit / patience / improvement tolerance	200 / 20 / 10^{-4}
Numerical precision	`float32`; no mixed precision
Random seeds	Global seed 0; patient-bootstrap seed 0

Cox fitting and baseline estimation use all available event and censoring times. The five- and ten-year horizons restrict prediction and scoring, not training follow-up.

4.5.2 Patient-Level Validation and Model Selection

Five repeats of five-fold cross-validation provide one held-out prediction per patient per repeat. Each outer-training partition reserves approximately 20% for a single inner-validation holdout, not an additional five-fold cycle. Compared models share assignments (Figure 4.2c; Algorithm 1).

Outer folds and inner holdouts are stratified jointly by ER status and target-event indicator. Genomic subsets inherit assignments by intersecting partitions with eligible identifiers. METABRIC outer-training and test sets contain 435–436 and 108–109 patients, respectively. Appendix A.3.4 specifies splitting functions and seeds.

Algorithm 1 Training and prediction within one outer split

Input: Outer-training patients D_{train}, inner-validation patients $D_{\text{val}} \subset D_{\text{train}}$, untouched test patients D_{test}, configurations C, horizon τ.

Output: Held-out curves and risks for D_{test}, or undefined selection.

$D_{\text{inner}} \leftarrow D_{\text{train}} \setminus D_{\text{val}}$
$P \leftarrow \textsc{FitPreparation}(D_{\text{inner}})$
$(Z_{\text{inner}}, Z_{\text{val}}) \leftarrow (P(D_{\text{inner}}), P(D_{\text{val}}))$
for each configuration $c \in C$ **do**
    $M_c \leftarrow \textsc{FitCandidate}(Z_{\text{inner}}, c)$
    $(s_c, e_c) \leftarrow \textsc{Validate}(M_c, Z_{\text{inner}}, Z_{\text{val}})$
end for
$C_{\text{valid}} \leftarrow \{c \in C : \text{selection score is valid and finite}\}$
if $C_{\text{valid}} = \varnothing$ **then**
    return undefined selection
end if
$c^\star \leftarrow \arg\min_{c \in C_{\text{valid}}} s_c$; $e^\star \leftarrow e_{c^\star}$
$P^\star \leftarrow \textsc{FitPreparation}(D_{\text{train}})$
$Z_{\text{train}} \leftarrow P^\star(D_{\text{train}})$
$M^\star \leftarrow \textsc{Refit}(Z_{\text{train}}, c^\star, e^\star)$
if $M^\star$ is Cox-type **then**
    Estimate its baseline hazards from complete training risk sets.
end if
for each patient $i \in D_{\text{test}}$ **do**
    $\widehat{S}_i(t) \leftarrow \textsc{PredictSurvival}(M^\star, P^\star(i))$; $\widehat{\psi}_i(\tau) \leftarrow 1 - \widehat{S}_i(\tau)$
end for

All partitions keep each patient's cores and modalities together. FitPreparation fits imputation, marker transformations, summary scaling, applicable mutation filters and niche prototypes. For epoch-trained models, FitCandidate retains checkpoints. Validate returns the best score s_c and selected epoch e_c, minimising full-risk-set inner-validation Cox loss for shared-head models or IBS for system benchmarks. For Cox-type benchmark IBS, Validate estimates candidate baselines on complete inner-training risk sets. Epoch arguments apply only to epoch-trained models.

For Cox-type benchmark IBS, each evaluated checkpoint uses baselines estimated from complete inner-training risk sets under its own stratification. ER-specific censoring survival is estimated on inner-validation patients. Candidates are excluded for undefined IPCW weights, non-finite predictions or non-finite IBS. Valid candidates minimise IBS.

If none is valid, no configuration or outer-test prediction is produced for that model and split. Such a split is not a completed fold for paired repeat-level comparisons. Successful selection is followed by refitting on all outer-training patients and, for Cox-type models, re-estimating baselines. Basel-native learning uses separate within-cohort partitions.

Held-out predictions from all outer folds are scored together within each repeat; the five metrics are then averaged. Curves are not averaged across repeats before scoring. Paired differences and bootstrap intervals follow the same order.

Shared-head configurations (Table 4.5) resolve exact ties lexicographically in favour of smaller $(d, d_{\mathrm{p}}, \lambda_{\beta})$ and refit from scratch for the selected epoch count (Appendix A.3). Benchmarks retain grid order for exact ties (Table A.6). Epoch-trained benchmarks restore the best IBS checkpoint, using patience 20 and improvement tolerance 10^{-4}.

4.5.3 Prediction Scales and Evaluation Metrics

The ranking score is horizon risk $\widehat{\psi}_i(\tau) = 1 - \widehat{S}_i(\tau)$. For Cox-type models, the fitted score and baseline place predictions across folds and ER strata on a common endpoint-specific probability scale. Raw scores are restricted to labelled within-fit, within-stratum analyses. Truncated concordance and IBS are co-primary (Table 4.6).

Table 4.6. Performance measures and prediction scales.

Measure	Prediction evaluated	Role
Truncated Uno-type C	Endpoint-matched horizon risk	Co-primary risk ordering
Integrated Brier score	DSS net-survival or separately fitted OS curves	Co-primary probability error
Harrell's C	Horizon risk over observed follow-up	Supplementary ordering
Time-dependent AUC and Brier score	Ten-year target-specific predictions	Supplementary horizon measures
Cox-type calibration slope	Frozen prognostic index	Score-spread diagnostic

Truncated concordance compares target events at or before τ with strictly later observed follow-up; prediction ties receive half credit (Uno et al., 2011). For an earlier event in patient i compared with i', the weight is

$$w_{ii'} = \left\{\widehat{G}_{s_i}(\widetilde{T}_i^-)\widehat{G}_{s_{i'}}(\widetilde{T}_i^-)\right\}^{-1}.$$

Here $\widehat{G}_s$ is reverse Kaplan–Meier censoring survival for the evaluated patients in ER stratum s. Event/censoring ties use the pre-event risk-set denominator. These estimates weight assessment without updating predictions.

The Brier score measures censoring-adjusted squared error in survival status (Graf et al., 1999), as defined in Equation (A.18). Target events receive weight $1/\widehat{G}_{s_i}(\widetilde{T}_i^-)$; known survival beyond t receives $1/\widehat{G}_{s_i}(t)$. Patients censored at or before t contribute neither term but remain in the denominator. IBS integrates from zero to τ and divides by τ, using trapezoidal integration at 0.05-year intervals: 201 points for ten years and 101 for five. Cox-type curves use right-continuous training-derived Breslow baselines.

DSS scoring censors other-cause deaths and follow-up end; OS censors only follow-up end. Net-DSS assessment assumes independent censoring within ER strata, an assumption distinct from cause-specific hazard estimation and not established by ER stratification.

Horizon support requires at least 20% alive and observed at τ and five target events overall, plus $\widehat{G}_s(\tau^-) \geq 0.05$ in each represented ER stratum. Horizons are fixed before performance comparison. These rules apply to inner-validation IBS, pooled out-of-fold predictions within each repeat, frozen Basel assessment and bootstrap resamples. Primary comparisons do not score individual outer folds separately.

Metrics are undefined if support fails, required weights are non-finite or the comparable-pair denominator is zero. Neither IPCW clipping nor a shorter horizon is used. Paired comparisons use the same patients and complete repeats for both models.

Time-dependent AUC compares cumulative cases with survival beyond the horizon (Blanche et al., 2013). Genomic analyses fit Cox-type calibration slopes to frozen indices with ER-specific baselines and, internally, outer-fold-specific baselines within each repeat (Riley et al., 2024). Slopes diagnose score spread, not complete absolute-risk calibration. Section 4.6.2 describes grouped OS-transfer calibration.

4.5.4 Paired Comparisons and Uncertainty Estimation

Uncertainty uses 1,000 valid patient-bootstrap resamples (Efron and Tibshirani, 1993). Sampling with replacement is stratified by ER and preserves stratum sizes. Each patient carries all repeat-specific predictions; compared models share draws, with censoring survival re-estimated within each resampled

stratum. Cross-validation metrics are calculated per repeat and then averaged; frozen transfer uses one fixed prediction per patient.

A draw is accepted only if all repeat-specific metrics required for the contrast are defined. Otherwise, both models discard it and share a replacement until 1,000 valid draws are obtained. Appendix A.3.10 specifies the generator and weighting conventions. Percentiles 2.5 and 97.5 give pointwise 95% intervals conditional on saved fits: preprocessing, selection and training are not repeated, and intervals do not cover the whole comparison family simultaneously.

Differences are NIMSurv minus comparator, or genomic augmentation minus its matched unaugmented procedure. Positive concordance and negative IBS differences favour the first model; the two metrics are interpreted separately.

4.5.5 Implementation

GraphSAGE is implemented directly in PyTorch. Appendix A.3 records the software, precision, checkpoint rules and benchmark search spaces used in the reported analysis.

4.6 Genomic Augmentation and Cross-Cohort Evaluation

The extensions examine additional patient information through staged genomic fitting and cohort transfer through a frozen source-trained OS predictor.

4.6.1 Matched Genomic Augmentation

RQ3 compares genomic and non-genomic procedures on the same 527 patients and inherited partitions. Each first fits a patient-information score, then learns tissue contributions around that fixed score, unlike the primary joint clinical–IMC model.

Inner selection. Each arm fits an ER-stratified linear Cox model to clinical or clinical–genomic inputs on inner-training patients. Its L2 penalty is selected by unpenalised full-risk-set Cox loss on the inner-validation holdout, separately for each arm and outer split, then retained for outer refitting (Appendix A.3.8). Training offsets are in-sample; validation and outer-test offsets are generated without fitting on those patients.

A separately initialised tissue function learns regional, pairing, availability and summary terms with ER interactions around the fixed offsets. Inner validation selects its configuration and epoch without updating preliminary coefficients.

Outer refit and prediction. The preliminary model is refitted on all outer-training patients and frozen while the selected tissue function is refitted. ER-specific baselines use the combined score before outer-test prediction. Preprocessing and mutation filtering respect these boundaries. Since both stages are refitted separately in each arm, the contrast adds genomics to a matched staged procedure, not an unchanged fitted tissue model.

4.6.2 Common-Input Evaluation and Frozen Transfer

Basel-native learning uses the native panel and phenotypes in repeated patient-level validation. Frozen transfer instead applies a METABRIC-trained five-year OS predictor without updating. Held-out source performance uses the same restricted inputs as transfer, not the native METABRIC panel.

Compatible inputs. Shared markers, in order, are SMA, CK5, CK8–18, CD68, HER2 (3B5), CD3, CD20, ER, Ki-67 and panCK. Phenotypes map to epithelial, T cell, B cell, myeloid, other immune, fibroblast/stromal, endothelial and other/unassigned. Clinical inputs are age, grade, PR, HER2 and missingness flags. Both cohorts use this specification without substituting native panels; matching names does not establish measurement equivalence.

Source-only selection and refitting. All selection uses METABRIC. The final shared-head configuration is the most frequent tuple across source inner selections. Modal ties use mean validation loss over splits selecting each tuple, then over all source splits if tied, and finally lexicographic $(d, d_{\mathrm{p}}, \lambda_{\beta})$ order, omitting absent dimensions. Benchmarks follow the same modal rule using IBS, with exact ties resolved by grid order. Where applicable, final epochs are the rounded-up mean selected epochs for the chosen configuration. Systems are refitted on all eligible source patients; Cox-type baselines use complete source risk sets. Table A.8 records the final settings.

Frozen target-cohort assessment. Input order, preprocessing, applicable niche prototypes, parameters and baselines remain fixed. Basel receives predictions without selection, refitting or recalibration; outcomes and censoring estimates are used only for five-year OS assessment and paired bootstrap.

Grouped calibration orders the 276 patients by frozen predicted risk into five groups of 55–56, breaking ties by ascending identifier. Mean predicted survival is compared with five-year Kaplan–Meier survival

and pointwise 95% complementary log–log Greenwood intervals (Section 5.5). This restricted-input OS experiment does not externally validate the native-panel ten-year net-DSS predictor.

4.7 Score Tracing and Surrogate Evaluation

RQ5 links fitted pairing scores to tissue in three steps: exact allocation to crossing edges, description by measured phenotypes and proteins, and patient-held-out reconstruction using marker-based surrogates.

4.7.1 Exact Decomposition of Pairing Scores

The additive tissue score separates regional, pairing, availability and summary terms. For a supported type pair, the pairing term is

$$\eta_{i,ab}^{\text{pair}} = \left\langle \boldsymbol{\Theta}_{ab} + \nu_i \boldsymbol{\Gamma}_{ab},\, \mathbf{C}_{i,ab} \right\rangle_{\text{F}},$$

where $\boldsymbol{\Theta}_{ab}$ and $\boldsymbol{\Gamma}_{ab}$ are the pairing coefficients and ER-negative deviations. The Frobenius inner product sums entrywise products. Holding features and coefficients fixed gives the edge allocation

$$\begin{aligned} a_{i,I,e} &= \frac{(\mathbf{x}_e - \bar{\mathbf{x}}_I)^\top (\boldsymbol{\Theta}_{ab} + \nu_i \boldsymbol{\Gamma}_{ab})(\mathbf{y}_e - \bar{\mathbf{y}}_I)}{N_{i,ab}}, \\ \eta_{i,ab}^{\text{pair}} &= \sum_r \sum_{I \in \mathcal{I}_{ab}^{(r)}} \sum_{e \in I} a_{i,I,e}. \end{aligned} \tag{4.7}$$

Substituting the interface and patient averages establishes the equality. It allocates a fixed log-risk term, not a change in survival probability or an edge-deletion effect. Availability terms and display offsets remain separate.

4.7.2 Contribution Summaries and Tissue-Case Selection

Type pairs are ranked by the standard deviation of raw contributions among ER-positive, repeat-1 out-of-fold patients, including unsupported-pair zeros. ER-negative displays use the same order without a between-stratum test. Availability counts and weighted unavailable-pair terms are reported separately. This ranks contribution variability, not biological importance.

Signed displays subtract each fit's ER-specific outer-training mean contribution, changing the plotted zero but not predictions. An unavailable pair can therefore display a non-zero value without an edge allocation. Risk tertiles use predicted ten-year net-DSS risk, not outcomes.

Two tissue examples use supported ER-positive patients nearest the tenth and ninetieth predicted-risk percentiles, without consulting outcomes. Selection then takes the largest retained core with non-zero

allocations, followed by the pair and interface with the largest absolute summed contribution; identifiers resolve ties. Tissue profiles count unique incident cells rather than NIRP's repeated appearances.

The separate CPH–NIMSurv prognostic example is retrospective and outcome-informed. MB-2933 and MB-0288 were selected across five repeats: both were ER-positive and held out together, with similar CPH risks but contrasting NIMSurv risks and ten-year DSS outcomes (Appendix A.3).

4.7.3 Measured Tissue Profiles and Score Reconstruction

Fidelity analysis considers five pairs selected from the ER-positive out-of-fold variability ranking. Niche profiles describe measured phenotypes and proteins after partitioning, without using outcomes or contribution signs to define them.

A ridge surrogate predicts each selected edge's frozen local score,

$$b_e = (\mathbf{x}_e - \bar{\mathbf{x}}_I)^\top (\boldsymbol{\Theta}_{ab} + \nu_i \boldsymbol{\Gamma}_{ab})(\mathbf{y}_e - \bar{\mathbf{y}}_I),$$

before division by $N_{i,ab}$. For measured markers p and p', the inputs are

$$t_{e,pp'} = (x^{\text{marker}}_{u_e,p} - \bar{x}^{L,\text{marker}}_{I,p})(x^{\text{marker}}_{v_e,p'} - \bar{x}^{R,\text{marker}}_{I,p'}),$$

using NIRP's endpoint-appearance lists for the means. All $36^2 = 1{,}296$ ordered marker products are included.

Surrogates are fitted by ER stratum and type pair, with each patient assigned unit total retained-edge weight. Three patient-level folds within outer training select scaling and the ridge penalty; outer-test patients assess reconstruction. Eligibility requires at least ten supporting training patients and five test patients. Appendix A.4.2 specifies scalers, penalties, solver and metric weights.

Four selected outer-test splits per pair give 20 attempted evaluations (Appendix A.3); Table 5.10 reports eligible evaluations. Predicted edge scores are averaged over each patient's retained edges to reconstruct the patient pairing score. Edge- and patient-level R^2 and RMSE measure fidelity. Across eligible evaluations, R^2 is summarised by median and interquartile range, and RMSE by median; a single evaluation has no interquartile range.

4.7.4 Niche-Type Alignment Across Fits

Hungarian matching aligns training-derived phenotype and marker profiles across fits with the same inputs and niche count (Kuhn, 1955). Repeat 1, Fold 1 provides the reference. Each fit's profiles use its own outer-training cells after smoothing and merging; Appendix A.4.3 defines scaling and assignment cost.

If alignment reverses a pair's left/right order, both residual and coefficient matrices are transposed, preserving their Frobenius inner product and pairing score. Predictions use native model coordinates; alignment affects only summaries and displays.

4.8 Chapter Summary

NIMSurv combines clinical inputs, tissue summaries, regional states and interface pairing. Matched comparisons assess prediction, genomic augmentation and transfer; score tracing and marker surrogates examine fitted tissue contributions. Chapter 5 reports these results.

CHAPTER 5

Results

The primary comparisons supported the complete NIMSurv framework and the use of paired-cell features. Further analyses examined genomic augmentation, prediction in Basel, and whether fitted tissue contributions could be located and reconstructed from measured proteins.

Unless stated otherwise, results concern ten-year model-based net disease-specific survival (DSS) in METABRIC. Higher Uno's concordance (C) indicates better risk ordering; lower integrated Brier score (IBS) indicates lower probability error over time. Reported 95% intervals are patient-bootstrap intervals; paired contrasts compare models on the same patients (Sections 4.5.3–4.5.4).

5.1 Cohort Characteristics and Tissue Coverage

5.1.1 Analytic Populations and Event Counts

METABRIC supplied the development cohort and a matched subset with genomic measurements; Basel supplied a second cohort for overall survival (OS) analyses. Figure 5.1 summarises patient selection, and Table 5.1 records the resulting patient, tissue and event counts. Events are reported over the available follow-up, rather than only within the prediction horizon.

5.1.2 Clinical Characteristics and Follow-up

In both cohorts, ER-negative patients had more high-grade tumours and shorter recorded event-or-censoring times (Table 5.2). The survival curves and risk sets show declining follow-up support, particularly late in the Basel ER-negative group (Figure 5.2).

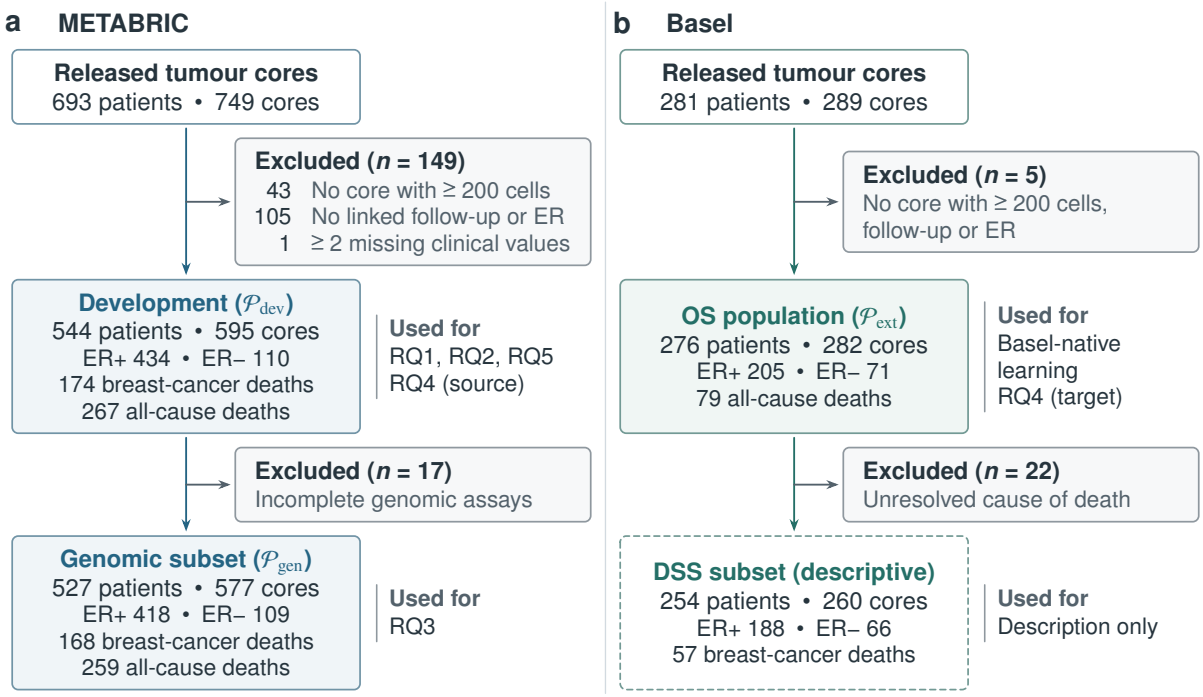


Figure 5.1. Patient and tissue flow: (a) METABRIC; (b) Basel; criteria in Section 4.1.2.

Table 5.1. Analytic populations and events.

A. Patient and tissue selection

Population	Patients	Cores	Cells
METABRIC: released tumour cores	693	749	1,066,966
At least 200 cells per core	650	704	1,062,219
Linked follow-up and ER information	545	596	926,466
METABRIC development ($\mathcal{P}_{dev}$)	544	595	925,425
Genomic candidates (three assays)	527	577	906,024
Matched genomic subset ($\mathcal{P}_{gen}$)	527	577	906,024
Basel: released tumour cores	281	289	767,833
Basel OS population ($\mathcal{P}_{ext}$)	276	282	760,211
DSS candidates (descriptive)	254	260	694,771

B. Events over available follow-up

Population	ER+ / ER–	Breast-cancer deaths	All-cause deaths
METABRIC development ($\mathcal{P}_{dev}$)	434 / 110	174 (128 / 46)	267 (205 / 62)
Genomic candidates	418 / 109	168 (122 / 46)	259 (197 / 62)
Matched genomic subset ($\mathcal{P}_{gen}$)	418 / 109	168 (122 / 46)	259 (197 / 62)
Basel OS population ($\mathcal{P}_{ext}$)	205 / 71	—	79 (58 / 21)
Basel DSS candidates	188 / 66	57 (41 / 16)	—

Table 5.2. Clinical characteristics by ER stratum.

	METABRIC		Basel	
Characteristic	**ER+ *n* = 434**	**ER− *n* = 110**	**ER+ *n* = 205**	**ER− *n* = 71**
Age at diagnosis, years	61.10 (51.57, 70.71)	57.06 (48.31, 67.55)	62.00 (52.00, 73.00)	66.00 (51.00, 76.00)
Tumour size, released scale	22.00 (18.00, 30.00)	25.00 (18.25, 35.00)	n/a	n/a
Grade 1	46 (10.6%)	4 (3.6%)	36 (17.6%)	2 (2.8%)
Grade 2	181 (41.7%)	10 (9.1%)	104 (50.7%)	12 (16.9%)
Grade 3	202 (46.5%)	96 (87.3%)	65 (31.7%)	57 (80.3%)
Grade: remainder	5 (1.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Positive nodes: 0	177 (40.8%)	57 (51.8%)	n/a	n/a
Positive nodes: 1–3	154 (35.5%)	29 (26.4%)	n/a	n/a
Positive nodes: ≥ 4	87 (20.0%)	23 (20.9%)	n/a	n/a
Positive nodes: remainder	16 (3.7%)	1 (0.9%)	n/a	n/a
PR positive	275 (63.4%)	10 (9.1%)	148 (72.2%)	1 (1.4%)
HER2 positive	37 (8.5%)	30 (27.3%)	29 (14.1%)	23 (32.4%)
Observed time, years	9.25 (5.24, 12.63)	5.53 (2.90, 11.73)	6.92 (4.92, 9.83)	3.50 (1.50, 6.00)
Breast-cancer deaths	128	46	—	—
All-cause deaths	205	62	58	21

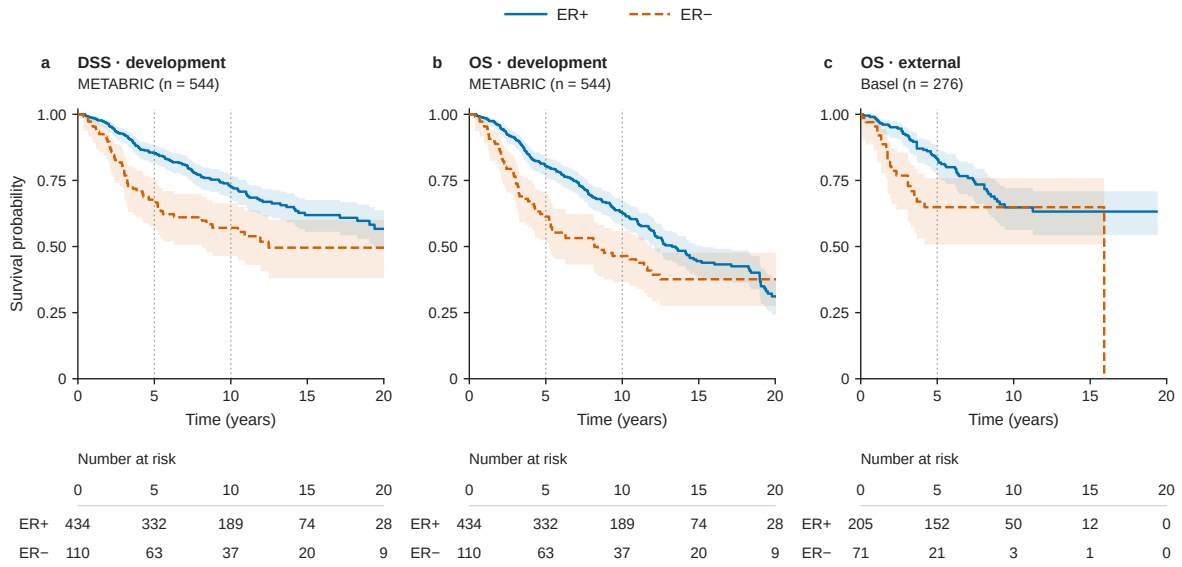


Figure 5.2. Kaplan–Meier curves and risk sets by ER stratum: (a) METABRIC DSS (other-cause deaths censored); (b) METABRIC OS; (c) Basel OS.

5.1.3 Niche Structure and Interface Availability

Most candidate interfaces were retained, yet individual niche-type pairs remained sparsely observed across patients (Table 5.3). Patients typically supported only a small fraction of the possible pairs, so high interface retention did not translate into broad coverage of each relationship.

Niche abundance also varied across fitted partitions (Appendix Table B.1). Table 5.8 introduces the working labels used in the tissue analyses.

Table 5.3. Interface retention and patient coverage.

Quantity	Reference fit
Interfaces retained ($n_I \geq 5$), %	77.71
Crossing edges per retained interface, median (IQR)	12 (8, 20)
Supported pairs per patient, out of 45, median (IQR)	6 (3, 9)
Pairs unsupported in more than half of patients	44 of 45

5.2 Primary METABRIC Evaluation

5.2.1 Framework Performance and Comparator Results

NIMSurv achieved the highest concordance in the primary evaluation (Table 5.4). It improved risk ordering and reduced integrated probability error relative to Clin+Comp and random survival forests (RSF), the strongest shared-head and system-level references for concordance. The paired intervals favoured NIMSurv for both metrics (Figure 5.3), supporting added predictive value even when the reference already included tissue measurements.

This advantage did not extend to every learned representation: Clin+Comp had higher concordance than either Clin+Flat or Clin+Node. The favourable result therefore concerned the complete framework, rather than graph learning or regional aggregation alone.

Table 5.4. Ten-year net-DSS prediction in METABRIC (n = 544).

Model	Uno's C ↑ 10 years	IBS ↓ 0–10 years
NIMSurv	**0.703**	**0.113**
Shared-head clinical and tissue references		
Clin	0.613	0.190
Clin+Comp	0.668	0.121
Clin+Sum	0.625	0.125
Clin+Flat	0.612	0.128
Clin+Node	0.606	0.126
System-level references		
CPH	0.574	0.211
CPH-strat	0.660	0.126
RSF	0.675	0.127
DeepSurv	0.613	0.130
DeepHit	0.593	0.129
BroadSurv	0.587	0.142
DAGSurv	0.598	0.128
AttenSurv-MH	0.576	0.189
DMGN-M	0.652	0.123
Refitted interface variants		
Support-only	0.652	0.135
Pooled-centred	0.670	0.132
Uncentred	0.692	**0.113**
Input-feature	0.642	**0.113**

5.2.2 Pairing, Centring and Endpoint-Feature Comparisons

The comparison with Clin+Node evaluated the full interface pathway, including readout availability. Against Support-only, retaining paired-cell features improved both concordance and IBS, with favourable paired intervals (Figure 5.3). Pairing therefore added predictive value beyond availability.

The evidence did not establish an advantage of local centring over pooled centring or uncentred products: the paired intervals included zero for both metrics. Context-encoded endpoints improved concordance over directly projected inputs, but the IBS difference remained unresolved.

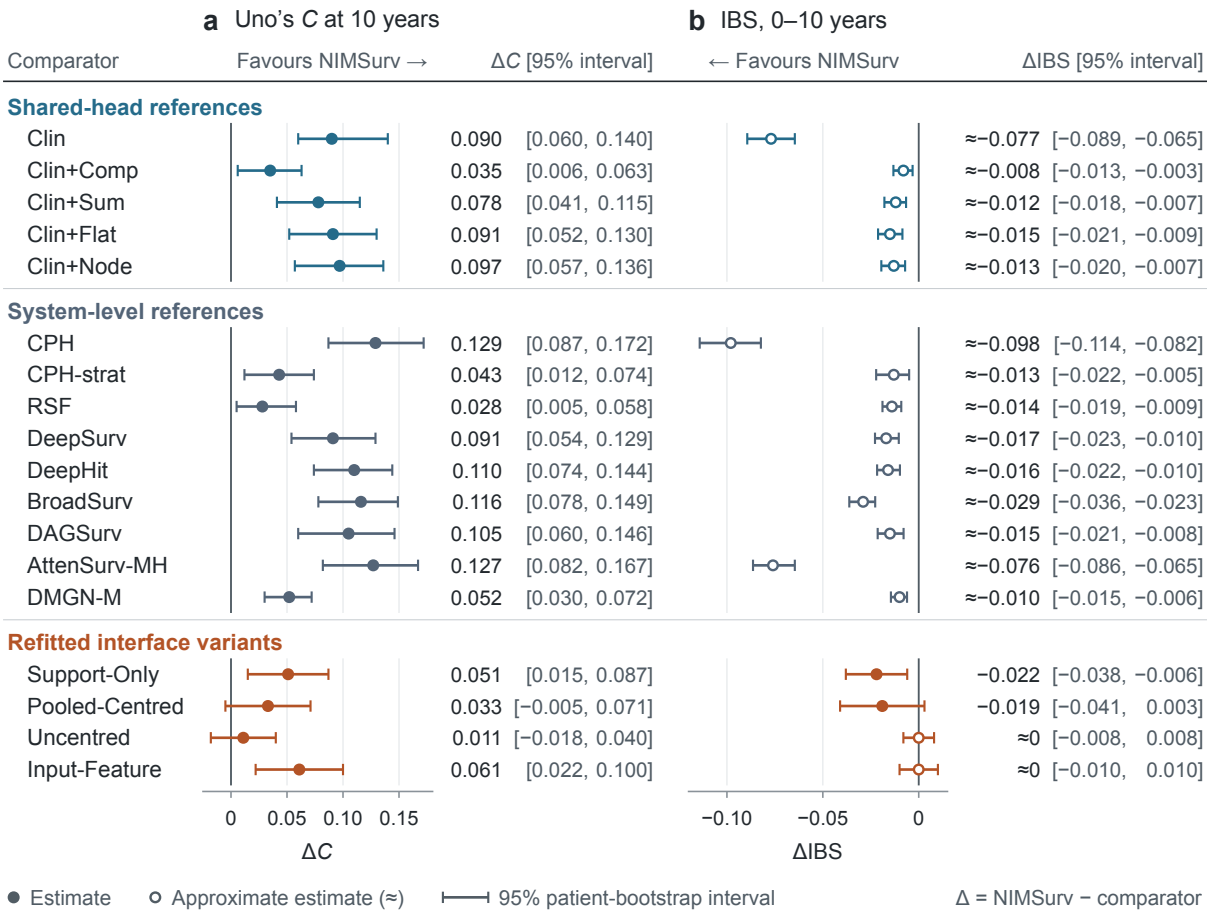


Figure 5.3. Paired net-DSS performance differences in METABRIC: (a) ten-year Uno's *C*; (b) 0–10-year IBS. Values: Appendix Tables B.3 and B.4.

5.2.3 Supplementary Performance Metrics

Supplementary metrics showed the same pattern: NIMSurv had the highest Harrell's concordance and ten-year AUC and the lowest ten-year Brier score (Appendix Table B.5).

5.3 Secondary Internal Analyses

5.3.1 Secondary METABRIC Survival Targets

For five-year DSS and ten-year OS, NIMSurv again had the highest concordance point estimates, with RSF the closest reported reference (Appendix Table B.6). These comparisons extend the evaluated outcomes within METABRIC.

5.3.2 Basel-Native Predictive Performance

When trained within Basel, NIMSurv had the highest five-year OS concordance, with paired intervals favouring it over all reported comparators, including Clin+Sum, the strongest reference for risk ordering (Table 5.5). Its IBS was also the lowest point estimate. This analysis evaluates learning from Basel's native measurements, separately from frozen transfer in Section 5.5.

Table 5.5. Basel-native five-year OS prediction (n = 276).

Model	Uno's C ↑ 5 years	IBS ↓ 0–5 years	Paired ΔC ↑ [95% interval]
NIMSurv	**0.757** [0.698, 0.808]	**0.0785** [0.0621, 0.0955]	*Reference*
Shared-head clinical and tissue references			
Clin	0.660 [0.599, 0.713]	0.0850 [0.0621, 0.0958]	+0.097† [+0.030, +0.160]
Clin+Sum	0.673 [0.620, 0.725]	0.0923 [0.0740, 0.1118]	+0.084† [+0.021, +0.144]
Clin+Node	0.588 [0.526, 0.643]	0.0995 [0.0792, 0.1207]	+0.169† [+0.099, +0.237]
System-level references			
CPH	0.639 [0.582, 0.698]	0.1818 [0.1575, 0.2060]	+0.117† [+0.049, +0.186]
CPH-strat	0.620 [0.566, 0.674]	0.1000 [0.0801, 0.1211]	+0.137† [+0.071, +0.203]
RSF	0.618 [0.552, 0.683]	0.0869 [0.0691, 0.1071]	+0.139† [+0.073, +0.201]
DeepSurv	0.607 [0.558, 0.652]	0.0989 [0.0798, 0.1202]	+0.150† [+0.082, +0.216]
DeepHit	0.562 [0.520, 0.603]	0.0961 [0.0787, 0.1151]	+0.195† [+0.131, +0.256]
BroadSurv	0.562 [0.521, 0.601]	0.1159 [0.0965, 0.1379]	+0.195† [+0.129, +0.259]
DAGSurv	0.522 [0.489, 0.555]	0.0910 [0.0711, 0.1134]	+0.235† [+0.164, +0.297]
AttenSurv-MH	0.548 [0.511, 0.585]	0.1780 [0.1732, 0.1832]	+0.209† [+0.142, +0.275]
DMGN-M	0.659 [0.614, 0.704]	0.0913 [0.0721, 0.1114]	+0.097† [+0.060, +0.133]

† marks favourable intervals excluding zero.

5.4 Matched Genomic Augmentation

On the matched patient subset, genomic augmentation modestly improved concordance in both clinical and tissue-inclusive models (Table 5.6). The paired intervals supported better risk ordering, but included zero for IBS. Calibration-slope estimates were close to one.

Table 5.6. Matched genomic augmentation for ten-year net DSS (n = 527).

A. Clinical models

Model	Uno's C ↑ 10 years	IBS ↓ 0–10 years	Calibration slope	Harrell's C ↑
Clin (matched)	0.605	0.212	1.028	0.610
Clin+Gen (matched)	**0.624**	**0.211**	1.002	**0.624**

B. Tissue-inclusive models

Model	Uno's C ↑ 10 years	IBS ↓ 0–10 years	Calibration slope	Harrell's C ↑
NIMSurv (staged)	0.705	0.113	1.017	0.710
NIMSurv+Gen (staged)	**0.723**	**0.111**	0.996	**0.734**

C. Paired augmentation differences

Contrast	Paired ΔC ↑ [95% interval]	Paired ΔIBS ↓ [95% interval]
Clin+Gen − Clin	+0.019† [+0.004, +0.035]	−0.001 [−0.006, +0.004]
NIMSurv+Gen − NIMSurv	+0.018† [+0.003, +0.033]	−0.002 [−0.007, +0.003]

† marks favourable intervals excluding zero.

5.5 Frozen Common-Input Transfer to Basel

Without refitting or recalibration, NIMSurv had favourable paired concordance intervals against every reported comparator in Basel (Table 5.7). Several IBS contrasts, including that with Clin+Sum, remained unresolved.

This transfer experiment concerns a separate common-input five-year OS predictor. The corresponding source-cohort results appear in Appendix Table B.7.

Table 5.7. Frozen common-input five-year OS prediction in Basel ($n = 276$).

Model	Uno's C ↑ 5 years	IBS ↓ 0–5 years	Paired ΔC ↑ [95% interval]	Paired ΔIBS ↓ [95% interval]
NIMSurv	**0.725**	**0.081**	*Reference*	*Reference*
Shared-head clinical and tissue references				
Clin	0.625	0.091	+0.100† [+**0.018**, +**0.184**]	−0.010† [−0.019, −0.002]
Clin+Sum	0.665	0.084	+0.061† [+0.003, +0.122]	−0.002 [−0.010, +0.005]
Clin+Node	0.633	0.086	+0.092† [+0.008, +0.174]	−0.004 [−0.013, +0.004]
System-level references				
CPH	0.654	0.099	+0.072† [+0.002, +0.147]	−0.017† [−0.030, −0.004]
CPH-strat	0.674	0.085	+0.052† [+0.002, +0.110]	−0.004 [−0.011, +0.003]
RSF	0.641	0.086	+0.084† [+0.020, +0.150]	−0.004 [−0.012, +0.003]
DeepSurv	0.677	0.084	+0.049† [+0.002, +0.123]	−0.002 [−0.011, +0.007]
DeepHit	0.580	0.090	+0.145† [+0.059, +0.235]	−0.009 [−0.020, +0.001]
BroadSurv	0.562	0.103	+0.163† [+0.087, +0.240]	−0.022† [−0.036, −0.010]
DAGSurv	0.526	0.090	+0.199† [+0.113, +0.291]	−0.009† [−0.018, 0.000]
AttenSurv-MH	0.500	0.159	+0.226† [+0.134, +0.320]	−0.078† [−0.090, −0.065]
DMGN-M	0.654	0.088	+0.071† [+0.023, +0.123]	−0.007† [−0.012, 0.000]

† marks favourable intervals excluding zero before rounding.

Grouped calibration remained uneven (Figure 5.4). Q1–Q3 lay near the diagonal, while predicted survival exceeded the Kaplan–Meier estimate in Q4 and fell below it in Q5, where uncertainty was wide.

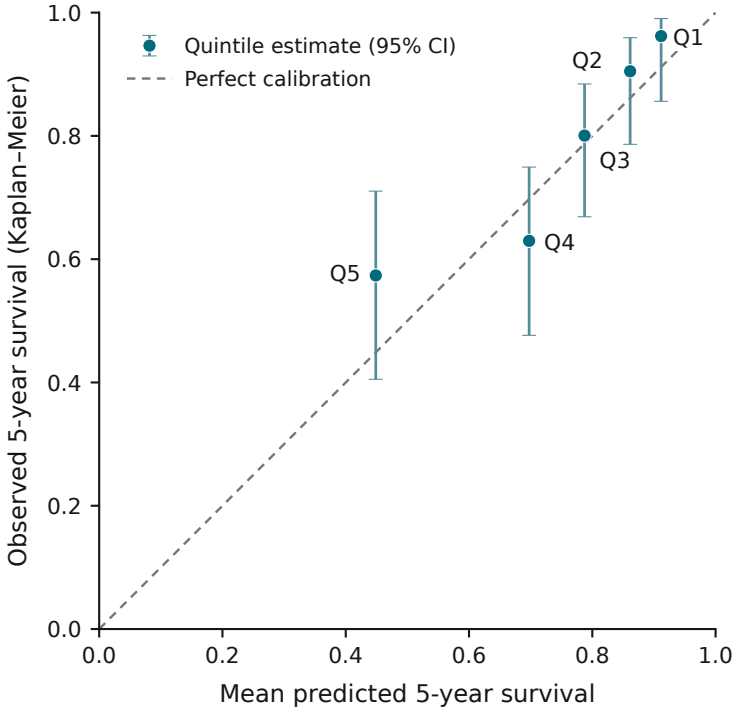


Figure 5.4. Frozen NIMSurv five-year OS calibration in Basel. Quintiles compare mean predicted with Kaplan–Meier survival (95% confidence intervals); the diagonal indicates perfect calibration.

5.6 Tissue Tracing and Reconstruction Fidelity

The tissue analyses distinguished locating a fitted contribution from explaining its value: selected contributions could be mapped to boundaries, but measured-marker models reconstructed them poorly.

5.6.1 Measured Niche-Type Profiles

The reference niche types included fibroblast-rich, myofibroblast-containing, mixed immune, epithelial and marker-defined profiles. Their working labels identify the numbered pairs below (Table 5.8); they describe mixtures rather than pure cell populations. Detailed compositions and marker summaries appear in Appendix Table B.2.

Table 5.8. Reference niche-type labels.

Type	Working label	Type	Working label
1	Fibroblast-rich	6	CK^{+} $CXCL12^{+}$
2	ER^{hi} $CXCL12^{+}$	7	Mixed immune
3	CK^{med} ER^{lo}	8	$CD15^{+}$
4	Myofibroblasts	9	Basal
5	$HER2^{+}$	10	$CK8–18^{hi}$ $CXCL12^{hi}$

5.6.2 Pairing Contributions and Patient Support

High variation in pairing contributions did not necessarily imply broad patient support (Table 5.9). Pair 3–6 had the largest reported standard deviation, whereas the highly ranked $HER2^{+}$ / basal pair (5–9) was rarely observed. These summaries include unavailable-pair zeros.

Several leading pairs had negative reference-centred means in the highest predicted-risk tertile (Figure 5.5). Their contributions were below the training reference despite the patients' high combined predictions.

Table 5.9. ER-positive pairing variation and patient support.

Pair	Pair SD (10^{-3})	Supported patients	Retained interfaces	Upper-risk centred mean (10^{-3})	Availability-term SD (10^{-3})
3–6	3.54	78/434	451	−0.404	0.85
2–9	3.33	26/434	57	−0.132	2.39
2–10	2.83	82/434	370	−0.244	0.86
2–4	2.22	109/434	343	−0.166	1.44
9–10	2.12	33/434	70	−0.062	3.91
5–9	2.05	3/434	5	0.463	1.02
6–10	1.97	70/434	436	0.013	0.17
2–6	1.96	54/434	209	0.065	3.12
2–8	1.81	10/434	55	−0.028	0.91
4–6	1.57	86/434	334	−0.123	3.44

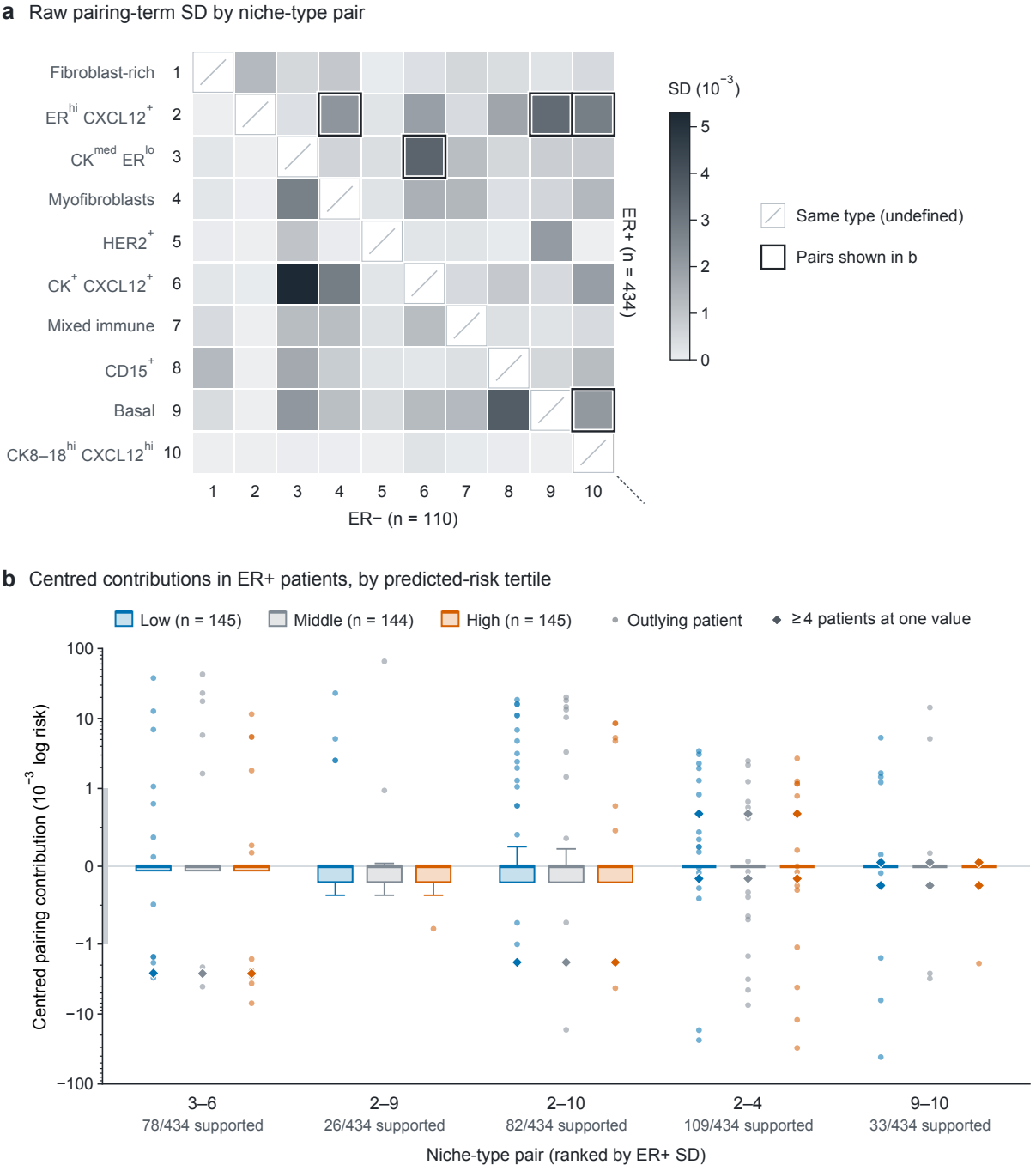


Figure 5.5. Held-out pairing contributions: (a) raw SD (ER-positive upper triangle; ER-negative lower); (b) top-five ER-positive pairs by predicted-risk tertile, centred on each fit's ER-specific training mean. Symmetric-log axis; grey bar: linear range ($\pm 10^{-3}$ log risk).

5.6.3 Boundary-Level Contribution Maps

Figures 5.6–5.7 locate selected contributions at boundaries in a lower- and a higher-predicted-risk patient. Both interfaces had positive summed raw contributions, showing that a positive local term can coexist with either overall risk level. The examples were selected without consulting outcomes; their phenotype and marker profiles describe the cells incident to the illustrated boundaries.

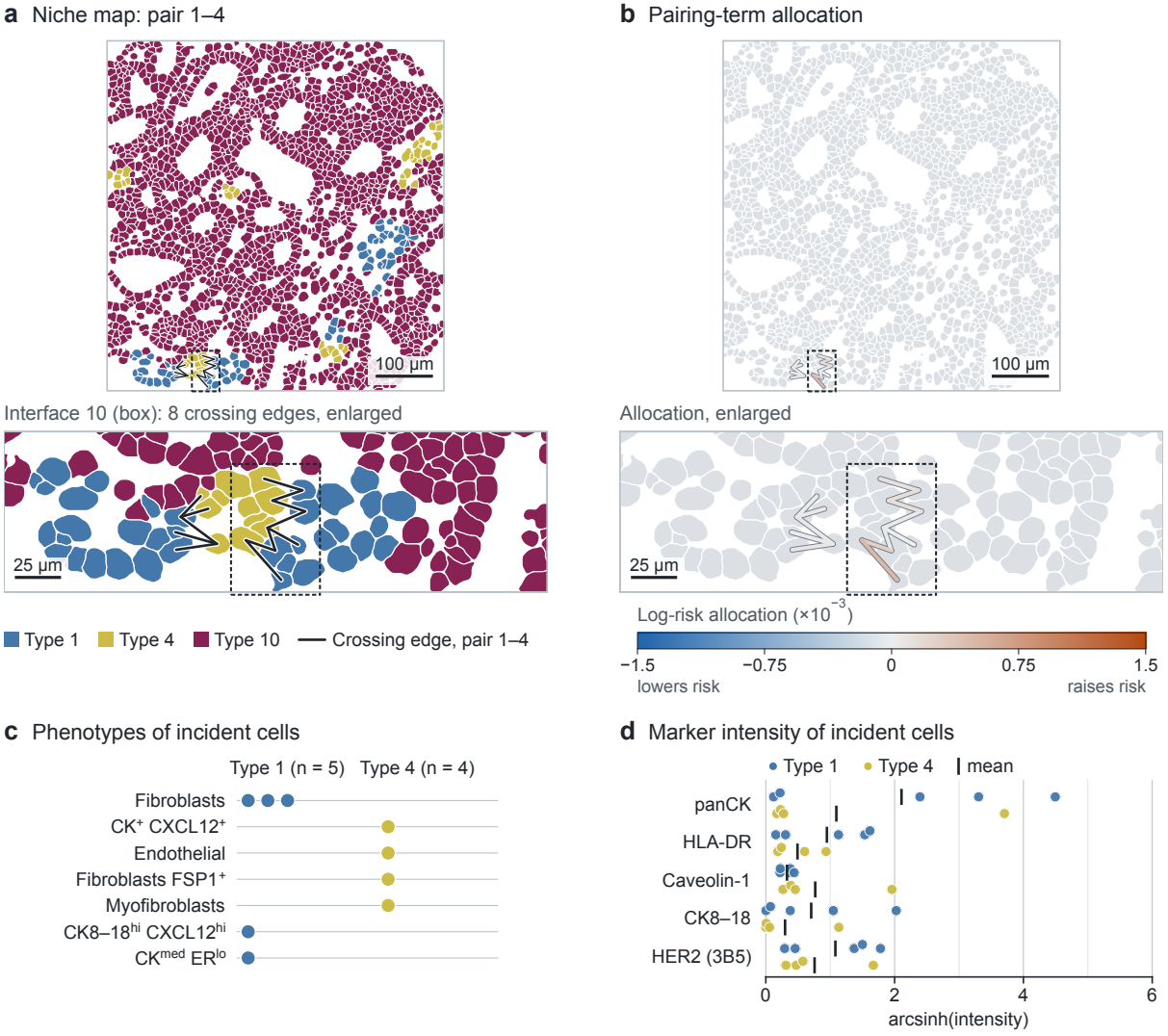


Figure 5.6. Tissue tracing for MB-0232 at a type 1–4 boundary: (a) niche map with the crossing edges of retained type 1–4 interfaces (black); (b) signed edge log-risk allocations; (c) phenotypes and (d) arcsinh marker intensities of the unique incident cells, one dot per cell (bars: means). Insets enlarge the boxed boundary.

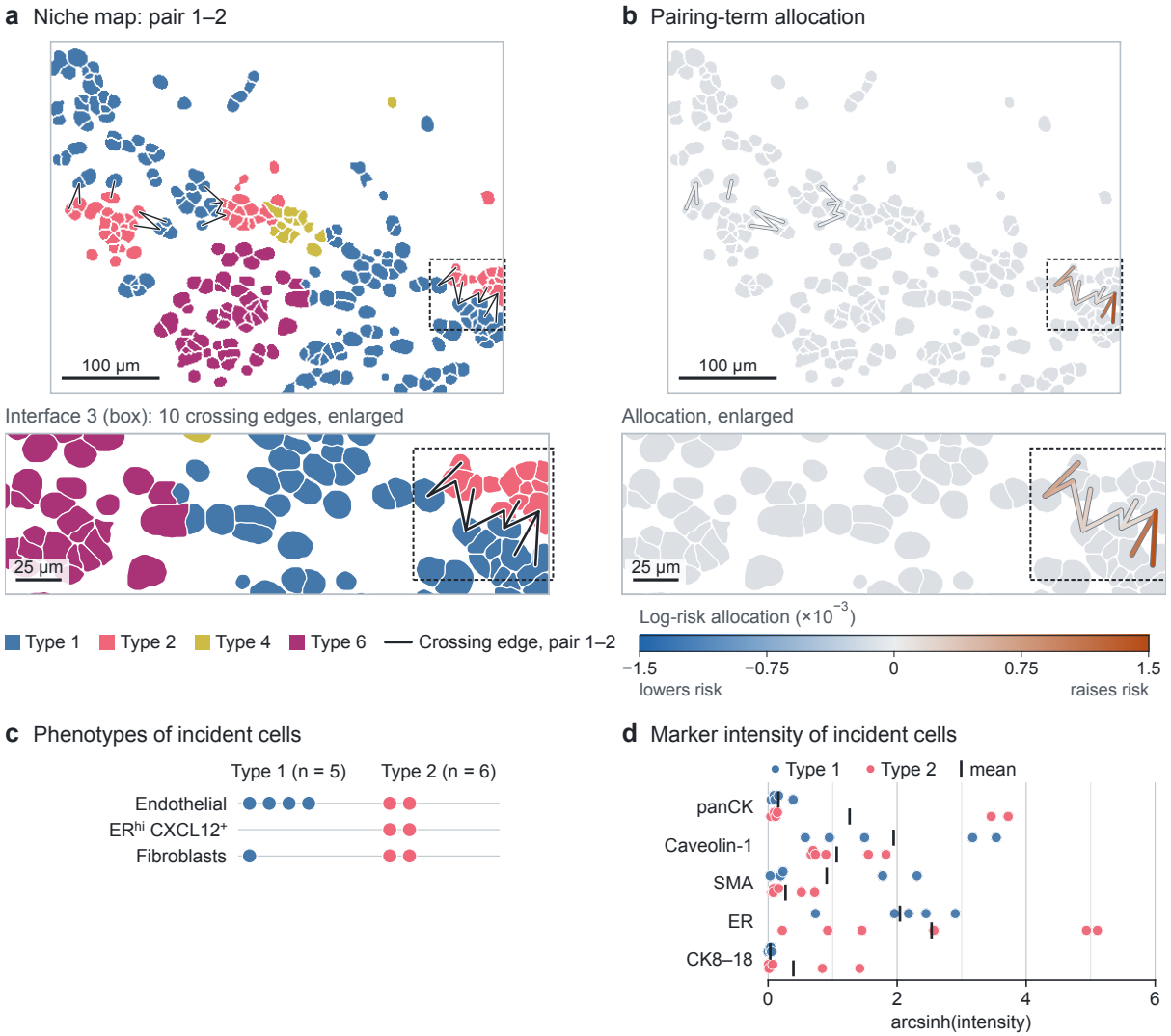


Figure 5.7. Tissue tracing for MB-0000 at a type 1–2 boundary; panels and scales as in Figure 5.6.

5.6.4 Marker-Based Score Reconstruction Fidelity

Measured-marker products did not reliably reconstruct the selected patient-level pairing scores (Table 5.10). Median patient-level R^2 was negative for every selected pair, although some individual evaluations were positive. Edge-level reconstruction was also limited. Evaluation coverage varied substantially between pairs, with especially limited support for pair 2–9.

Table 5.10. Held-out marker-based reconstruction of pairing scores.

Pair	Eligible evaluations	Test patients	Edge R^2 Median (IQR)	Patient R^2 Median (IQR)
3–6	3	6–31	0.001 (0.001, 0.064)	−0.045 (−0.154, −0.031)
2–9	1	6	−0.054	−0.200
2–10	4	17–21	0.033 (−0.001, 0.075)	−0.003 (−0.048, 0.042)
2–4	4	21–24	0.098 (0.060, 0.132)	−0.009 (−0.185, 0.147)
9–10	4	5–10	−0.096 (−0.163, −0.028)	−0.656 (−0.946, −0.358)

Test-patient counts are ranges; IQR requires multiple evaluations.

5.6.5 Case Study: Contrasting Risk Estimates

The two ER-positive cases illustrate how the fitted models can assign contrasting patient-level risks (Figure 5.8). MB-2933 and MB-0288 were held out together. CPH, the simpler clinical-plus-summary reference, assigned similar ten-year net-DSS risks of 3.34% and 3.06%, respectively. NIMSurv assigned risks of 8.22% and 51.30%, showing substantially greater separation between the patients.

The NIMSurv ordering was consistent with their observed outcomes. MB-2933 had no breast-cancer death within ten years and later died from another cause at 13.49 years; MB-0288 experienced breast-cancer death at 5.32 years. The IMC images, niche maps and signed edge allocations provide tissue-level context for these predictions, allowing selected pairing contributions to be inspected within each sampled core.

These cases were selected retrospectively with knowledge of outcomes. They illustrate contrasting fitted predictions and their connection to tissue organisation; they do not provide independent evidence of predictive performance on new patients.

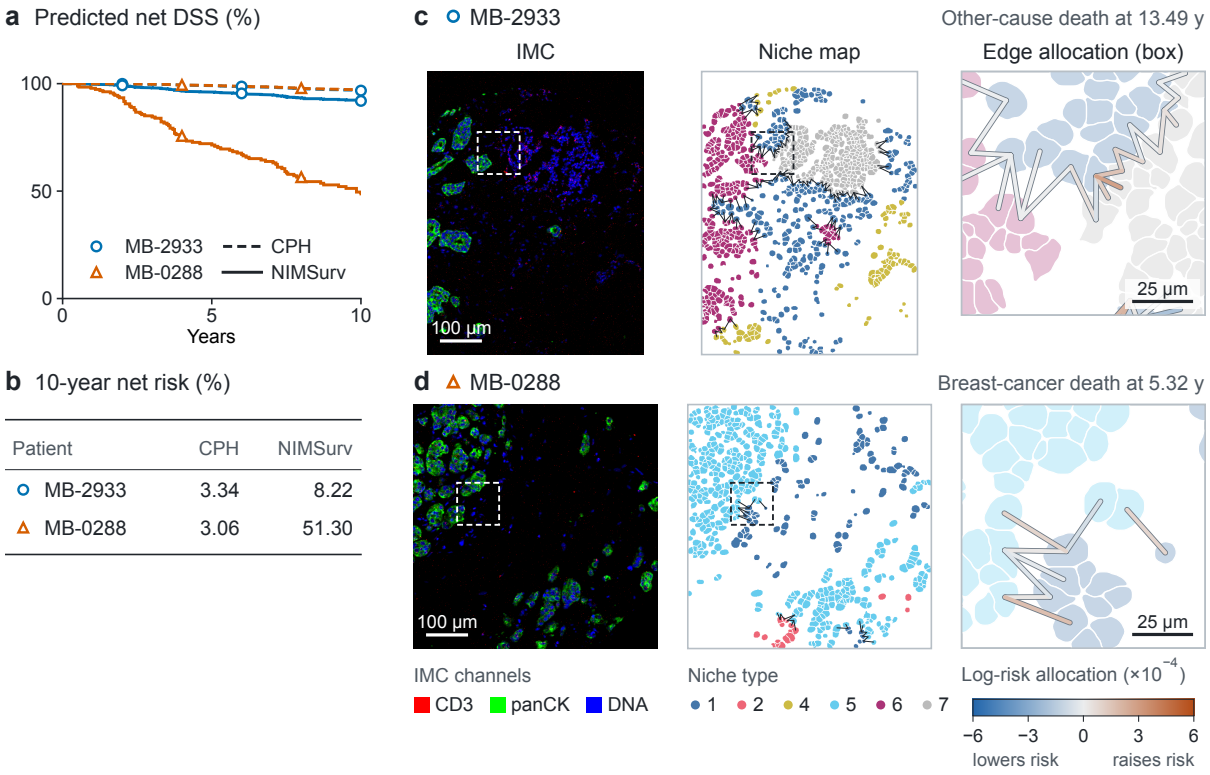


Figure 5.8. MB-2933 (blue circles) and MB-0288 (orange triangles): (a) predicted net-DSS curves (CPH dashed; NIMSurv solid); (b) ten-year net risk; (c–d) IMC image, niche map with retained crossing edges (black) and the boxed region enlarged, cells tinted by niche type and edges coloured by signed log-risk allocation (one scale for both patients).

5.7 Chapter Summary

The complete framework improved primary prediction (RQ1), and paired-cell features added value beyond interface availability (RQ2). Genomics modestly improved risk ordering (RQ3). Basel-native learning and frozen common-input OS transfer extended the evaluation, although transfer calibration was uneven (RQ4). Tissue tracing located selected contributions, but measured-marker products reconstructed them poorly (RQ5). Chapter 6 considers the implications.

CHAPTER 6

Discussion, Limitations and Future Work

6.1 Principal Findings

The principal finding is that the complete NIMSURV procedure improved development-stage prediction of ten-year model-based net disease-specific survival (DSS) in METABRIC beyond the evaluated clinical and tissue-summary references. The advantage over CLIN+COMP extended to both risk ordering and integrated probability error (RQ1). Component comparisons also supported retaining paired boundary-cell states (RQ2).

The additional analyses clarify the scope of these findings. Genomic augmentation modestly improved risk ordering on matched patients (RQ3). Evaluation within Basel also supported the framework's predictive value when trained on a second cohort's measurements. Frozen transfer of a separate common-input five-year overall-survival (OS) predictor retained useful risk ordering, but showed uneven calibration (RQ4). Finally, selected pairing contributions could be traced exactly to tissue boundaries, whereas measured-marker surrogates did not reliably reconstruct their values (RQ5).

Together, these findings distinguish three questions that are often combined: whether a complete learning procedure improves prediction, which information a representation retains, and how faithfully its fitted contributions can be explained. NIMSURV provides an integrated setting for examining these questions, while NIRP makes boundary-cell pairing an explicit and traceable part of the patient representation. Predictive gains, mathematical characterisation and explanatory fidelity therefore form complementary, rather than interchangeable, evidence.

6.2 Added Value Beyond Clinical and Tissue Summaries

The comparison with CLIN+COMP provides the most direct answer to RQ1 because this reference already includes clinical predictors, cell composition and mean protein measurements. Relative to it, NIMSURV increased concordance by 0.035 and reduced integrated Brier score (IBS) by approximately 0.008, with favourable paired intervals for both measures (Figure 5.3). Thus, the complete procedure provided predictive value beyond composition and mean-marker summaries. The agreement between the two

metrics is important: better risk ordering was accompanied by lower average error in the predicted survival curves. This strengthens the predictive finding. Because the models differ in several tissue features and in their fitting, the gain belongs to the complete procedure rather than to boundary pairing alone.

The pattern across the simpler representations helps explain why that distinction matters. Clin+Sum, Clin+Flat and Clin+Node all had lower concordance point estimates than Clin+Comp. Performance therefore did not improve steadily as more summaries, learned cell features or regional structure were added. One possible explanation is that pairing helps both by retaining useful relationships in the patient readout and by directing the shared encoder towards features useful for survival prediction. Differences in regularisation and optimisation provide alternative explanations. The results suggest that the successful combination of representation and learning procedure matters more than architectural complexity alone.

The system-level comparisons provide a complementary perspective. Random survival forests, the strongest reported system-level reference for risk ordering, also had lower concordance and higher IBS than NIMSurv. These comparisons matched patients, outcomes and partitions, but not prediction functions, optimisation or tuning budgets (Table A.6). The shared-head comparisons therefore provide the more direct evidence for added value within a common survival-model family.

DMGN and SpaCEy provide the closest breast IMC prediction precedents. DMGN integrates image regions with patient variables and includes comparisons using patient variables and cell-type proportions, whereas SpaCEy's breast survival experiments evaluate cell-graph features against tissue-only references (Fu et al., 2023; Rifaioglu et al., 2026). Against this background, the present contribution is an integrated clinical–tissue framework in which interface-level pairing is represented explicitly and evaluated alongside informative tissue-summary and regional references. It extends the question from whether tissue learning is useful to how boundary relationships can be retained and examined within patient-level prediction.

6.3 Predictive Role of Boundary-Cell Pairing

6.3.1 Pairing in Refitted Prediction Models

The Support-Only comparison provides more focused evidence for the interface pathway. Retaining paired-state features increased concordance by 0.051 and reduced IBS by 0.022, with favourable paired intervals (Figure 5.3). Because the comparator retains indicators of whether a boundary readout is available, those indicators alone do not account for the observed advantage. The result supports using paired states, rather than availability alone.

NIRP's contribution is to make a particular relationship explicit at readout. The cell encoder already passes information across regional boundaries, so contextual embeddings can contain information from both sides. Connected-region analysis provides a basis for defining these boundaries, while second-order graph pooling and COVET provide precedents for aggregating within-node or within-cell feature products (Bhate et al., 2022; Wang and Ji, 2023; Haviv et al., 2025). NIRP instead forms products between the states of distinct crossing-edge endpoints, centres them within each connected interface, and aggregates them by niche-type pair and patient. Its specificity lies in the observational unit, grouping and reference used for those products.

The empirical comparison nevertheless concerns learning procedures. Removing the pairing block changes the training signals reaching the shared encoder, as well as coefficient estimation and model selection. Its benefit may therefore arise from the readout, the features learned under that readout, or their combination. Similarly, the concordance gain over Input-Feature supports learned endpoint descriptions, but the comparison changes both neighbourhood aggregation and nonlinear feature transformation at the interface endpoints. It does not isolate message passing, and its IBS contrast remains unresolved. These results support pairing within the complete learning process while motivating the fixed-feature comparisons proposed in Section 6.8.

6.3.2 Centring and Retained Information

The centring analysis clarifies what the pairing representation measures. For fixed endpoint features and interfaces, Proposition 4.1 separates within-interface pairing from differences between interface means. Local centring retains the former and removes the latter from the pairing statistic. Pooled centring retains both, while uncentred products additionally retain the pooled mean-state product. Local centring is therefore a choice to describe paired deviations relative to each boundary's own endpoint means, rather than simply a preprocessing step intended to improve prediction.

That choice has a clear interpretation but an empirical trade-off. Removing mean-state contrasts focuses the statistic on within-boundary variation, yet those contrasts may themselves contain prognostic information. The comparisons did not establish local centring as better than pooled centring or uncentred products. Intervals including zero leave the predictive choice unresolved; they do not establish equivalence. The mathematical characterisation remains useful because it identifies precisely which information differs between the alternatives.

Retaining more components also need not improve the fitted predictor. The pooled and uncentred variants combine the components before applying a coefficient matrix, rather than assigning each component independent coefficients. They therefore do not test whether within-interface, between-interface and mean-state information would be more useful as separate predictors. The next methodological question is consequently which components should be represented and estimated separately, not whether one centring reference is intrinsically preferable.

6.4 Genomic Augmentation and Cross-Cohort Evaluation

6.4.1 Added Predictive Value of Genomic Information

Genomic augmentation improved risk ordering on matched patients both with and without learned tissue. In the tissue-inclusive comparison, concordance increased by 0.018, while the IBS difference remained unresolved (Table 5.6). Matching the patients rules out a change in the evaluation population as the explanation for this gain. The result suggests that the evaluated genomic measurements provide useful predictive information alongside the tissue representation. Although breast IMC studies link molecular profiles with cellular organisation (Ali et al., 2020; Danenberg et al., 2022), such correspondence does not make the measurements interchangeable for prediction.

This inference applies to the matched staged procedures. Each tissue model adapts to its own fixed clinical or clinical–genomic score, so the contrast includes that adaptation rather than adding genomics to an unchanged tissue model. RQ3 is therefore supported for additional risk ordering within the matched design.

6.4.2 Basel-Native Learning and Frozen Transfer

The two Basel analyses address different forms of generalisation. Basel-native learning tests whether the framework can learn from another cohort's measurements. Its concordance gain of 0.084 over Clin+Sum, the strongest reported reference for ordering, supports that use (Table 5.5). Frozen transfer asks the more specific question of whether a fitted source-cohort predictor remains useful without adapting to the target cohort.

For the separate common-input five-year OS model, frozen transfer to Basel achieved concordance of 0.725 and a paired gain of 0.061 over the transferred Clin+Sum reference, without refitting or recalibration (Table 5.7). This indicates that useful risk ordering survived the cohort change under the restricted input

scheme. The source-matched evaluation in Table B.7 provides context, but subtracting source and target performance would combine population differences with differences in fitting stage, rather than estimate a paired transfer effect.

Probability prediction was less convincing. IBS gains against the summary and regional references remained unresolved, and grouped calibration was uneven (Figure 5.4). Thus, input matching and successful risk ordering were not sufficient to establish reliable survival probabilities across the target population. This distinction makes explicit calibration assessment central to further evaluation of transportability (Riley et al., 2024; Collins et al., 2024). The transfer result concerns the common-input OS predictor; the primary ten-year net-DSS model still requires independent evaluation with its own outcome, horizon and measurement specification.

6.5 Score Traceability and Explanatory Fidelity

6.5.1 Interpretive Scope of Tissue Maps

The additive representation gives model inspection a defined target: a selected patient-level pairing term can be allocated exactly to the boundary connections used to calculate it. The incident cells can then be described through their measured phenotypes and proteins. This links a fitted numerical contribution to an identifiable tissue location, without requiring that location to explain the whole prediction. Indeed, a positive boundary term can occur in patients with different overall risks because other tissue terms, clinical information and the fitted baseline survival also contribute to the prediction.

This traceability describes the fitted calculation, rather than a biological mechanism or an intervention effect. Local allocations use contextual cell states and interface means; changing a cell or deleting an edge may alter those quantities as well as its displayed contribution. Likewise, ranking pairs by contribution variation combines supported values, unavailable-readout zeros and outputs from different fits. It identifies calculations worth inspecting, but does not establish biological importance or stable biomarkers. The maps are therefore a basis for examining where selected scores arise.

6.5.2 Limits of Marker-Based Score Reconstruction

The surrogate experiment asks whether measured-marker products can reproduce selected learned pairing scores on held-out patients. Median patient-level R^2 was negative for every selected pair (Table 5.10), providing no consistent evidence of faithful reconstruction by the tested surrogates. The extent of the

shortfall varied: the median was close to zero for pair 2–10 (−0.003), but substantially lower for pair 9–10 (−0.656), and some individual evaluations were positive. The finding is therefore a lack of reliable reconstruction across the selected pairs, rather than uniform failure of the same magnitude.

Several features of the experiment may contribute. The encoder receives both marker measurements and phenotype labels, and its learned states incorporate nonlinear neighbourhood information. The surrogate instead receives centred products of endpoint markers, without directly receiving phenotype labels or an explicit neighbourhood representation. It differs from the learned model in both its functional form and the inputs made directly available to it. Patient support is also limited and unequal: pair 2–9 had only one eligible evaluation, involving six test patients. Omitted contextual descriptions, limited surrogate flexibility and estimation error therefore remain competing explanations.

The present experiment cannot distinguish these explanations, but it does establish a useful boundary on interpretation. Recognisable phenotypes at a highlighted boundary do not show that measured-marker products explain its fitted score. Exact localisation and faithful reconstruction must be assessed separately. The current results support the former, while leaving richer measured-tissue explanations and their stability after refitting open; this limits the explanatory claim rather than the separate patient-level prediction comparisons.

6.6 Limitations and Scope of Inference

6.6.1 Development-Stage Evidence and Conditional Uncertainty

Patient-level partitioning, training-only preprocessing and matched comparisons are strengths of the final evaluation design (Section 4.5.2). Keeping each patient's cores and modalities together separates the patients used for fitting and configuration selection from those used for assessment within that procedure. However, architecture and tissue construction had already been developed through preliminary METABRIC experiments before the final model definitions, thresholds and search spaces were fixed. The final cross-validation therefore evaluates a specified procedure after methodological development on the same cohort, rather than independently evaluating the entire development process. Cohort-guided design choices may still contribute to optimistic performance estimates, even when fitting and selection within the final procedure respect the patient partitions. Repeated cross-validation does not remove this possibility or provide independent replications in new populations.

The information available for fitting is also limited by patient and event counts. The primary cohort contains 544 patients and 174 breast-cancer deaths over the available follow-up, including 46 among ER-negative patients. Each outer-training partition consequently contains only 36–37 ER-negative events. The evaluated full-panel configurations contain 11,465–33,737 trainable parameters. Parameter sharing, projection and penalties constrain estimation, so the raw parameter count alone does not establish overfitting. Nevertheless, the limited event information makes regularisation and stability assessment particularly important for ER-dependent tissue effects and sparsely observed pairs. Additional cells and edges enrich the tissue description without increasing the number of independent patient outcomes.

The bootstrap intervals address a narrower source of uncertainty. Resampling patients preserves paired predictions, but preprocessing, configuration selection and training are not repeated (Section 4.5.4). The intervals are conditional on the saved fits and do not capture the variability of fitting the procedure again or quantify optimism from earlier cohort-guided design choices. They are also pointwise, rather than simultaneous intervals for the full family of comparisons.

6.6.2 Tissue Coverage and Representation Stability

The representation describes sampled cores rather than the whole tumour. Boundary availability depends on tissue sampling, cell segmentation and annotation, and the construction of regions and retained interfaces. Patients supported a median of only six of the 45 possible niche-type pairs, and 44 pairs were unavailable in more than half of patients (Table B.1). Availability indicators distinguish an unobserved readout from an observed zero, but cannot determine whether a boundary is biologically absent or simply unsampled.

This sparsity connects the measurement limitations to the event-count limitations above. Rare pairs provide few supported patient records for estimating their coefficients or evaluating surrogates, even when many crossing edges are available in those records. Niche abundance also varied across fitted partitions. Profile-based alignment provides a correspondence between fitted categories, but numerical labels should not be assumed to identify exactly the same biological entity after refitting. The tracing examples establish inspectability within a fitted model; reproducibility across tissue samples, region definitions and new fits remains to be assessed.

6.6.3 Survival Targets and Probability Interpretation

The primary prediction is model-based net DSS. Treating other-cause deaths as censoring does not make its complement the cumulative incidence of breast-cancer death in the presence of competing mortality

(Section 2.3.2). The ER-stratified Cox-type model also assumes proportional hazards within strata, while the reported censoring adjustment assumes independent censoring within ER strata. Flexible tissue features do not remove these modelling and assessment assumptions. Formal proportional-hazards tests and alternative censoring-adjustment analyses were not performed.

Accordingly, lower IBS supports lower probability error under the stated evaluation target and assumptions, not accurate probabilities for a different clinical question. Neither the genomic calibration-slope point estimates nor the grouped Basel plot provides a complete assessment of absolute-risk calibration. Independent evaluation needs to match the intended outcome and horizon, and assess calibration with uncertainty alongside discrimination (Riley et al., 2024).

6.7 Potential Clinical Application and Translational Requirements

The potential application of NIMSurv is to supplement breast cancer prognostic assessment with spatial tissue information. At this stage, the framework is a candidate for prognostic research; a specific decision pathway or threshold for clinical action has not been established. In a proposed workflow, clinical variables and IMC measurements from sampled tumour cores would enter a fixed pipeline, producing a patient-level survival estimate for review alongside established prognostic assessments.

NIRP could complement this assessment by locating selected fitted pairing contributions for pathologist-led examination and hypothesis generation. Such inspection is useful even before a biological explanation is established. A prognostic association does not identify benefit from a particular treatment (Ballman, 2015).

Translation would require reproducible tissue sampling, IMC measurements and cell annotations, together with a practicable assay workflow. The common-input OS experiment motivates reduced-panel studies. Independent predictive evaluation would first need to establish added value over relevant prognostic assessments and adequate calibration for the intended target. Only after specifying the decision to be supported could benefit at clinically relevant thresholds be evaluated (Riley et al., 2024; Vickers and Elkin, 2006). These requirements motivate the priorities below.

6.8 Priorities for Future Research

The first priority is independent evaluation of a fixed pipeline in patients who played no part in its development, using compatible measurements and a prespecified survival target. This would test whether the primary gain persists beyond METABRIC, with the clinical-plus-composition reference and relevant

established predictors assessed on the same patients. Calibration curves and numerical summaries with uncertainty should accompany discrimination and probability error. Any subsequent recalibration should be reported as model updating, separately from the initial frozen evaluation. For a prespecified clinical decision, net benefit should also be assessed at relevant thresholds (Riley et al., 2024; Collins et al., 2024; Vickers and Elkin, 2006).

The second priority is a controlled test of the pairing representation. Within each training partition, the encoder, endpoint projection, regional definitions and other feature blocks could be fixed, with candidate readouts evaluated under a common survival-head specification and selection protocol. The origin of those fixed features matters: an encoder trained jointly with NIRP may already favour that readout. A useful design would therefore include measured-feature controls and features learned without the candidate interface readouts, with all data-dependent construction confined to training patients. Comparing within-interface, between-interface and mean-state components separately and jointly would then test their predictive contributions conditional on each specified feature source. Repeating preprocessing, selection and fitting would additionally assess variability omitted by bootstrap intervals conditional on saved predictions.

The third priority is to test whether tissue interpretation survives changes in model and measurement. Building on the existing profile-based alignment (Section 4.7.4), repeated fits should assess the stability of signed pairing contributions. Sensitivity to core sampling, cell annotation and interface construction would test whether the same findings recur. Surrogate studies could separately vary the inclusion of phenotype and neighbourhood descriptions and the flexibility of the reconstruction model, helping to distinguish insufficient inputs from limited representational capacity. All such comparisons would require held-out patients, adequate support for each pair and explicit reconstruction error.

Together, these priorities address the three levels of evidence developed in this thesis: predictive value beyond development, the contribution of a specified pairing representation, and faithful, reproducible explanation of its fitted scores. They provide a path from an explicit and traceable tissue representation towards a predictor whose practical prognostic value can be assessed independently.

CHAPTER 7

Conclusion

This thesis develops NIMSURV, a multimodal graph learning framework that brings the spatial organisation of tumour tissue into patient-level breast cancer survival prediction. By integrating contextual cell features, tissue regions and their interfaces with clinical information and predefined tissue summaries, the framework connects the cellular detail of sampled tissue with the broader task of patient prognosis.

A distinct methodological contribution is Niche-conditioned Interface Residual Pooling (NIRP), which explicitly represents learned cell states paired across tissue boundaries. Its mathematical characterisation distinguishes within-interface pairing from differences between interface means and clarifies the information retained by alternative centring schemes. An exact score decomposition further traces fitted pairing contributions to individual boundary connections, providing a precise link between the patient-level calculation and the tissue from which it arises.

In development-stage evaluation of ten-year model-based net disease-specific survival in METABRIC, the complete NIMSURV procedure improved concordance by 0.035 and reduced integrated Brier score by approximately 0.008 relative to a clinical reference incorporating cell composition and mean marker measurements. Improvements in both risk ordering and integrated probability error support the framework's added predictive value beyond these evaluated summaries. Component comparisons further supported the usefulness of paired-state features within refitted prediction models.

The additional analyses broadened this evidence across modalities and cohort settings. Matched genomic augmentation improved risk ordering, supporting the value of combining genomic information with clinical and tissue representations. Basel-native evaluation supported learning in a second cohort, while frozen transfer of a separate common-input five-year overall-survival predictor retained useful risk ordering when applied from METABRIC to Basel without refitting or recalibration.

Tissue maps gave the fitted pairing contributions a spatial context, locating selected terms at identifiable boundaries and describing the associated cells through measured phenotypes and proteins. Alongside patient-held-out surrogate evaluation, this work provides a concrete basis for examining computational traceability and explanatory fidelity as complementary aspects of tissue-based prediction.

These contributions motivate independent evaluation with compatible measurements, prespecified survival targets and explicit calibration assessment. Controlled readout comparisons with cell features held fixed, richer measured-tissue surrogates and stability assessment across refits offer further directions for developing the representation and its interpretation.

Together, NIMSurv and NIRP give boundary-cell relationships a defined place in survival prediction and a visible path back to tissue. By bringing prediction and computational traceability into the same framework, this thesis advances a spatially informed approach to prognostic modelling - one that connects the organisation of cells, the boundaries between tissue regions and the patient-level outcomes they are used to predict.

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Appendix A

Supplementary Methods and Mathematical Details

A.1 Input Encodings and Cross-Cohort Mapping

Table A.1 lists marker (P), phenotype (A), clinical (C) and summary (Q) dimensions. Cell inputs concatenate markers and one-hot phenotypes: 68/57/18 coordinates for METABRIC/Basel/common inputs. METABRIC has 16 epithelial and 16 TME phenotypes; Basel has 27 metaclusters. Genomics adds at most 26 coordinates.

Table A.1. Native and common-input dimensions.

Input scheme	***P***	***A***	***C***	***Q***
METABRIC native	36	32	13	199
Matched genomic subset	36	32	13	199
Basel native	30	27	8	188
Common-input METABRIC / Basel	10	8	8	149

Common markers, in order: SMA, CK5, CK8–18, CD68, HER2 (3B5), CD3, CD20, ER, Ki-67 and panCK. Table A.2 fixes phenotype order and Basel mapping. Native Nd145 retains its isotope identity; CD45/CD45RO, CD31–vWF/vWF-cocktail and Nd145 correspondences are excluded from transfer.

Table A.2. Phenotype mapping for frozen transfer.

Common category	**Basel metacluster identifiers**
Epithelial	14–27
T cell	3, 5
B cell	1
Myeloid	4, 6
Other immune	2
Fibroblast / stromal	8–13
Endothelial	7
Other / unassigned	None assigned

A.1.1 Source Annotations and Patient Linkage

METABRIC cell identifiers are `image_number` and `object_number`, centroids `cell_x`/`cell_y`, and phenotypes `cellPhenotype`. Core-to-patient fields `patient_id`/`metabric_id` link to `brca_metabric.PATIENT_ID`; cores remain distinct (Danenberg et al., 2022; Ali et al., 2020; Curtis et al., 2012; Pereira et al., 2016). Basel uses released annotations (Jackson et al., 2020) and the exclusions in Section 4.1.2. Before splitting, source dictionaries resolve invalid values where possible; otherwise values become missing. Duplicates are reconciled; entirely unobserved training fields remain non-estimable.

METABRIC OS uses `OS_MONTHS`/12 and `OS_STATUS`; causes use `Died of Disease`/`Died of Other Causes`. Basel months are converted to years. DSS counts breast-cancer deaths and censors other causes at death and survivors at last follow-up; unresolved deaths enter OS only (Section 4.1.2). Populations are $\mathcal{P}_{\text{dev}}$, $\mathcal{P}_{\text{gen}} \subseteq \mathcal{P}_{\text{dev}}$ and $\mathcal{P}_{\text{ext}}$.

A.1.2 Clinical and Genomic Encoding

METABRIC inputs are age (years), $\log(\text{size} + 1)$ (millimetres), grade (1–3), node indicators (1–3 and ≥ 4; zero-node reference), PR, HER2 and six missingness flags. Basel/common models use age, grade, PR, HER2 and four flags. Training medians impute age, size and grade before transformation/standardisation; training modes impute nodal category, PR and HER2. Flags precede imputation. Positive PR/HER2 is one; binary inputs remain unscaled. ER defines strata or, for applicable unstratified benchmarks, an input predictor.

Genomic eligibility requires PAM50, IntClust, copy number and panel-level callable sequencing. Binary, unscaled coordinates comprise PAM50 (LumA, LumB, Her2, Basal, claudin-low; Normal-like reference), IntClust (2, 3, 4ER+, 4ER−, 5–10; cluster 1 reference), mutations (*TP53*, *PIK3CA*, *GATA3*, *CDH1*, *MAP3K1*, *KMT2C*; $\geq 5\%$ training prevalence), and copy-number indicators (*ERBB2*, *MYC*, *CCND1*, *FGFR1* gains at +2; *PTEN* loss at −2). Mutation one denotes a non-synonymous call; zero requires callable wild type. Maximum dimension: $5 + 10 + 6 + 5 = 26$.

A.1.3 Quality Control and Marker Numerics

Before the retained-cell threshold, source flags remove normal, in-situ and non-tumour tissue, imaging/segmentation failures and invalid coordinate metadata (METABRIC: `is_normal`, `is_dcis`). Non-finite or negative marker values become zero. Channels are clipped to $[0, q_{0.999}]$ (linearly interpolated training-cell percentile), arcsinh-transformed (cofactor one) and standardised using training cells; SD

$< 10^{-8}$ gives zero. Validation/test transformations and column order remain fixed. Physical-area density is omitted.

A.1.4 Ordered Tissue-Summary Schema

Table A.3 fixes column order and pooling: $Q = A + P + 10 + 45 + 72 + 2 + 2 = A + P + 131$. Coarse phenotype contacts use the eight categories in Table A.2, 36 unordered pairs, including same-type pairs.

Table A.3. Tissue-summary schema.

Block	Size	Patient-level aggregation
Phenotype composition	A	Cell-count-weighted phenotype fractions.
Marker means	P	Cell-count-weighted means after marker preprocessing.
Niche abundance	10	Cell-count-weighted fractions of final niche labels.
Niche contacts	45	All-edge-weighted fractions for canonical pairs $a < b$.
Phenotype contacts and flags	72	36 contact z-scores followed by 36 undefined-value indicators, each pooled by cell count.
Mixing and flag	2	All-edge-weighted mixing ratio; unweighted fraction of cores with an undefined ratio.
Sampling	2	Usable-core count; indicator that any retained core has no graph edges.

Niche-contact denominators count all retained undirected edges. Phenotype-contact z-scores use 500 uniform label permutations on the fixed core graph (seed 20000+ numerical core identifier; SD denominator 499). Zero reference variance, including absent categories, gives score zero and flag one; cell-count pooling permits fractional flags.

Mixing is the epithelial–immune/immune–immune edge ratio, with T, B, myeloid and other immune cells treated as immune. A zero denominator gives ratio zero and flag one. Patient ratios use all-edge weights; flags use the unweighted fraction of undefined cores. Edgeless cores contribute zero to edge-weighted summaries; entirely edgeless patients receive zero summaries and the no-edge flag. All summary columns are standardised on training patients (Section 4.3.5).

A.2 Mathematical Definitions and Properties

A.2.1 Cell Encoding and Regional Pooling

With $\mathbf{h}_v^{(0)} = [\mathbf{x}_v; \mathrm{onehot}(\ell_v)]$, two GraphSAGE layers give

$$\mathbf{h}_v^{(k)} = \mathrm{LN}\Big(\mathrm{ReLU}\Big[W_k^{\mathrm{self}}\mathbf{h}_v^{(k-1)} + W_k^{\mathrm{nb}}\bar{\mathbf{h}}_{\mathcal{N}(v)}^{(k-1)} + \mathbf{b}_k\Big]\Big), \quad k = 1, 2. \tag{A.1}$$

Isolated-cell neighbour means are zero; $d \in \{32, 64\}$. Normalisation/dropout follow Section A.3.3. For instance j,

$$\boldsymbol{\rho}_j^{(0)} = \left[|V_j|^{-1} \sum_{v \in V_j} \mathbf{h}_v ; \log |V_j| \right] \in \mathbb{R}^{d+1}, \tag{A.2}$$

$$\boldsymbol{\rho}_j^{(1)} = \mathrm{ReLU}\left[U^{\mathrm{self}} \boldsymbol{\rho}_j^{(0)} + U^{\mathrm{nb}} \sum_{j' \in \mathcal{N}_{\mathrm{niche}}(j)} \frac{c_{jj'}}{\sum_{j'' \in \mathcal{N}_{\mathrm{niche}}(j)} c_{jj''}} \boldsymbol{\rho}_{j'}^{(0)} + \mathbf{b}_{\mathrm{reg}} \right] \in \mathbb{R}^{d}. \tag{A.3}$$

Crossing-edge counts $c_{jj'} = c_{j'j}$ weight neighbours; isolated instances receive zero neighbour input. Within core r,

$$\mathbf{z}_{r,a}^{\mathrm{node}} = \begin{cases} \dfrac{\sum_{j:t(j)=a} |V_j| \boldsymbol{\rho}_j^{(1)}}{\sum_{j:t(j)=a} |V_j|}, & a \text{ present}, \\ \mathbf{0}, & a \text{ absent}. \end{cases} \tag{A.4}$$

Append $\alpha_{r,a} = \mathbb{I}(a \text{ absent})$ for $K(d+1)$ core coordinates, then pool states and indicators as $\mathbf{z}_i^{\mathrm{node}} = \sum_r w_{ir} \mathbf{z}_r^{\mathrm{node}}$, $w_{ir} = |V_{ir}| / \sum_{r'} |V_{ir'}|$.

A.2.2 Interface Orientation, Projection and Aggregation

For instances with $t(j) < t(j')$, each undirected crossing edge enters once:

$$I_{jj'} = \{(u, v) : \{u, v\} \in E,\ u \in V_j,\ v \in V_{j'}\}, \qquad n_I = |I_{jj'}|. \tag{A.5}$$

NIRP requires $n_I \geq 5$; smaller interfaces remain in graphs and contact summaries. A shared bias-free projection $\Phi \in \mathbb{R}^{d_{\mathrm{p}} \times d}$, $d_{\mathrm{p}} \in \{4, 8\}$, gives

$$\mathbf{x}_e = \Phi \mathbf{h}_{u_e}, \qquad \mathbf{y}_e = \Phi \mathbf{h}_{v_e}, \qquad e = (u_e, v_e) \in I. \tag{A.6}$$

Each cell appears once per incident crossing edge in products and means:

$$\bar{\mathbf{x}}_I = \frac{1}{n_I} \sum_{e \in I} \mathbf{x}_e = \frac{1}{n_I} \sum_u d_u^L \Phi \mathbf{h}_u, \qquad \bar{\mathbf{y}}_I = \frac{1}{n_I} \sum_{e \in I} \mathbf{y}_e = \frac{1}{n_I} \sum_v d_v^R \Phi \mathbf{h}_v. \tag{A.7}$$

Here d_u^L, d_v^R count endpoint appearances. Uniform re-pairing of fixed endpoint lists has expectation

$$\mathbb{E}_\pi \left[\frac{1}{n_I} \sum_{e \in I} \mathbf{x}_e \mathbf{y}_{\pi(e)}^\top \right] = \bar{\mathbf{x}}_I \bar{\mathbf{y}}_I^\top. \tag{A.8}$$

The local readout subtracts this reference: $C_I = n_I^{-1} \sum_{e \in I} (\mathbf{x}_e - \bar{\mathbf{x}}_I)(\mathbf{y}_e - \bar{\mathbf{y}}_I)^\top$. With distinct-cell feature matrices F_L, F_R, biadjacency B and appearance counts $\mathbf{n}_L = B\mathbf{1}$, $\mathbf{n}_R = B^\top \mathbf{1}$, equivalently

$$C_I = \frac{1}{n_I} F_L^\top \left(B - \frac{\mathbf{n}_L \mathbf{n}_R^\top}{n_I} \right) F_R, \qquad n_I = \mathbf{1}^\top B \mathbf{1}. \tag{A.9}$$

For retained interfaces $\mathcal{I}_{ab}^{(r)}$ in core r,

$$N_{ab}^{(r)} = \sum_{I \in \mathcal{I}_{ab}^{(r)}} n_I, \qquad C_{ab}^{(r)} = \begin{cases} \dfrac{\sum_{I \in \mathcal{I}_{ab}^{(r)}} n_I C_I}{N_{ab}^{(r)}}, & N_{ab}^{(r)} > 0, \\ 0, & N_{ab}^{(r)} = 0. \end{cases} \tag{A.10}$$

Set $N_{i,ab} = \sum_r N_{ab}^{(r)}$ and $C_{i,ab} = N_{i,ab}^{-1} \sum_r N_{ab}^{(r)} C_{ab}^{(r)}$ when supported, otherwise zero. With $m_{i,ab} = \mathbb{I}(N_{i,ab} = 0)$,

$$\mathbf{z}_i^{\text{int}} = [\text{vec}(C_{i,ab}); m_{i,ab}]_{1 \le a < b \le K} \in \mathbb{R}^{\binom{K}{2}(d_{\text{p}}^2+1)}. \tag{A.11}$$

All d_{p}^2 entries and coefficients follow type-pair/left-feature/ right-feature order, right features fastest. The flag distinguishes unavailable from observed-zero residuals.

A.2.3 Pooled Centring and Fixed-Feature Properties

For fixed features, interfaces and weights, let $N = \sum_I n_I > 0$, $\pi_I = n_I/N$, $\mu_x = \sum_I \pi_I \bar{\mathbf{x}}_I$ and $\mu_y = \sum_I \pi_I \bar{\mathbf{y}}_I$. Expanding $\mathbf{x}_e - \mu_x = (\mathbf{x}_e - \bar{\mathbf{x}}_I) + (\bar{\mathbf{x}}_I - \mu_x)$ and similarly for $\mathbf{y}_e$ gives

$$\begin{aligned} C_{i,ab}^{\text{pool}} &= \frac{1}{N} \sum_{I \in \mathcal{I}_{i,ab}} \sum_{e \in I} (\mathbf{x}_e - \mu_x)(\mathbf{y}_e - \mu_y)^\top \\ &= \underbrace{\sum_I \pi_I C_I}_{C_{i,ab}} + \underbrace{\sum_I \pi_I (\bar{\mathbf{x}}_I - \mu_x)(\bar{\mathbf{y}}_I - \mu_y)^\top}_{\mathcal{B}_{i,ab}}. \end{aligned} \tag{A.12}$$

Mixed terms vanish because interface-centred deviations sum to zero. The uncentred moment is $M_{i,ab}^{\text{pool}} = C_{i,ab} + \mathcal{B}_{i,ab} + \mu_x \mu_y^\top$.

For fixed features, joint endpoint reordering, consistent cell relabelling and sidewise constant shifts preserve C_I; one-sided re-pairing may not. Side reversal transposes residuals and coefficients. Constant endpoints give zero, and averaging may cancel opposing residuals. The induced matrix $\Phi^\top(\Theta_{ab} + \nu_i \Gamma_{ab})\Phi$ has rank $\le d_{\text{p}}$; compensating projection/coefficient rescaling preserves predictions. For N_r cells and E_r^{ret} retained crossing edges, projection and accumulation cost $O(N_r d d_{\text{p}} + E_r^{\text{ret}} d_{\text{p}}^2)$, excluding grouping.

A.2.4 Efron Objective, Baselines and Score Components

For fitting set $\mathcal{A}$, define $R_s(t) = \{i \in \mathcal{A} : s_i = s, \tilde{T}_i \geq t\}$, $D_s(t) = \{i \in \mathcal{A} : s_i = s, \tilde{T}_i = t, \Delta_i = 1\}$, $D_{\mathcal{A}} = \sum_{s,t} |D_s(t)|$, $\Sigma_s^R(t) = \sum_{i \in R_s(t)} e^{\eta_i}$ and $\Sigma_s^D(t) = \sum_{i \in D_s(t)} e^{\eta_i}$. The event-normalised Efron loss is

$$\mathcal{L}_{\text{Cox}}(\mathcal{A}) = -\frac{1}{D_{\mathcal{A}}} \sum_s \sum_{t:|D_s(t)|>0} \left[\sum_{i \in D_s(t)} \eta_i - \sum_{\ell=0}^{|D_s(t)|-1} \log\left\{ \Sigma_s^R(t) - \frac{\ell}{|D_s(t)|} \Sigma_s^D(t) \right\} \right], \tag{A.13}$$

for $D_{\mathcal{A}} > 0$. Training risk sets are batch-restricted; validation risk sets are complete. Genomic first-stage scores remain fixed offsets. Candidate and refitted baselines use complete inner- and outer-training sets, respectively. Breslow increments are

$$\Delta \widehat{H}_{0,s}(t) = \frac{|D_s(t)|}{\sum_{i \in R_s(t)} \exp(\eta_i)}. \tag{A.14}$$

Baseline estimation requires training target events in the stratum; OS uses a separate all-cause fit.

For $\mathbf{v}_i = [\mathbf{z}_i^{\text{node}}; \mathbf{z}_i^{\text{int}}; \mathbf{q}_i]$, coefficients are $\boldsymbol{\beta}_V + \nu_i \boldsymbol{\gamma}_V$, $\nu_i = \mathbb{I}(s_i = \text{ER}^-)$, with regularised deviations and ER-specific baselines. Tissue, clinical and optional genomic scores add; the tissue score decomposes as

$$\eta_i^{\text{tissue}} = \eta_i^{\text{node}} + \eta_i^{\text{pair}} + \eta_i^{\text{support}} + \eta_i^{q}, \tag{A.15}$$

with pair-specific terms

$$\begin{aligned} \eta_{i,ab}^{\text{pair}} &= \langle \Theta_{ab} + \nu_i \Gamma_{ab}, C_{i,ab} \rangle_F, \\ \eta_{i,ab}^{\text{support}} &= (\vartheta_{ab} + \nu_i \varrho_{ab}) m_{i,ab}, \\ \eta_{i,ab}^{\text{int}} &= \eta_{i,ab}^{\text{pair}} + \eta_{i,ab}^{\text{support}}. \end{aligned} \tag{A.16}$$

Raw pairing terms equal the summed edge allocations in Equation (4.7); support terms and display offsets remain separate.

A.2.5 Horizon-Specific Concordance and Integrated Brier Score

For horizon risk $\hat{\psi}_i(\tau) = 1 - \hat{S}_i(\tau)$, define

$$\begin{aligned} \chi_{ii'}(\tau) &= \mathbb{I}(\tilde{T}_i < \tilde{T}_{i'}, \Delta_i = 1, \tilde{T}_i \leq \tau), \\ \kappa_{ii'} &= \mathbb{I}(\hat{\psi}_i > \hat{\psi}_{i'}) + \tfrac{1}{2}\mathbb{I}(\hat{\psi}_i = \hat{\psi}_{i'}). \end{aligned}$$

ER-specific evaluation censoring survival $\hat{G}_s$ gives

$$\widehat{C}_\tau = \frac{\sum_{i \neq i'} \chi_{ii'}(\tau) w_{ii'} \kappa_{ii'}}{\sum_{i \neq i'} \chi_{ii'}(\tau) w_{ii'}}, \qquad w_{ii'} = \{\hat{G}_{s_i}(\tilde{T}_i^-) \hat{G}_{s_{i'}}(\tilde{T}_i^-)\}^{-1}. \tag{A.17}$$

Reverse Kaplan–Meier treats target events as censored; DSS censoring includes other-cause deaths and follow-up end. For n_{eval} patients,

$$\widehat{\mathrm{BS}}(t) = \frac{1}{n_{\text{eval}}} \sum_{i=1}^{n_{\text{eval}}} \left[\frac{\hat{S}_i(t)^2 \mathbb{I}(\tilde{T}_i \le t, \Delta_i = 1)}{\hat{G}_{s_i}(\tilde{T}_i^-)} + \frac{\{1 - \hat{S}_i(t)\}^2 \mathbb{I}(\tilde{T}_i > t)}{\hat{G}_{s_i}(t)} \right]. \tag{A.18}$$

Patients censored by t contribute zero but remain in the denominator. DSS uses $\hat{S}_{1,\text{net}}$; OS uses its separate all-cause fit. Then

$$\widehat{\mathrm{IBS}}(0, \tau) = \frac{1}{\tau} \int_0^\tau \widehat{\mathrm{BS}}(t)\, \mathrm{d}t, \tag{A.19}$$

using a net subscript for DSS and 0.05-year trapezoidal steps. Censoring estimates weight fixed predictions. Harrell's concordance uses horizon risk; raw-score diagnostics remain within fit and ER stratum. Support rules are in Section A.3.10.

A.3 Implementation Settings

A.3.1 Graph Construction and Prototype Settings

Core graphs are simple, undirected Delaunay graphs without self-loops or cross-core edges. Coordinates use 1 μm pixels; duplicate centroids retain the lower identifier. Degeneracies use Qhull `QJ`; edges $> 30\,\mu$m are removed. Composition uses each cell and its 19 Euclidean nearest neighbours, resolving distance ties by identifier.

Prototype fitting samples ≤ 300 cells per training core without replacement, phenotype-proportionally with largest-remainder rounding (seed $1000r + f$). Square-root proportions enter Lloyd k-means: $K = 10$, ten k-means++ starts, 300 iterations, tolerance 10^{-4} (Lloyd, 1982; Arthur and Vassilvitskii, 2007). Minimum within-cluster sum of squares selects the fit; ties retain the first start. Nearest-prototype ties use the lowest index. Measured profiles describe types (Danenberg et al., 2022; Schürch et al., 2020).

A.3.2 Deterministic Instance Merging

One synchronous pass assigns the modal pre-pass label among each cell and its neighbours; ties retain the current label, otherwise the lowest maximiser. Same-label connected components take their smallest cell identifier. Non-isolated components with < 5 cells are merged in size/identifier order into the neighbour with most crossing edges, breaking ties by type then component identifier. The source adopts the destination label; newly connected same-label components merge immediately. All counts, identifiers

and adjacency update after each merge. Processing stops when every remaining component below five cells is isolated.

A.3.3 Neural Layers and Parameter Counts

GraphSAGE/regional self maps have biases; neighbour maps and Φ do not. Both layer normalisations use $\epsilon = 10^{-5}$ and learned scale/shift; dropout 0.1 follows only the first GraphSAGE normalisation and only during training.

For $F = P + A$ and $D_V = \dim(\mathbf{v}_i)$, the two GraphSAGE layers, regional layer and projection have $2Fd + 3d$, $2d^2 + 3d$, $2d(d + 1) + d$ and dd_{p} parameters. Each GraphSAGE count includes d biases and $2d$ normalisation parameters. Hence

$$N_{\mathrm{neural}} = 2Fd + 4d^2 + 9d + dd_{\mathrm{p}}, \qquad N_{\mathrm{head}} = C + 2D_V.$$

The intercept-free head contains $\boldsymbol{\beta}_C, \boldsymbol{\beta}_V, \boldsymbol{\gamma}_V$; baseline steps are estimated separately. Table A.4 uses $F = 68$, $C = 13$, $K = 10$, $Q = 199$.

Table A.4. NIMSurv parameter counts.

d	d_{p}	D_V	Neural	Head	Total
32	4	1,294	8,864	2,601	11,465
32	8	3,454	8,992	6,921	15,913
64	4	1,614	25,920	3,241	29,161
64	8	3,774	26,176	7,561	33,737

A.3.4 Patient Partitions

For zero-based r, $f \in \{0, \ldots, 4\}$, shuffled five-fold `StratifiedKFold` uses seed $100 + r$; the 20% inner holdout uses `StratifiedShuffleSplit`, seed $1000 + 10r + f$. Both stratify by ER/event status and keep each patient's cores/modalities together. Genomic subsets inherit parent assignments. Of 128 ER-positive and 46 ER-negative DSS events, outer-test folds contain 25–26/9–10 and outer-training folds 102–103/36–37 (means 102.4/36.8; total 139.2).

A.3.5 Training and Benchmark Settings

Tables A.5–A.6 specify implementation and tuning; benchmark selection minimises inner-validation IBS.

Table A.5. Implementation settings.

Item	Specification
Environment	Python 3.12.3; PyTorch 2.8.0+cu128; PyTorch Geometric 2.8.0.post1; scikit-survival 0.28.0; lifelines 0.30.3; pycox 0.3.0; NumPy 2.1.2; pandas 2.3.3; scikit-learn 1.9.1; SciPy 1.18.1. GraphSAGE is implemented directly in PyTorch.
Precision/seeds	float32; global and patient-bootstrap seeds 0.
Validation	Patient-level 5×5 cross-validation; one 20% inner holdout per outer-training partition.
Shared-head fitting	Joint encoder/readout/head training; batch-restricted Cox loss and Adam (Table 4.5).
Shared-head selection	Minimum unpenalised full-risk-set validation Cox loss; replace checkpoints only for improvement $> 10^{-4}$. Restore the best state; exact ties favour smaller $(d, d_{\mathrm{p}}, \lambda_\beta)$ lexicographically. Refit from scratch for the selected epochs.
Benchmark selection	Minimum finite inner-validation IBS; exact ties retain grid order. Iterative models score every epoch, require improvement $> 10^{-4}$ and use patience 20. Restore the best checkpoint and refit for its epoch count; no valid candidate gives undefined selection.

Table A.6. Benchmark grids; selection minimises inner-validation IBS.

Model	Candidates and fixed settings	*n*
CPH	Ridge $\alpha \in \{10^{-4}, 10^{-2}, 1, 10, 100, 1000\}$; ER predictor.	6
CPH-strat	Ridge $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$; ER strata.	5
RSF	Minimum leaf size $\{5, 15, 30\}\times$ maximum features $\{\texttt{sqrt}, 0.5\}$; 500 trees, library-default maximum depth.	6
DeepSurv	Weight decay $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$; MLP 64–32, ELU, dropout 0.2, Adam learning rate 3×10^{-4}.	5
DeepHit	`DeepHitSingle`; $\alpha_{\mathrm{pycox}} \in \{0.2, 0.5\}\times$ weight decay $\{10^{-4}, 10^{-3}\}$; $M = 20$ fixed time bins; other fixed settings below and loss in Equation (A.20).	4
BroadSurv	Enhancement nodes $\{100, 500, 1000\}\times$ ridge $\{10^{-4}, 10^{-2}, 1\}$; 5 time intervals, 10×10 mapping nodes.	9
DAGSurv	KL weight $\beta \in \{0.1, 1, 10\}$; hidden sizes 64/32, Adam learning rate 10^{-3}, batch size 64.	3
AttenSurv-MH	(Learning rate, Bi-LSTM size): $(5 \times 10^{-5}, 2)$ and $\{10^{-3}, 10^{-2}\} \times \{2, 16, 64\}$.	7
DMGN-M	Adam learning rate $\{3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}\}\times$ auxiliary-loss weight $\{1, 4\}$; ResNet-50 + GAT, weight decay 10^{-4}, batch size 8, at most 100 epochs, patience 20.	6

DeepHit details. `DH-1`–`DH-4` order $(\alpha_{\mathrm{pycox}}, \lambda_{\mathrm{wd}})$ lexicographically over $\{0.2, 0.5\} \times \{10^{-4}, 10^{-3}\}$; thus `DH-2` is $(0.2, 10^{-3})$.

Training-fitted `pycox.preprocessing.label_transforms.LabTransDiscreteTime` uses $M = 20$ equidistant bins on $[0, \max \tilde{T}_{\mathrm{train}}]$ for every candidate. `MLPVanilla` has hidden widths 64–32 (as in DeepSurv/DAGSurv), ReLU, batch normalisation ($\epsilon = 10^{-5}$, momentum 0.1), dropout 0.2 after each

hidden layer and an M-dimensional linear output. Training uses at most 200 epochs, batch size 64, Adam learning rate 10^{-3}, $\sigma = 0.1$, patience 20 and IBS improvement $> 10^{-4}$; regularisation is weight decay and dropout, without an additional gradient penalty. Records: `configs/benchmarks/deephit_grid.yaml` and `archive/benchmarks/deephit_cv_selection_2026.csv`.

A.3.6 Benchmark Implementations

CPH/RSF use scikit-survival's `CoxPHSurvivalAnalysis`/ `RandomSurvivalForest`; CPH-strat uses lifelines' `CoxPHFitter`; DeepSurv and `DeepHitSingle` use pycox. BroadSurv, DAGSurv, AttenSurv-MH and DMGN-M use local PyTorch implementations (Wu et al., 2021; Sharma et al., 2021; Sun et al., 2021; Fu et al., 2023).

Table A.7. System-level inputs; tabular preprocessing is training-fitted.

System	Inputs and ER handling
CPH	Clinical predictors and tissue summaries; ER predictor.
CPH-strat	Clinical predictors and tissue summaries; ER-specific baselines.
RSF, DeepSurv, DeepHit, BroadSurv, DAGSurv, AttenSurv-MH	Clinical predictors and tissue summaries; ER predictor where applicable.
DMGN-M	Marker-image regions and clinical inputs; ER predictor; Cox-type head with training-derived baselines.

The default pycox 0.3.0 `DeepHitSingle` loss is (Kvamme, 2024)

$$\mathcal{L}_{\mathrm{DH}} = \alpha_{\mathrm{pycox}}\mathcal{L}_{\mathrm{NLL}} + (1 - \alpha_{\mathrm{pycox}})\mathcal{L}_{\mathrm{rank}}(\sigma), \qquad 0 \le \alpha_{\mathrm{pycox}} \le 1. \tag{A.20}$$

Here α_{pycox} weights likelihood; one gives likelihood-only fitting. Tabular models use the clinical imputation and summary standardisation in Chapter 4.

DMGN-M uses clipped, arcsinh-transformed marker tensors and up to 16 non-overlapping $64 \times 64\,\mu$m regions per core. A learned 1×1 convolution maps markers to three channels before an ImageNet-initialised ResNet-50. A two-layer MLP encodes clinical inputs; a two-layer GAT fuses image and clinical embeddings.

A.3.7 Batch Allocation

$n_{\mathrm{batch}} = \lceil n_{\mathrm{train}}/64 \rceil$: six batches for 348 METABRIC inner-training patients, seven for 435–436 outer-training patients, and four for 220–221 Basel outer-training patients. Each ER stratum is split approximately equally before corresponding subsets are combined.

A.3.8 Preliminary Model for Genomic Augmentation

Matched arms first fit ER-stratified linear Cox scores $\eta_i^{(1)} = \boldsymbol{\beta}_C^\top \mathbf{c}_i$, adding $\boldsymbol{\beta}_G^\top \mathbf{g}_i$ for genomics. The objective is Equation (A.13) plus $\lambda_1(\|\boldsymbol{\beta}_C\|_2^2 + \|\boldsymbol{\beta}_G\|_2^2)$, omitting absent genomics. Full-batch L-BFGS starts at zero (500 iterations; tolerance 10^{-7}); $\lambda_1 \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$ minimises unpenalised full-risk-set inner-validation Cox loss. It supplies in-sample training and held-out validation offsets, then is refitted on all outer-training patients and frozen during tissue refitting. Combined scores determine baselines and outer-test predictions.

A.3.9 Final Settings for Frozen Transfer

Common-input five-year OS settings use source-only inner selection: modal shared-head tuple, ties by mean inner criterion then the lower-dimensional rule; epochs are the rounded-up mean for that tuple. Benchmarks retain their selection procedures. Table A.8 gives final settings; others follow Tables 4.5 and A.6. Final fits use all 544 source patients. Input order, mappings, preprocessing, prototypes, parameters and baselines remain frozen in Basel.

Table A.8. Frozen-transfer model settings.

Model	Final configuration	Epochs
NIMSurv	$d = 32$, $d_{\text{p}} = 4$, $\lambda_\beta = 10^{-2}$	54
Clin	$\lambda_\beta = 10^{-2}$	41
Clin+Sum	$\lambda_\beta = 10^{-2}$	46
Clin+Node	$d = 32$, $\lambda_\beta = 10^{-2}$	49
CPH	Ridge penalty 10^{-4}	—
CPH-strat	Ridge penalty 10^{-4}	—
RSF	Minimum leaf size 15; `max_features = sqrt`	—
DeepSurv	Weight decay 10^{-3}	62
DeepHit	`DeepHitSingle`; configuration `DH-2` ($\alpha_{\text{pycox}} = 0.2$, weight decay 10^{-3}). Fixed settings as in Table A.6.	55
BroadSurv	500 enhancement nodes; L2 penalty 10^{-2}	—
DAGSurv	KL weight $\beta = 1$	48
AttenSurv-MH	Learning rate 10^{-3}; Bi-LSTM size 16	44
DMGN-M	Adam learning rate 10^{-4}; auxiliary-loss weight 4	38

A.3.10 Scoring Support and Paired Bootstrap

Support is checked in inner-validation holdouts, pooled out-of-fold patients per repeat, frozen target cohorts and bootstrap samples. At the fixed horizon, $\geq 20\%$ must remain alive/observed, ≥ 5 target events must have occurred, and every represented ER stratum must have $\widehat{G}_s(\tau^-) \geq 0.05$. Failed support, non-finite inputs or zero comparable-pair denominator give undefined metrics.

NumPy PCG64 (seed 0) resamples patients within ER strata, preserving stratum sizes and all repeat-specific predictions; censoring survival is re-estimated per draw. Metrics are scored per repeat then averaged; transfer uses one prediction per patient. Contrasts share draws and jointly redraw any required undefined score. Percentiles 2.5/97.5 of 1,000 valid draws give pointwise 95% intervals conditional on saved fits.

A.4 Tissue Visualisation and Case Records

A.4.1 Display Reference and Figure Conventions

For outer-training patients $\mathcal{A}_s$ in ER stratum s,

$$b_{ab,s} = \frac{1}{|\mathcal{A}_s|} \sum_{i \in \mathcal{A}_s} \eta^{\text{pair}}_{i,ab}, \qquad \tilde{\eta}^{\text{pair}}_{i,ab} = \eta^{\text{pair}}_{i,ab} - b_{ab,s_i}. \tag{A.21}$$

Display centring preserves predictions; unsupported-pair zeros become $-b_{ab,s_i}$, with offsets kept separate from edge allocations.

Figure 5.2 shows ER-specific Kaplan–Meier curves/risk sets for METABRIC DSS/OS and Basel OS. Figure 5.5 shows repeat-1 out-of-fold raw SDs for 45 aligned pairs: ER-positive above the diagonal ($n = 434$), ER-negative below ($n = 110$), diagonals undefined. ER-positive variation ranks pairs. Top-five boxplots use predicted ten-year net-DSS risk tertiles (145/144/145 patients), centred on each fit's training mean: median/IQR, whiskers to observations within 1.5 IQRs, individual outliers. Units are 10^{-3} log risk; the symmetric-log axis is linear on $[-1, 1]$.

Table A.9. Tissue-linked analyses.

Analysis	Quantity
Score decomposition	Regional, pairing, support and summary contributions.
Contribution variability	Repeat-1 out-of-fold SD, ER-positive ranking and patient support.
Tissue tracing	Two outcome-blind ER-positive cases (Table A.11).
Measured-feature fidelity	Tissue profiles and held-out score reconstruction for five ranked pairs.
Profile alignment	Training-derived phenotype/marker correspondence across fits.

A.4.2 Marker-Based Surrogate Fitting

For pair (a, b) and ER stratum, the frozen edge target is $b_e = (\mathbf{x}_e - \bar{\mathbf{x}}_I)^\top(\Theta_{ab} + \nu_i\Gamma_{ab})(\mathbf{y}_e - \bar{\mathbf{y}}_I)$. Allocations are $a_{i,I,e} = b_e/N_{i,ab}$; patient means recover raw pairing contributions. Inputs are all $36^2 = 1{,}296$ ordered marker products,

$$t_{e,pp'} = (x^{\text{marker}}_{u_e,p} - \bar{x}^{L,\text{marker}}_{I,p})(x^{\text{marker}}_{v_e,p'} - \bar{x}^{R,\text{marker}}_{I,p'}). \tag{A.22}$$

using the outer model's marker preprocessing and NIRP's repeated endpoint appearances for means. Edge weights $w_e = 1/N_{i,ab}$ give each patient unit total weight. Patient-grouped three-fold validation (`GroupKFold`) selects unscaled versus weighted-standardised inputs and ridge $\alpha \in \{10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4\}$ by weighted MSE. Weighted `StandardScaler` is fitted within each inner fold; SD $< 10^{-8}$ gives zero. Ridge uses an unpenalised intercept and Cholesky solver, then refits all supported outer-training patients. Test-edge R^2/RMSE use the same weights; patient metrics equally weight patient means. Zero target variance gives undefined R^2.

Evaluation inventory. Testing uses ER-positive patients in Repeat 1, Folds 2–5, excluding reference Fold 1 so each evaluation follows cross-fit niche alignment (Sections 4.7.2, 4.7.4 and A.4.3). This common, deterministic rule was fixed before inspecting support or fidelity; no extra sampling seed or tie rule is needed beyond outer seed 100. One-based (r, f) corresponds to code indices $(r-1, f-1)$. Eligibility requires ≥ 10 supported training and ≥ 5 test patients.

Table A.10. Surrogate support in Repeat 1. Entries are training/test patient counts; asterisks mark exclusions, all due only to fewer than five test patients. F_1 gives reference-fold test support; it is not a surrogate evaluation.

Pair	F_1	F_2	F_3	F_4	F_5	Eligible
3–6	18	61/31	59/19	48/6	45/4*	3/4
2–9	9	20/6	21/4*	19/4*	18/3*	1/4
2–10	9	63/21	64/18	65/17	62/17	4/4
2–4	20	85/24	87/22	86/22	88/21	4/4
9–10	4	25/10	26/8	27/6	24/5	4/4

Across all five folds, support totals are 78, 26, 82, 109 and 33 (Table 5.9); eligible test ranges are 6–31, 6, 17–21, 21–24 and 5–10. Of 20 attempts, 16 qualify (3/1/4/4/4 by pair), as reported in Tables 5.10 and B.8.

A.4.3 Niche Alignment Cost

The 25 outer fits align to Repeat 1, Fold 1 within each input scheme. Profiles use outer-training cells after smoothing/merging. For fit m, type a, $\widetilde{\mathbf{p}}_{m,a} \in \mathbb{R}^A$ contains phenotype proportions standardised across its ten types (constant coordinates zero); $\widetilde{\boldsymbol{\mu}}_{m,a} \in \mathbb{R}^P$ contains mean training-standardised markers. The cost is

$$c_m(b,a) = \left[\frac{\|\widetilde{\mathbf{p}}_{m,b} - \widetilde{\mathbf{p}}_{\mathrm{ref},a}\|_2^2}{A} + \frac{\|\widetilde{\boldsymbol{\mu}}_{m,b} - \widetilde{\boldsymbol{\mu}}_{\mathrm{ref},a}\|_2^2}{P} \right]^{1/2}.$$

Hungarian assignment minimises $\sum_{a=1}^{10} c_m(\sigma_m(a), a)$ with rows/columns in native-identifier order. The solver is `scipy.optimize.linear_sum_assignment`. Dimension normalisation equally weights phenotype/marker blocks. Alignment changes abundance summaries/displays, not prediction coordinates. Reversed pairs transpose both residuals and coefficients, preserving $\langle \Theta^\top, C^\top \rangle_F = \langle \Theta, C \rangle_F$.

A.4.4 Tissue-Tracing and Prognostic Cases

Outcome-blind tracing selects supported ER-positive patients nearest the 10th/90th predicted ten-year net-DSS risk percentiles, then the largest retained core with non-zero allocations, and the pair/interface with largest absolute summed contribution; identifiers break ties. Profiles count unique incident cells. Maps use source coordinates and keep raw pairing allocations separate from support terms and display offsets.

Table A.11. Tissue-tracing case records.

	Lower predicted risk	**Higher predicted risk**
Patient	MB-0232	MB-0000
Validation repeat / fold	1 / 2	1 / 5
Core identifier	82	64
Niche-type pair	1–4	1–2
Selected interface	10	3
Raw interface log-risk term	$\approx 1.26 \times 10^{-3}$	$\approx 5.14 \times 10^{-3}$
Unique cells on the two sides	5 / 4	5 / 6
Displayed phenotype classes	7	3
Displayed marker contrasts	5	5

The separate, outcome-informed prognostic example (Figure 5.8) selects ER-positive MB-2933 and MB-0288 across five repeats, held out together with similar clinical-plus-summary CPH risks (3.34/3.06%), contrasting NIMSurv risks (8.22/51.30%) and ten-year DSS outcomes. Overview/inset scale bars are 100/25 source-image pixels.

Appendix B

Supplementary Results

Unless stated otherwise, results are METABRIC ten-year model-based net DSS. Intervals are patient-paired bootstrap (Section A.3.10); differences are NIMSurv minus comparator, with positive ΔC and negative ΔIBS favouring NIMSurv.

B.1 Niche Partition and Measured Profiles

Table B.1. Niche partition and interface availability.

A. Tissue regions and retained interfaces

Quantity	Reference fit	Fit range
Cells relabelled by smoothing, %	3.92	3.42–3.92
Instances per core, median (IQR)	20 (12, 30)	18–20
Instances merged by the five-cell rule, %	26.77	25.53–26.96
Instance size, cells, median (IQR)	13 (6, 32)	13–14
Candidate interfaces per core, median (IQR)	20 (9.5, 39.5)	18–20
Interfaces retained ($n_I \geq 5$), %	77.71	77.71–79.93
Crossing edges per retained interface, median (IQR)	12 (8, 20)	12–13
Supported pairs per patient, median (IQR), out of 45	6 (3, 9)	5–6
Pairs unsupported in > 50% of patients	44	44–44
Cores with undefined mixing ratio	44	—
Undefined contact z-score entries	7,911	—

B. Niche abundance (percentage of all-patient cells)

Type	Reference	Fit range	Type	Reference	Fit range
1	10.2	10.1–11.2	6	11.3	4.0–17.5
2	8.4	8.0–8.6	7	12.8	10.8–13.9
3	16.1	10.4–19.8	8	6.5	6.3–6.6
4	8.2	7.8–9.5	9	11.1	4.7–14.0
5	5.0	4.9–5.0	10	10.4	10.3–15.6

Table B.2. Reference niche profiles.

Type / working name	Dominant phenotypes	Highest mean markers (SD units)	Cells (%)
1: Fibroblast-rich	Fibroblasts (51.4%) Endothelial (9.0%) Myofibroblasts (8.2%)	HLA-DR (−0.02) CD163 (−0.03) CD68 (−0.06)	10.82
2: $ER^{hi}CXCL12^{+}$	$ER^{hi}CXCL12^{+}$ (48.4%) $CK^{med}ER^{lo}$ (10.1%) $CK8\text{-}18^{hi}CXCL12^{hi}$ (7.1%)	ER (1.31) PD-1 (1.00) c-Caspase3 (0.80)	8.99
3: $CK^{med}ER^{lo}$	$CK^{med}ER^{lo}$ (39.0%) $CK^{lo}ER^{lo}$ (18.9%) Fibroblasts (7.8%)	HER2 (3B5) (0.16) panCK (0.15) ER (0.06)	15.92
4: Myofibroblasts	Myofibroblasts (31.6%) Endothelial (25.0%) Fibroblasts (12.6%)	Caveolin-1 (1.02) SMA (0.79) PDGFRB (0.79)	8.38
5: $HER2^{+}$	$HER2^{+}$ (59.9%) Myofibroblasts (10.6%) Fibroblasts (4.6%)	HER2 (D8F12) (2.85) HER2 (3B5) (1.67) SMA (0.40)	5.45
6: $CK^{+}CXCL12^{+}$	CK^{+} $CXCL12^{+}$ (30.9%) $CK^{med}ER^{lo}$ (16.0%) Fibroblasts (11.3%)	panCK (0.45) CXCL12 (0.23) CK8-18 (0.21)	12.43
7: Mixed immune	B cells (19.8%) $CD8^{+}$ T cells (13.7%) Macrophages (12.0%)	HLA-ABC (1.09) HLA-DR (1.04) B2M (1.03)	12.29
8: $CD15^{+}$	$CD15^{+}$ (54.7%) $CK^{med}ER^{lo}$ (6.1%) Myofibroblasts (5.4%)	CD15 (2.28) panCK (0.63) CD57 (0.39)	5.84
9: Basal	Basal (19.8%) $CK^{lo}ER^{lo}$ (18.0%) Ep $CD57^{+}$ (7.9%)	CK5 (1.08) CD57 (0.78) Ki-67 (0.38)	9.92
10: $CK8\text{–}18^{hi}$ $CXCL12^{hi}$	$CK8\text{-}18^{hi}CXCL12^{hi}$ (49.0%) CK^{+} $CXCL12^{+}$ (12.1%) $CK^{med}ER^{lo}$ (7.2%)	CK8-18 (1.25) CXCL12 (1.17) panCK (0.95)	9.96

B.2 Detailed Primary Comparisons

Table B.3. Ten-year net-DSS prediction in METABRIC.

Model	Uno's C ↑ 10 years	IBS ↓ 0–10 years	Paired ΔC ↑ [95% interval]	Paired ΔIBS ↓ [95% interval]
NIMSurv	**0.703**	**0.113**	*Reference*	*Reference*
Shared-head clinical and tissue references				
Clin	0.613	0.190	+0.090† [+0.060, +0.140]	≈ −0.077† [−0.0894, −0.0646]
Clin+Comp	0.668	0.121	+0.035† [+0.006, +0.063]	≈ −0.008† [−0.0132, −0.0032]
Clin+Sum	0.625	0.125	+0.078† [+0.041, +0.115]	≈ −0.012† [−0.0179, −0.0066]
Clin+Flat	0.612	0.128	+0.091† [+0.052, +0.130]	≈ −0.015† [−0.0212, −0.0085]
Clin+Node	0.606	0.126	+0.097† [+0.057, +0.136]	≈ −0.013† [−0.0195, −0.0071]
System-level references				
CPH	0.574	0.211	+0.129† [+0.087, +0.172]	≈ −0.098† [−0.1142, −0.0823]
CPH-strat	0.660	0.126	+0.043† [+0.012, +0.074]	≈ −0.013† [−0.0222, −0.0050]
RSF	0.675	0.127	+0.028† [+0.005, +0.058]	≈ −0.014† [−0.0189, −0.0091]
DeepSurv	0.613	0.130	+0.091† [+0.054, +0.129]	≈ −0.017† [−0.0228, −0.0104]
DeepHit	0.593	0.129	+0.110† [+0.074, +0.144]	≈ −0.016† [−0.0217, −0.0098]
BroadSurv	0.587	0.142	+0.116† [+0.078, +0.149]	≈ −0.029† [−0.0362, −0.0227]
DAGSurv	0.598	0.128	+0.105† [+0.060, +0.146]	≈ −0.015† [−0.0214, −0.0079]
AttenSurv-MH	0.576	0.189	+0.127† [+0.082, +0.167]	≈ −0.076† [−0.0864, −0.0646]
DMGN-M	0.652	0.123	+0.052† [+0.030, +0.072]	≈ −0.010† [−0.0145, −0.0061]

† marks favourable intervals excluding zero.

Table B.4. Boundary-pairing comparisons for ten-year net DSS.

Model	Uno's C ↑ 10 years	IBS ↓ 0–10 years	Paired ΔC ↑ [95% interval]	Paired ΔIBS ↓ [95% interval]
NIMSurv	**0.703**	**0.113**	*Reference*	*Reference*
Interface variants				
Support-only	0.652	0.135	+0.051† [+0.015, +0.087]	−0.022† [−0.038, −0.006]
Pooled-centred	0.670	0.132	+0.033 [−0.005, +0.071]	−0.019 [−0.041, +0.003]
Uncentred	0.692	**0.113**	+0.011 [−0.018, +0.040]	≈ 0 [−0.008, +0.008]
Input-feature	0.642	**0.113**	+0.061† [+0.022, +0.100]	≈ 0 [−0.010, +0.010]

† marks favourable intervals excluding zero.

B.3 Supplementary Metrics and Survival Targets

Table B.5. Additional ten-year net-DSS metrics in METABRIC.

Model	Harrell's C ↑ Observed follow-up	AUC ↑ 10 years	Brier score ↓ 10 years
NIMSurv	**0.706**	**0.734**	**0.181**
Shared-head clinical and tissue references			
Clin	0.626	0.633	0.281
Clin+Comp	0.670	0.706	0.189
Clin+Sum	0.622	0.656	0.198
Clin+Flat	0.613	0.644	0.204
Clin+Node	0.604	0.636	0.202
System-level references			
CPH	0.578	0.582	0.321
CPH-strat	0.661	0.689	0.201
RSF	0.676	0.711	0.188
DeepSurv	0.610	0.637	0.207
DeepHit	0.589	0.612	0.207
BroadSurv	0.591	0.594	0.245
DAGSurv	0.595	0.625	0.203
AttenSurv-MH	0.576	0.611	0.236
DMGN-M	0.652	0.675	0.201

Table B.6. Concordance for secondary METABRIC outcomes.

Model	Uno's C ↑ DSS, 5 years	Uno's C ↑ OS, 10 years
NIMSurv	**0.735** [0.686, 0.781]	**0.695** [0.662, 0.728]
System-level references		
CPH	0.601 [0.548, 0.655]	0.576 [0.543, 0.610]
RSF	0.711 [0.663, 0.758]	0.681 [0.650, 0.714]
DeepSurv	0.629 [0.588, 0.673]	0.596 [0.566, 0.624]
DeepHit	0.602 [0.561, 0.644]	0.585 [0.556, 0.610]
BroadSurv	0.610 [0.578, 0.642]	0.588 [0.565, 0.611]
DAGSurv	0.611 [0.570, 0.655]	0.593 [0.564, 0.619]
AttenSurv-MH	0.579 [0.545, 0.614]	0.576 [0.548, 0.602]
DMGN-M	0.667 [0.627, 0.704]	0.680 [0.650, 0.710]

B.4 Source-Cohort Transfer Evaluation

Table B.7 reports held-out METABRIC five-year OS performance with Basel-compatible inputs; frozen Basel results are in Table 5.7.

Table B.7. Common-input five-year OS prediction in METABRIC.

Model	Uno's C ↑ 5 years	IBS ↓ 0–5 years	Paired ΔC ↑ [95% interval]	Paired ΔIBS ↓ [95% interval]
NIMSurv	**0.664**	**0.082**	*Reference*	*Reference*
Shared-head clinical and tissue references				
Clin	0.564	0.092	+0.101† [+0.013, +0.188]	−0.010† [−0.019, −0.002]
Clin+Sum	0.635	0.086	+0.030† [+0.008, +0.066]	−0.003† [−0.006, 0.000]
Clin+Node	0.628	0.086	+0.036† [+0.003, +0.076]	−0.003† [−0.006, 0.000]
System-level references				
CPH	0.612	0.113	+0.052† [+0.008, +0.097]	−0.031† [−0.039, −0.023]
CPH-strat	0.657	0.088	+0.007† [+0.002, +0.009]	−0.005† [−0.010, −0.001]
RSF	0.627	0.086	+0.038† [+0.003, +0.078]	−0.003† [−0.007, 0.000]
DeepSurv	0.587	0.089	+0.078† [+0.037, +0.122]	−0.007† [−0.011, −0.003]
DeepHit	0.562	0.088	+0.103† [+0.062, +0.144]	−0.006† [−0.011, −0.002]
BroadSurv	0.586	0.101	+0.079† [+0.039, +0.118]	−0.018† [−0.024, −0.012]
DAGSurv	0.552	0.088	+0.113† [+0.066, +0.161]	−0.006† [−0.010, −0.002]
AttenSurv-MH	0.542	0.166	+0.122† [+0.074, +0.172]	−0.084† [−0.093, −0.074]
DMGN-M	0.651	0.086	+0.013† [+0.010, +0.038]	−0.003† [−0.006, −0.001]

† marks favourable intervals excluding zero before rounding.

B.5 Marker-Surrogate Reconstruction Error

Median RMSE supplements Table 5.10; support and exclusions appear in Table A.10.

Table B.8. Marker-based reconstruction error for held-out pairing scores.

Pair	Edge RMSE Median	Patient RMSE Median
3–6	0.01422	0.00507
2–9	0.04273	0.02581
2–10	0.01898	0.00585
2–4	0.02258	0.00974
9–10	0.02080	0.01037

Appendix C

Literature Search and Screening Record

This record supports Chapter 3; archive files are listed in Section C.5.

C.1 Database Search Histories

Imaging searches combine platform, representation and outcome blocks with `AND`, and within-block terms with `OR` (Table 3.1). Initial searches: 15–16 March 2026; update: 1 November 2026; coverage: 1 January 2014–1 November 2026. English filters apply where supported. Exact queries and source-specific filters are retained below.

Table C.1. Database queries and retrieval counts. Counts are initial + update; the combined total is 355 + 37 = 392.

Source and record	Exact query; applied filters
PubMed / MEDLINE 109 + 9 `pubmed_main_2026.bib`	`("imaging mass cytometry"[tiab] OR "multiplexed imaging"[tiab] OR "multiplex immunofluorescence"[tiab] OR "MIBI"[tiab] OR "CODEX"[tiab] OR "cyclic immunofluorescence"[tiab] OR "spatial proteomics"[tiab]) AND ("graph neural network"[tiab] OR "cell graph"[tiab] OR "cellular neighbourhood"[tiab] OR "cellular neighborhood"[tiab] OR "niche"[tiab] OR "tissue microenvironment"[tiab] OR "spatial statistics"[tiab] OR "cell-cell interaction"[tiab]) AND ("survival"[tiab] OR "prognosis"[tiab] OR "prognostic"[tiab] OR "hazard"[tiab] OR "recurrence"[tiab] OR "outcome"[tiab])` *Filters:* 2014-01-01 to 2026-11-01; English.
Scopus 151 + 13 `scopus_main_2026.bib`	`TITLE-ABS-KEY(("imaging mass cytometry" OR "multiplexed imaging" OR "multiplex immunofluorescence" OR "MIBI" OR "CODEX" OR "cyclic immunofluorescence" OR "spatial proteomics") AND ("graph neural network" OR "cell graph" OR "cellular neighbo*rhood" OR "niche" OR "tissue microenvironment" OR "spatial statistics" OR "cell-cell interaction") AND ("survival" OR "prognosis" OR "prognostic" OR "hazard" OR "recurrence" OR "outcome"))` *Filters:* `PUBYEAR > 2013`; `LANGUAGE(English)`; document types ar, cp, ip.

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Table C.1 (continued)

Source and record	Exact query; applied filters
arXiv (API / advanced search) 36 + 6 `arxiv_main_2026.bib`	`abs:("imaging mass cytometry" OR "multiplexed imaging" OR "spatial proteomics" OR "CODEX" OR "multiplex immunofluorescence") AND abs:("graph neural network" OR "cell graph" OR "niche" OR "microenvironment" OR "spatial") AND abs:("survival" OR "prognosis" OR "prognostic" OR "hazard" OR "outcome")` *Filters:* Submitted 2014-01-01 to 2026-11-01; cs.LG, cs.CV, q-bio, stat.AP.
bioRxiv / medRxiv 59 + 9 `biorxiv_main_2026.bib`	`("imaging mass cytometry" OR "multiplexed imaging" OR "spatial proteomics" OR "CODEX") AND ("graph neural network" OR "cell graph" OR "cellular neighborhood" OR "niche") AND ("survival" OR "prognosis" OR "prognostic" OR "outcome")` *Filters:* Posted 2014-01-01 to 2026-11-01; latest version retained.

C.2 Methodological Searches

Four searches omit the imaging-platform block. Initial searches: 20–21 March 2026; all updated 1 November 2026. Table C.2 retains each recorded query, platform, date, filter, count and export.

Table C.2. Methodological queries and retrieval counts (n = 132).

Strand and record	Exact query; applied filters
1. Second-order / bilinear pooling and edge-conditioned graph learning Scopus, arXiv; 20 March 2026 n = 46; `strand1_pooling.csv`	`TITLE-ABS-KEY(("second-order pooling" OR "bilinear pooling" OR "covariance pooling" OR "edge-conditioned" OR "bipartite modularity") AND ("graph neural network" OR "graph pooling" OR "representation learning" OR "spatial"))` *Filters:* English; peer-reviewed or foundational arXiv preprints.
2. Competing risks and calibration in survival prediction PubMed, Scopus; 20 March 2026 n = 34; `strand2_comprisk.csv`	`TITLE-ABS-KEY(("competing risks" OR "cause-specific hazard" OR "subdistribution hazard" OR "net survival") AND ("prediction model" OR "concordance" OR "Brier score" OR "calibration"))` *Filters:* English; biostatistics and clinical methodology.
3. Variance and comparison of cross-validated estimates Scopus, Google Scholar; 21 March 2026 n = 28; `strand3_cvvar.csv`	`TITLE-ABS-KEY(("cross-validation" OR "resampled" OR "bootstrap") AND ("variance" OR "comparing classifiers" OR "paired comparison" OR "generalization error"))` *Filters:* English; statistical learning and evaluation.

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Table C.2 (continued)

Strand and record	Exact query; applied filters
4. Fidelity and faithfulness of graph explanations Scopus, arXiv; 21 March 2026 n = 24; `strand4_fidelity.csv`	`TITLE-ABS-KEY(("graph neural network" OR "saliency" OR "attribution") AND ("explanation" OR "explainability") AND ("fidelity" OR "faithfulness" OR "sanity checks" OR "surrogate"))` *Filters:* English; 2017–2026.

C.3 Citation Tracking

Eight seeds (Section 3.1.2) underwent backward reference screening and forward Google Scholar tracking, last checked 1 November 2026: 62 records, 27 overlaps and 35 unique records.

Table C.3. Citation tracking for the seed papers.

Seed	Role	Retrieved	Overlap	Unique
Jackson et al. (2020)	Breast IMC atlas	11	6	5
Ali et al. (2020)	Breast IMC phenogenomics	9	5	4
Danenberg et al. (2022)	Breast IMC TME structures	12	6	6
Fu et al. (2023)	Multimodal IMC survival (DMGN)	8	3	5
Rifaioglu et al. (2026)	Cell-graph IMC survival (SpaCEy)	6	2	4
Wang and Ji (2023)	Second-order graph pooling	5	2	3
Haviv et al. (2025)	Covariance environment (COVET)	5	1	4
Bhate et al. (2022)	Connected-instance motifs	6	2	4
Total		62	27	35

C.4 Screening Flow and Full-Text Exclusions

Duplicates/overlaps (n = 174): 128 cross-database/preprint–journal, 19 methodological/database and 27 citation-tracking. Included records (n = 91): 17 biological associations, 13 held-out survival/progression prediction studies and 61 methodological/clinical foundations. Edgren Schüllerqvist et al. (2025, 2027) count as one study across versions.

C.5 Electronic Archive and File Inventory

Archive: `archive/literature_search/` (queries, exports, deduplication mappings and screening logs).

Table C.4. Retrieval counts and screening flow.

Stage	Database	Methods	Citations	Total
Identified	392	132	62	586
Duplicates/overlaps removed	−128	−19	−27	−174
Titles/abstracts screened	264	113	35	412
Titles/abstracts excluded	−196	−64	−14	−274
Full texts assessed	68	49	21	138
Full texts excluded	−31	−11	−5	−47
Included in synthesis	37	38	16	91

Table C.5. Full-text exclusion reasons. Full decisions and metadata: `fulltext_exclusions_2026.csv`.

Reason	*n*	Log identifiers and scope
Non-spatial analysis	18	`EXCL-01-18`: bulk, marginal abundance or dissociated expression without spatial graph, neighbourhood or distance modelling.
Single-marker / non-transferable assay	12	`EXCL-19-30`: single-plex IHC or routine staining without multiplexed single-cell resolution or transferable spatial graphs.
Non-human / non-transferable study	7	`EXCL-31-37`: descriptive murine/organoid imaging without patient survival or generalisable graph pooling.
Insufficient methods	6	`EXCL-38-43`: abstracts/summaries lacking mathematical definitions, cohort partitions or held-out protocols.
Superseded preprint	4	`EXCL-44-47`: arXiv/bioRxiv versions replaced by Rifaioglu (2026), Tsang (2026), Xie (2026) and Haviv (2025).
Total	47	`EXCL-01-47`

Table C.6. Search and screening file inventory.

Files	Format	Content
Four exports in Table C.1	BibTeX	Initial/update results (n = 392).
`strand1_pooling.csv`–`strand4_fidelity.csv`	CSV	Queries and metadata (n = 132).
`seed_citation_tracking_2026.csv`	CSV	Eight-seed tracking and overlap flags (n = 62).
`title_abstract_screening_log.csv`	CSV	Deduplicated provenance, decisions and eligibility (n = 412).
`fulltext_exclusions_2026.csv`	CSV	Full-text decisions (n = 138): 91 included; 47 excluded with primary codes (`EXCL-01-47`).